

Explainable AI Unravels Sepsis Heterogeneity via Coagulation-Inflammation Profiles for Prognosis and Stratification

Corresponding Author: Dr Bailin Niu

Parts of this Peer Review File have been redacted as indicated to remove third-party material.

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The study looks at a significant problem. While prior literature has suggested various algorithms that can predict sepsis very well and, at times, better than physicians, this paper has three unique contributions. First, the model is built using external datasets and validated using a separate hospital dataset which presents some form of external validation for the model. Second, the development of a scoring system presents a possible translational use of this algorithm in clinical environments where the operating of real-time algorithms is not feasible. Third, it looks at biomarkers and provides some explanatory attempts of predicting sepsis, which is something challenging.

There are, however, some points to consider in the revision.

The writing, although clear, does not highlight the key findings of the paper. The results section focused excessively on the methods employed and did not provide sufficient clinical explanation as to why these biomarkers matter in explaining sepsis prediction. For example, SHAP plots are good for visualizing the importance of the predictors in the model, but in the main body of the paper, more should be presented to explain the clinical aspects of the findings. The methodological reporting should go into the methods section which also needs further revisions.

I find the explanatory part of sepsis prediction something that is significant, but this current form does not provide sufficient write-up in this area. I hope the authors can provide more details that may contribute to the clinical understanding of sepsis onset.

In terms of the reporting of the methods, here, less is perhaps more. The authors have presented a multitude of methods to support their results. Although in general, I do believe in the findings from a reviewer perspective, the large number of methods employed with limited reporting makes it exacting for the reviewer to establish the correctness of the execution of these methods. To illustrate, in the text, the authors wrote "[...] radar plots of subtypes identified by unsupervised clustering methods, including GMM, MiniBatchKMeans, K-means, hierarchical agglomerative clustering (HAC), and Birch [...]". The writing here presents more of a report of a collection of methods with limited ability for the reviewers to verify and perhaps add limited value to the paper other than to signal to the review team that extensive work has been put into the paper. A similar writing style is pervasive in the entire methods section, e.g., "[...] MIMIC-III and MIMIC-IV are merged, and SMOTE is employed to balance the class distribution [...]" For example, what is the extent of SMOTE (percentage, number of neighbors used, etc.)? Even though I downloaded all the datasets in this submission, reading this article does not permit any reviewer to completely comprehend the inner workings of the analysis based on the reporting.

I suggest the authors do one of the two. 1) In the main paper, report a shorter list of analyses employed, but do that in a more comprehensive and clearer manner, complete with a STROBE or modified CONSORT diagram to showcase the data used. This presentation will lend more credibility to the results. If you want to showcase additional analyses reported here, maybe provide some additional supplement you might want to host on GitHub. These supplementary analyses may have minute computational errors or bugs – which is normally the case, but it does not fall within the purview of NComms review process.

2) Continue with your existing reporting of all the collection of analyses, but in your next review, you perhaps will have to consider providing more details of how the analyses for each technique is done, including the software codes in an executable format. This will also make the review process more exacting.

(Remarks on code availability)

I have downloaded the datasets, but the current reporting make it onerous and not possible for one to review the analysis in detail.

Reviewer #2

(Remarks to the Author)

1. What are the noteworthy results?

This study has some key findings.

(A) High Prognostic Accuracy of SepsisFormer

- SepsisFormer (Transformer-based model) achieves an AUC of 0.9301, outperforming traditional risk scoring models like qSOFA, SOFA, and APACHE II.
- The model demonstrates generalizability across multiple cohorts (MIMIC-III, MIMIC-IV, eICU-CRD, and a local ICU dataset).

(B) SMART Risk Stratification Model Improves Sepsis Classification

- SMART (a scoring system based on seven coagulation-inflammatory markers and patient age) achieved an AUC of 0.7360, surpassing most established clinical scoring systems.
- The study identifies four risk levels (Mild, Moderate, Severe, Dangerous), with clear mortality risk stratification (~5%, 15%, 30%, and 50% mortality rates, respectively).

(C) Identification of Two Sepsis Subphenotypes (CIS1 and CIS2) with Different Outcomes

- CIS2 patients exhibited higher mortality rates (~32.23%) compared to CIS1 (25.84%), and a stronger inflammatory-coagulation response.
- The study demonstrates that patients with CIS2 benefit more from anticoagulant therapy (e.g., heparin), while CIS1 shows moderate benefits.

(D) Transcriptomic Analysis Identifies Sepsis Biomarkers

- Five transcriptomic biomarkers (STAT5B, MTHFR, HPSE, AAK1, MX1) were found to be significantly linked to sepsis prognosis.
- The study suggests these biomarkers influence sepsis prognosis through immune and coagulation pathways.

2. Will the work be significant to the field and related fields?

Yes, this work is significant to the field.

Relevance to Sepsis Research:

- The study contributes to sepsis mortality prediction and risk stratification, both of which are essential for clinical decision-making in ICUs.
- The identification of coagulation-inflammatory subtypes (CIS1 and CIS2) aligns with ongoing efforts to better phenotype sepsis patients.

Relevance to Machine Learning & Explainable AI in Healthcare:

- SepsisFormer demonstrates that deep learning models can achieve superior predictive performance while maintaining interpretability through XAI methods.
- SMART scoring provides a structured risk assessment tool that could be clinically deployed.

Potential Applications in Other Fields:

- The approach can be extended to other critical conditions (e.g., septic shock, ARDS, multi-organ failure) where heterogeneity complicates prognosis.
- The explainability techniques used here could be applied to other AI-driven healthcare models.

Comparison to Existing Literature:

SepsisFormer's AUC of 0.9301 is notably higher than previously reported AI-based sepsis models (e.g., prior deep learning models in literature typically range from 0.85–0.91).

Risk stratification studies, such as those defining MARS1-4 or α , β , γ , δ subtypes, have not explicitly linked coagulation-inflammatory markers to treatment response.

However, the study does not compare CIS1 and CIS2 to other existing sepsis subtyping frameworks (e.g., MARS1-4, SRS1-2, hyper-/hypo-inflammatory states).

3. Does the work support the conclusions and claims, or is additional evidence needed?

Most claims are supported by strong experimental data.

Well-supported claims:

SepsisFormer outperforms existing scoring systems → Supported by ROC curves, AUC analysis, external validation across multiple datasets.

SMART provides a clinically relevant stratification system → Clear association between SMART's risk levels and increasing mortality.

CIS2 patients benefit more from anticoagulants → Kaplan-Meier survival analysis demonstrates significant mortality reduction in CIS2 patients receiving anticoagulation.

Less-supported claims:

CIS1/CIS2 are novel sepsis subtypes → The study does not compare them to previous subtyping models (e.g., MARS1-4, α -

δ).
Transcriptomic biomarkers are clinically relevant for sepsis prognosis → No external validation of transcriptomic findings in an independent dataset.

Recommendation:

- Benchmark CIS1/CIS2 against prior subtyping frameworks.
- Validate transcriptomic biomarkers in a completely independent dataset.

4. Are there any flaws in the data analysis, interpretation, and conclusions?

Overall, the methodology is robust, but a few concerns require attention.

(A) External Validation of SepsisFormer Needs Clarification

How the data were split for training, validation, and testing is unclear.

Recommendation: Clearly define cross-validation strategy and how model generalizability was assessed.

(B) SMART's Performance Varies Across Datasets

SMART's AUC ranges from 0.6222 to 0.7360—why does it perform worse in some cohorts?

Recommendation: Provide dataset-level breakdown and discuss factors influencing variability.

(C) Statistical Comparisons Are Missing for Model Performance

No DeLong's test or confidence intervals are reported to compare AUCs of different models.

Recommendation: Perform statistical comparisons to confirm that SepsisFormer is significantly better than traditional methods.

5. Is the methodology sound? Does the work meet the expected standards in your field?

Yes, the study follows best practices for machine learning in medicine, but it requires clarifications.

Areas for Improvement:

No external validation of transcriptomic biomarkers → Validation in an independent cohort is needed.

No prospective validation of SepsisFormer in real-time ICU settings.

6. Is there enough detail provided in the methods for the work to be reproduced?

The methodology is well-documented but requires additional clarification in some areas.

Details that are clear and reproducible:

- * Feature selection process for SMART.
- * SHAP-based explainability analysis.
- * Survival analysis methodology.

Details that require clarification:

- * How were CIS1 and CIS2 validated against existing subphenotypes?
- * How was model performance compared across datasets (cross-validation strategy)?
- * What hyperparameters were used in training SepsisFormer?

Recommendation:

- * Provide hyperparameter details for SepsisFormer.
- * Clarify subphenotype validation and external benchmarking.

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)

1. What are the noteworthy results?

- A prognosis model for sepsis that leverages Transformer models and domain adaptation technique that achieves high predictive accuracy and a scorecard model for risk stratification.

- The actual subphenotypes and risk levels classified by the findings from the model experiments, and how these actually capture patient heterogeneity. Although, I could appreciate this contribution later in the paper.

- An in depth analysis of and evidence for the heterogeneity of sepsis patients based on subphenotypes and risk levels.

2. Will the work be of significance to the field and related fields? How does it compare to the established literature? If the work is not original, please provide relevant references.

- The performance of existing AI prediction/prognosis models is mentioned in the introduction but does not include the techniques used in those models, which can help the reader to appreciate how the proposed models in this work contribute and differ from existing literature (in terms of input data, model choice, training paradigms, etc.).

- In the domain adaptive generation, the method for oversampling is claimed to be novel but also says that it uses existing domain adaptation methods. It is not clear what the novelty of the method is. Besides, the domain adaptation is briefly mentioned in the abstract and discussed again in the results under the Prognostic prediction performance of SepsisFormer based on coagulation–inflammatory markers header. The last paragraph of the introduction can include this component, as it seems relevant considering the heterogeneity and class imbalance problems it addresses.

- The transcriptomic biomarker evidence analysis and study of anticoagulant treatment effect seem crucial for better understanding sepsis, reveal insights about heterogeneity, and validate algorithmic model development. However, I think

that the presentation of these approaches can be clarified because it wasn't clear what experiments actually took place in this study or were using measurements from existing public datasets (E.g., the Heparin vs control studies, or PCR studies), and how the biomarkers were actually selected.

3. Does the work support the conclusions and claims, or is additional evidence needed?

- As I was reading the paper, I struggled to understand the support for some of the claims in the results section since it wasn't clear the criteria used for doing further classifications or analyses. For example, how does further categorizing the predictors into clinically relevant ranges (not clear how these are defined) facilitate clinical decision making? And then, why further analyzing the two subtypes across the five key categories of sepsis predictors? This comment is also related to the clarity presenting the methods.

- From my understanding and interpretation, I agree with the paper that explainable AI was used to uncover patient heterogeneity, I am not sure if the model SepsisFormer was built to be explainable. Many parts use modules from Transformers, a complex deep learning architecture, that as mentioned, can be beneficial for modeling the complex relationships between the predictors. In fact, the explainability technique used in the model directly is a post-hoc technique and this can be acknowledge for clarity. Besides, it is mentioned that the model is heterogeneity-aware because of the domain-adaptive generator, but this sounds like a data augmentation strategy for training.

- Claims around improving patient outcomes in the introduction: this has to be evaluated through the application in clinical decision-making and translational research.

- What is meant by clinically relevant in this statement: "XAI method achieved clinically relevant sensitivity and accuracy in prognostic prediction with only eight routine markers."

4. Are there any flaws in the data analysis, interpretation and conclusions? - Do these prohibit publication or require revision?

- Even though the components of SepsisFormer along with their functionality are explained, the process for making these algorithmic choices is not clear. Was there any ablation study to select the parameters or assess the contribution of some parts (e.g., the domain adaptation)? This will provide stronger evidence for the models proposed in this paper.

- The discussion section currently presents medical literature and definitions in a list-like format. While this demonstrates a strong understanding of sepsis from a clinical perspective (which is valuable for algorithm development and interpretation of the results), the manuscript would benefit from a more analytical approach. I recommend expanding the discussion to include comparative analyses of the different models for the prognosis and risk stratification task, the different domain adaptation strategies used, or any other pertinent comparisons.

- Considering the multiple insights found in this study, the paper is lacking an overall conclusion about the approach followed. This can also be beneficial to reinforce the main message and contribution of the paper.

5. Is the methodology sound? Does the work meet the expected standards in your field?

In general, the selected methods seem appropriate for the problems to be addressed in the paper. However, I found the following concerns:

- In the introduction it says that the methods use only routine laboratory blood measurements and how this provides an alternative to blood transcriptomic-based methods. Therefore, I found confusing when later in the paper an analysis of transcriptomic biomarkers is introduced (which involved additional methodological steps). Please clarify if I misunderstood this part.

- The analyses presented in the paper are extensive, but the rationale for each type is missing. For example, why does it matter to see the contribution of the five categories of predictors? And their associations? Why is it important to analyze the relationships between the 1-N transformer heads and lab measurements? Including these justifications can also clarify the choice for the visualization and/or analysis technique.

- Not clear how the subtypes alpha and beta in the variable-level explainability analysis differ from the subphenotypes if they are both identified based on distinctions in mortality rates.

- Throughout the paper, it is stated that eight markers were identified (7 coagulation-inflammatory measures + age) and used for prognosis modeling, risk stratification, and further identifying subphenotypes, but the rationale for selecting this subset of predictors is unclear. The use of routine coagulation-inflammatory measures is motivated in the introduction, but looking at Figure 2 it seems like there are more than 7 measures.

- For the evaluation of the risk stratification model, it could be beneficial to clarify what ground truth is used for the evaluation of this task.

6. Is there enough detail provided in the methods for the work to be reproduced?

Considering the details and codes provided in the supplementary materials, probably yes. However, I found additional experiments, like fine-tuning or pertaining the prognosis models, which was never mentioned in the main manuscript.

(Remarks on code availability)

Code was not available in the submission. Some pseudo codes were presented in the supplementary materials.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

In my previous review I have raised the following points:

1. The clarity and framing of the main message/ contribution of the paper
2. Issues relating to the reporting of the methodology and sample
3. Greater transparency towards the methods employed (e.g. datasets and codes)

Glad to share that these three issues were addressed in this latest revision.

The results of these paper are noteworthy as indicated in my earlier review and the ability to translate the algorithm developed into a clinical profiling score has field implications.

Good effort of showcasing the execution of the codes in the videos.

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)

The authors have adequately addressed the concerns, and the manuscript is now suitable for publication.

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)

The main concerns were addressed in the revision, including the performance comparisons to existing models, including both of the main model and the domain-adaptation strategy. Implementation details and definitions were also provided for the latter method, which helped to understand how it differs from existing techniques. However, in the description of the domain adaptation in the supplementary method, the justification that analyzed class imbalance in mortality outcomes is not theoretical but rather empirical I think, since you based the analysis on your data samples. Further methodological details were provided for the transcriptomic analysis and study of HTEs effect. Ablation studies are also presented and clarifications around techniques used and claims made throughout the manuscript.

Besides, the authors extended the interpretation of the explainability analysis and how this information is useful to understand the heterogeneity of the disease and relevant biomarkers for sepsis prognosis. They also commented on the implications of having a model that predicts risk-stratified levels and the corresponding analyses.

In the results, there is connection between findings and medical relevance (e.g., found importance of coagulation-inflammatory indicators vs functional indicators of specific organs and how the former better reflects the pathophysiology of sepsis), and an application use case of the tool available online was demonstrated.

The paper still needs some organization edits, especially in the presentation of the results:

- Some analyses are listed but then presented later. E.g., 1) the correlation between predictors mentioned in line 145 and then presented in line 176, 2) finding a subset of laboratory measurements within one category (part (a) in line 155) but the clustering results of part (b) are presented first, followed by the correlation analysis.

- Improve the introduction of the transcriptomic-level explanations and transcriptome analysis: It is clarified that this was to show the important roles of coagulation-inflammation related indicators, but there are some aspects that can be clarified. where was the gene information gathered from? Briefly mention the methods involved in the paragraph in line 207.

- It is still unclear to me how the SEVEN predictors were selected. In the model-level explainability only 3 are mentioned (APTT, WBC, and age) and the importance of coagulation-inflammation predictors, but there at least 10 from what I can see in Figure 2c and 8 in Figure 2g. The overall selection process was illustrated in a figure in the response letter, but I may be missing some details to fully understand the criteria for picking those 7 variables. I also noticed that the functions of these 7 coagulation-inflammation markers is discussed in detail in the paragraph starting in line 362 and they are part of the online platform.

- Domain adaptation comparisons are introduced out of the blue in page 7. Explained the results using comparisons to other methods.

- Revise claim about an XAI tool: Sepsis former on its own is not an XAI tool. The analysis used post-hoc XAI tools to understand contributors to predictions and relationships between inputs.

- The use of unsupervised clustering is better explain to find the subphenotypes, but I am still confused about why there are two analyses for this, i.e., first to identify alpha and beta groups using 36 predictors and then CIS1 and CIS2 using only the seven coagulation-inflammation predictors? As another reviewer mentioned, not all analyses have to be part of the manuscript because it becomes hard to appreciate what are the main take aways.

- Regarding the radar plot in the assessment of HTEs: It seems like all predictors are included in this analysis. I thought that the SMART model only used 8 predictors. Please clarify why variables from the other categories were included and the purpose of this analysis. As another reviewer mentioned, not all analyses have to be part of the manuscript because it becomes hard to appreciate what are the main take aways.

Discussion:

Overall, the claims in the discussion are supported by the results: how explainability analyses were done to study sepsis heterogeneity, how the different patient populations stratified by subphenotype and risk levels had different clinical characteristics and mortality rates, and the insights from the combined analysis of HTEs' effect.

The clarifying sentence at the end of the third paragraph of the discussion can be included in the paragraph opening to mention the transition to a new analysis.

Included other works that have done subtyping, how the findings and techniques differ, and acknowledge that comparisons are hard between identified subphenotypes.

I suggest beginning the paragraph in line 400 with this sentence before you contrast the results with other works that have done subtyping: "Another strength is the ability to distinguish subphenotypes and risk levels of septic patients using only coagulation-inflammatory markers and patient age". And then in the next paragraph discuss the effects of HTEs for the subgroups.

Minor comments:

Introduce some acronyms in the main paper for clarity: HTE in the introduction, MMID-SMOTE in the results, the machine learning abbreviations (even if they are well known).

Line 425 "and so on" may not be clear for all readers and sounds informal.

(Remarks on code availability)

Even though the code is provided, the documentation can be improved and a step by step description on what codes to run to generate what results. The README file does not have enough instructions and does not mention which dependencies are needed. The names of the scripts can also be more informative. For the transcriptomic analysis results, the folder only contains data files, so I wonder if the code to recreate these results is missing.

Version 2:

Reviewer comments:

Reviewer #3

(Remarks to the Author)

Overall, the revision provides satisfactory responses to the comments I included in the previous iteration. Language clarifications were made in the method's description and justifications (for domain adaptation, correlation analyses, clustering, and explainability). Improvements were made in the organization of the results. The paper now explains the selection of variables for risk models and other analyses.

(Remarks on code availability)

The documentation of the repository has been improved with more clear instructions on how to run the code.

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Explainable AI-driven heterogeneity using coagulation–inflammatory markers improves prognosis prediction, risk stratification, and anticoagulant treatment effects for sepsis

Corresponding Author: Niu Bailin

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Reviewer #1 (Remarks to the Author):

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The writing, although clear, does not highlight the key findings of the paper. The results section focused excessively on the methods employed and did not provide sufficient clinical explanation as to why these biomarkers matter in explaining sepsis prediction. For example, SHAP plots are good for visualizing the importance of the predictors in the model, but in the main body of the paper, more should be presented to explain the clinical aspects of the findings. The methodological reporting should go into the methods section which also needs further revisions.

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clustering methods, including GMM, MiniBatchKMeans, K-means, hierarchical agglomerative clustering (HAC), and Birch [...]”. The writing here presents more of a report of a collection of methods with limited ability for the reviewers to verify and perhaps add limited value to the paper other than to signal to the review team that extensive work has been put into the paper. A similar writing style is pervasive in the entire methods section, e.g., “[...] MIMIC-III and MIMIC-IV are merged, and SMOTE is employed to balance the class distribution [...]” For example, what is the extent of SMOTE (percentage, number of neighbors used, etc.)? Even though I downloaded all the datasets in this submission, reading this article does not permit any reviewer to completely comprehend the inner workings of the analysis based on the reporting.

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- Overall, the methodology is robust, but a few concerns require attention.
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- The methodology is well-documented but requires additional clarification in some areas.
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 - * SHAP-based explainability analysis.
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- Recommendation:
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Reviewer #3 (Remarks to the Author):

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- The transcriptomic biomarker evidence analysis and study of anticoagulant treatment effect seem crucial for better understanding sepsis, reveal insights about heterogeneity, and validate algorithmic model development. However, I think that the presentation of these approaches can be clarified because it wasn't clear what experiments actually took place in this study or were using measurements from existing public datasets (E.g., the Heparin vs control studies, or PCR studies), and how the biomarkers were actually selected.

3. Does the work support the conclusions and claims, or is additional evidence needed?

- As I was reading the paper, I struggled to understand the support for some of the claims in the results section since it wasn't clear the criteria used for doing further classifications or analyses. For example, how does further categorizing the predictors into clinically relevant ranges (not clear how these are defined) facilitate clinical decision making? And then, why further analyzing the two subtypes across the five key categories of sepsis predictors? This comment is also related to the clarity presenting the methods.

- From my understanding and interpretation, I agree with the paper that explainable AI was used to uncover patient heterogeneity, I am not sure if the model SepsisFormer was built to be explainable. Many parts use modules from Transformers, a complex deep learning architecture, that as mentioned, can be beneficial for modeling the complex relationships between the predictors. In fact, the explainability technique used in the model directly is a post-hoc technique and this can be acknowledge for clarity. Besides, it is mentioned that the model is heterogeneity-aware because of the domain-adaptive generator, but this sounds like a data augmentation strategy for training.

- Claims around improving patient outcomes in the introduction: this has to be evaluated through the application in clinical decision-making and translational research.

- What is meant by clinically relevant in this statement: “XAI method achieved clinically relevant sensitivity and accuracy in prognostic prediction with only eight routine markers.”

4. Are there any flaws in the data analysis, interpretation and conclusions? - Do these prohibit publication or require revision?

- Even though the components of SepsisFormer along with their functionality are explained, the process for making these algorithmic choices is not clear. Was there any ablation study to select the parameters or assess the contribution of some parts (e.g., the domain adaptation)? This will provide stronger evidence for the models proposed in this paper.

- The discussion section currently presents medical literature and definitions in a list-like format. While this demonstrates a strong understanding of sepsis from a clinical perspective (which is valuable for algorithm development and interpretation of the results), the manuscript would benefit from a more analytical approach. I recommend expanding the discussion to include comparative analyses of the different models for the prognosis and risk stratification task, the different domain adaptation strategies used, or any other pertinent comparisons.

- Considering the multiple insights found in this study, the paper is lacking an overall conclusion about the approach followed. This can also be beneficial to reinforce the main message and contribution of the paper.

5. Is the methodology sound? Does the work meet the expected standards in your field?

In general, the selected methods seem appropriate for the problems to be addressed in the paper. However, I found the following concerns:

- In the introduction it says that the methods use only routine laboratory blood measurements and how this provides an alternative to blood transcriptomic-based methods. Therefore, I found confusing when later in the paper an analysis of transcriptomic biomarkers is introduced (which involved additional methodological steps). Please clarify if I misunderstood this part.

- The analyses presented in the paper are extensive, but the rationale for each type is missing. For example, why does it matter to see the contribution of the five categories of predictors? And their associations? Why is it important to analyze the relationships between the 1-N transformer heads and lab measurements? Including these justifications can also clarify the choice for the visualization and/or analysis technique.

- Not clear how the subtypes alpha and beta in the variable-level explainability analysis differ from the subphenotypes if they are both identified based on distinctions in mortality rates.

- Throughout the paper, it is stated that eight markers were identified (7 coagulation-inflammatory measures + age) and used for prognosis modeling, risk stratification, and further identifying subphenotypes, but the rationale for selecting this subset of predictors is unclear. The use of

routine coagulation-inflammatory measures is motivated in the introduction, but looking at Figure 2 it seems like there are more than 7 measures.

- For the evaluation of the risk stratification model, it could be beneficial to clarify what ground truth is used for the evaluation of this task.

6. Is there enough detail provided in the methods for the work to be reproduced?

Considering the details and codes provided in the supplementary materials, probably yes. However, I found additional experiments, like fine-tuning or pertaining the prognosis models, which was never mentioned in the main manuscript.

Reviewer #3 (Remarks on code availability):

Code was not available in the submission. Some pseudo codes were presented in the supplementary materials.

Response to Reviewers' Comments

We are incredibly grateful to the reviewers for their insightful and constructive comments, which have been very helpful in further improving the manuscript and enhancing its robustness, logic, and presentation of work highlights. The following is our detailed and thorough response to each comment one by one. We hope to address your concerns and obtain your approval.

Reviewer #1 (Remarks to the Author):

***Q1:** The study looks at a significant problem. While prior literature has suggested various algorithms that can predict sepsis very well and, at times, better than physicians, this paper has three unique contributions. First, the model is built using external datasets and validated using a separate hospital dataset which presents some form of external validation for the model. Second, the development of a scoring system presents a possible translational use of this algorithm in clinical environments where the operating of real-time algorithms is not feasible. Third, it looks at biomarkers and provides some explanatory attempts of predicting sepsis, which is something challenging.*

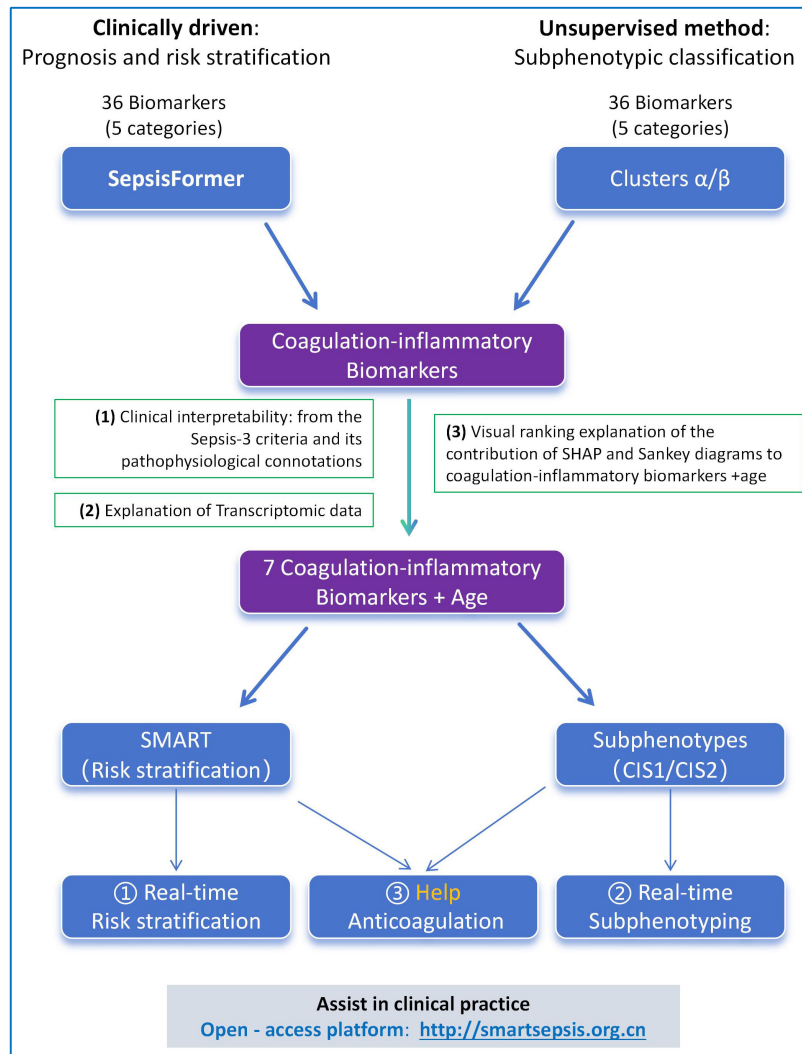
Response: We sincerely appreciate your positive comments on the significance of our study and the recognition of our three unique contributions. We are very grateful for your constructive suggestions. We have carefully read them, discussed them, and made one-on-one revisions and responses. All the relevant revised parts in the manuscript have been marked in red. We hope to obtain your approval.

***Q2:** There are, however, some points to consider in the revision. The writing, although clear, does not highlight the key findings of the paper. The results section focused excessively on the methods employed and did not provide sufficient clinical explanation as to why these biomarkers matter in explaining sepsis prediction. For example, SHAP plots are good for visualizing the importance of the predictors in the model, but in the main body of the paper, more should be presented*

310 *to explain the clinical aspects of the findings. The methodological reporting should go*
311 *into the methods section which also needs further revisions.*

312 **Response:** We are very grateful for your constructive comments. We fully understand
313 your concerns regarding "highlighting the key findings of the paper" and the "results
314 section lacking in-depth explanation of biomarker clinical significance, with an
315 overemphasis on methods." These insights are crucial for improving the quality of our
316 manuscript. Therefore, we have implemented comprehensive revisions across
317 multiple sections of the manuscript, including the **Abstract, Introduction, Results,**
318 **and Discussion** sections, with a primary focus on the following aspects.

319 **1. Addition of Supplementary Figure 1.** We have included Supplementary Figure 1
320 to complement Figure 1 in the main text, providing an integrated overview of the
321 model architecture, key biomarker selection process, and the overall study framework.
322 This figure is intended to help readers quickly and intuitively grasp the scientific
323 rationale, methodological design, and experimental procedures of this study.



(New) Supplementary Figure 1. Integrated framework: An overview of models, methods, biomarkers, and clinical applications in our study.

2. Emphasizing Key Findings and Their Clinical Relevance. The core findings of our study are now more clearly summarized in the **Abstract**, **Introduction**, and **Discussion** section. We particularly highlight the clinical importance of the identified coagulation-inflammatory biomarkers for sepsis risk stratification, clinical subphenotype identification, and guiding anticoagulant therapy, thereby underscoring their translational value. (Page 2, 4–5, 12)

Revised in the manuscript:

“Our work, therefore, offers a novel set of simple, real-time executable tools for sepsis heterogeneity, demonstrating the considerable potential to significantly enhance sepsis clinical practice globally, particularly in resource-constrained healthcare

settings.” (Page 2)

“In this study, we developed a Transformer-based prognostic model (SepsisFormer), an automated risk stratification system (Sepsis Mortality and Risk Tool, SMART), and identified two distinct subphenotypes (CIS1/CIS2). An open-access sepsis risk assessment platform (<http://smartsepsis.org.cn>) was established to provide real-time outputs of risk levels (Mild, Moderate, Severe, Dangerous) and subphenotypes. Both SMART and subphenotyping require only patient age and seven routine coagulation-inflammatory markers. These markers were selected and validated using multi-view explainability analyses across EHR variables, models, and transcriptomic levels. SepsisFormer's prognostic performance was comprehensively benchmarked against a wide range of machine learning and deep learning models. In parallel, SMART's risk stratification capability was systematically evaluated in comparison with established clinical scoring systems. To investigate the influence of SMART-derived risk levels and the identified CIS1/CIS2 subphenotypes on clinical outcomes, we conducted further heterogeneous treatment effects analyses, specifically examining the efficacy of heparin anticoagulation. Overall, this work, by combining XAI and coagulation-inflammatory markers, deeply explores sepsis heterogeneity and develops high-performance, real-time tools for clinical practice.”

(Page 4)

“The 35 laboratory measurements collectively capture multi-organ dysfunction spanning five categories: coagulation-inflammatory, hepatic, renal, blood gas, and oxygen transport (Erythrocyte), while essentially conforming to the connotation of Sepsis-3 criteria (excluding neurological markers). However, their full clinical adoption is hindered by prohibitive costs, large blood volume requirements, and implementation barriers in resource-constrained settings. To overcome these challenges, explainability analyses were performed across five categories to achieve two clinical goals: (a) Identify a sepsis mechanism relevant and clinically feasible subset of laboratory measurements within one category for real-time and cost-effective application; (b) Select an efficient unsupervised clustering method to

uncover clinically meaningful sepsis subphenotypes, essential for understanding disease heterogeneity and guiding clinical insights⁴” (Page 5)

“ In summary, this study introduces the prognostic model **SepsisFormer**, the automated risk stratification tool **SMART**, and the subphenotypes **CIS1/CIS2** to characterize sepsis heterogeneity. Through multi-level explainability analyses, **age** and seven routine coagulation-inflammatory markers were identified as key predictors for sepsis diagnosis and prognosis. **SepsisFormer** achieved an AUC of **0.9301**, outperforming state-of-the-art models, while **SMART** reached an AUC of **0.7360**, exceeding conventional clinical scores and effectively stratifying mortality risk. Notably, **CIS2** patients showed higher mortality and distinct coagulation-inflammatory profiles compared to **CIS1**. Integrating subphenotyping with risk stratification uncovered heterogeneity in anticoagulation treatment effects, supporting more precise and safer therapeutic decision-making in sepsis management. An open-access, web-based platform (<http://smartsepsis.org.cn>) facilitates real-time risk stratification and subphenotype identification using only seven low-cost, routinely available coagulation-inflammatory biomarkers. Its simplicity, accessibility, and practical applicability make it a promising tool for improving sepsis management worldwide, particularly in resource-constrained healthcare settings.” (Page 12)

3. Deepening the Clinical Interpretation of the Results

(1) We have expanded our analysis of the clinical relevance of coagulation and inflammation biomarkers. The **Results** section now presents the distribution of values for the 7 coagulation-inflammatory indicators, with detailed explanations of their roles in the pathophysiology of sepsis. In the **second paragraph of the Discussion**, we provide an in-depth clinical analysis of all 7 markers, describing the implications of different ranges and their potential to inform clinical outcomes. (Page 5-6, 11)

Revised in the manuscript:

“ At the model-level explanation scope, the SHapley Additive exPlanations (SHAP) analysis indicated that APTT, WBC, and patient age are important predictors of sepsis outcomes, regardless of the perspective (global, local, or hierarchical cumulative

contribution). These factors significantly influence whether a patient's condition will deteriorate or improve, providing valuable guidance for clinicians to focus on these key indicators during patient assessment. The SHAP summary plot further quantified the contributions of predictors such as APTT to the prediction outcomes, highlighting the importance of predictor ranking in model prediction (Fig. 2g). The decision plot, with predicted values below -0.40 or above 0.40, showed how the contributions of predictors have different impacts on sepsis outcomes in specific patient populations. The Sankey diagram shows that as Transformer depth increases, the cumulative contributions of the eight predictors slightly increase yet remain generally stable, indicating consistent feature integration across layers.” **(Page 6)**

(2) Regarding the identified sepsis subphenotypes, we have substantially expanded the third paragraph of the Discussion. This section now thoroughly discusses the clinical implications, potential for real-time assessment, and comparative analysis with other known subphenotypes. The discussion begins with "One of the important studies that directly applied transcriptomic data to the subtyping of sepsis came from the MARS consortium in 2017..." and aims to enhance the clinical translatability of the subphenotype findings **(Page 11)**.

Revised in the manuscript:

“One of the important studies that directly applied transcriptomic data to the subtyping of sepsis came from the MARS consortium in 2017⁴⁴. They enrolled a total of 787 cases of sepsis patients in a discovery cohort and two validation cohorts. Through machine learning and analysis of differentially expressed genes, sepsis was classified into four subtypes, MARS1-4. Among them, MARS1 had the poorest prognosis, with a mortality rate as high as 35%. This classification at the level of gene expression is also named endotype. In 2019, Seymour CW et al.⁷ conducted phenotypic analysis on 20,189 patients with sepsis using statistics, machine learning, and simulation tools. Twenty-nine variables, including cardiovascular, hematopoietic, hepatic, coagulation-inflammatory, neurological, pulmonary, renal, and so on, were used. Finally, they divided the patients into four phenotypes: α , β , γ , and δ . Among them, the mortality rate of phenotype α was the lowest at approximately 5%, while the

mortality rates of type β , γ and δ were 13%, 24% and 40% respectively. From the perspective of mortality rates, our risk stratification results are similar to those of the Seymour CW's phenotypic classification. However, we revealed the heterogeneity of sepsis from two aspects: the stratification of risks and the subtypes. Moreover, the clinical indicators we use are fewer and more beneficial for clinical practice, which not only reduces the consumption of patients' blood samples but also saves costs, and it is also conducive to real-time dynamic assessment. Furthermore, the Seymour CW team has not yet developed a scoring system or a shared platform that can be universally implemented by clinicians. Most importantly, although many sepsis subtypes have been published at present, due to the different goals, the use of different clinical indicators, different machine learning models, or black-box algorithms, there are varying degrees of differences among these subtypes, ultimately resulting in low comparability or overlap among them (like MARS1-4, SRS1-2, Hyper or hypo inflammatory, and SENECA subtypes) ⁸, and no shared classification tools or platforms have been simultaneously proposed. However, it is still necessary to further conduct comparative studies on our classification and the existing subtype classifications.” (Page 11)

4. Refining the Description of Methodology. In line with your suggestion, we have relocated the redundant methodological descriptions from the Results section to a dedicated “Methods” section and thoroughly revised them. This section now contains comprehensive technical details and statistical procedures, supplemented by detailed information in **Supplementary Method 1, 3, 4**, to ensure greater rigor and logical coherence.

We believe these revisions have significantly improved the manuscript's clarity, clinical relevance, and scientific merit. We are grateful for your valuable suggestions that contributed to these important improvements.

Q3: *I find the explanatory part of sepsis prediction something that is significant, but this current form does not provide sufficient write-up in this area. I hope the authors can provide more details that may contribute to the clinical understanding of sepsis*

onset.

Response: We truly appreciate your insightful comment regarding the explanatory part of sepsis prediction. We completely agree that more detailed information in this area is crucial for enhancing the clinical understanding of sepsis onset. In response to your comment, we enriched and improved the relevant parts as follows:

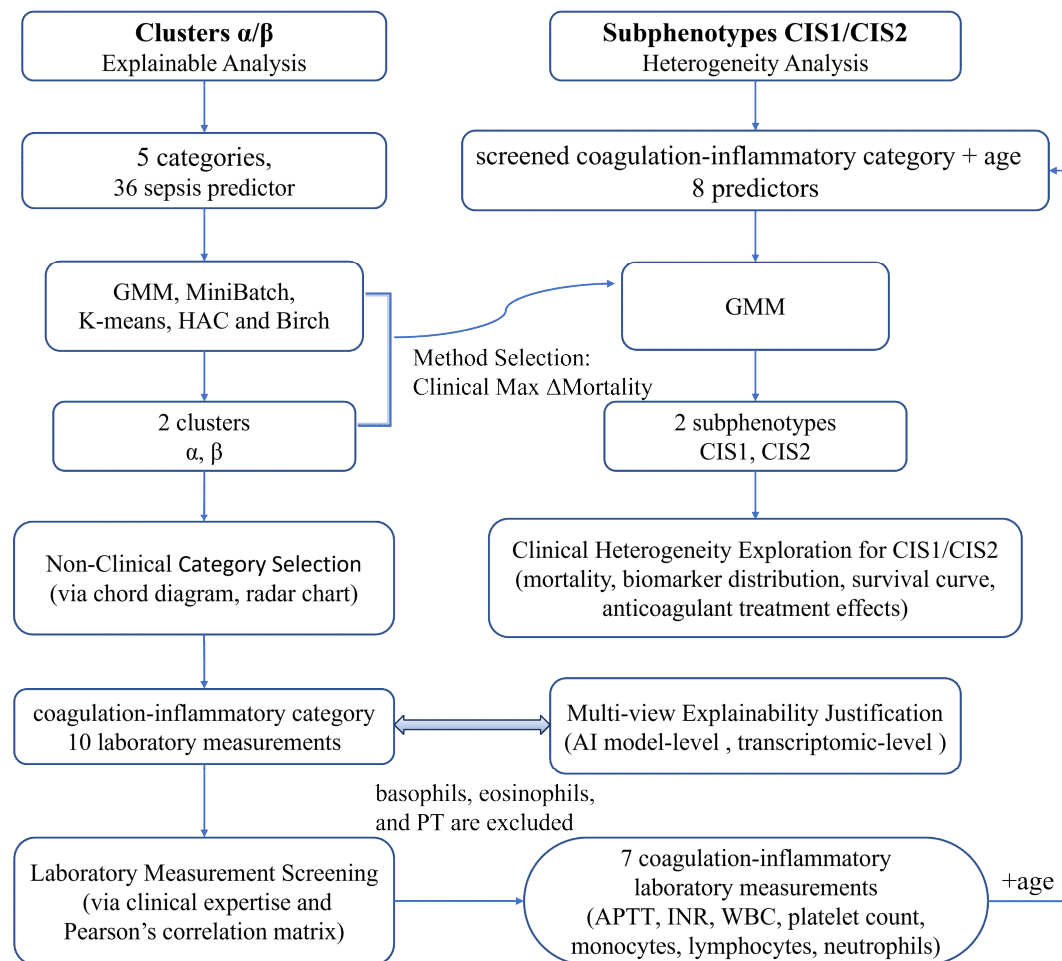
1.We refined the explainability methods used in our study.

Revised in the manuscript:

“Multi-view explainability analyses were conducted from three perspectives: two post-hoc approaches (cluster-informed EHR variable-level and AI model-level) and a transcriptomic-level analysis inspired by multi-omics principles. To select key categories that effectively mirror sepsis pathophysiology and are practical for clinical application, we employed unsupervised clustering methods, including Gaussian Mixture Model (GMM), MiniBatchKMeans, K-means, Hierarchical Agglomerative Clustering (HAC), and Birch. Chord diagrams and radar plots enhance the interpretability of unsupervised clustering results by revealing the variable patterns defining each cluster. Clinical expertise and Pearson's correlation matrix were employed to analyze the associations between individual laboratory measurements within each category.” (Page 14-15)

2.Incorporate the Research Framework for Cluster and Subphenotype Analyses.

We've further clarified the research ideas and the objectives of the study regarding the use of interpretable methods. This framework highlights explicitly the key distinctions in purpose and methodology between non-clinical and clinical analyses (Supplementary Fig.2).



(New) Supplementary Figure 2. Research framework for cluster and subphenotype.

3. Elaboration on Coagulation-Inflammatory Markers in Cluster Analysis

(1) Enriching Chart Interpretation Details: In the revised manuscript, we added more detailed explanations when describing the chord diagrams and radar charts. For the chord width of coagulation-inflammatory markers in the chord diagram, we explained how this visual feature directly reflects the significant difference between the two clusters. We added the content in the results section as follows:

Revised in the manuscript:

“Clusters α and β were automatically derived using unsupervised clustering methods using 36 sepsis predictors, requiring no prior information (e.g., mortality, disease outcomes, or treatment medications), based on the optimal number of clusters (Supplementary Table 4). Five different unsupervised methods—Gaussian Mixture Model (GMM), MiniBatchKMeans, K-means, Hierarchical Agglomerative Clustering (HAC), and Birch—generated highly aligned clusters, a consistency across distinct

approaches that confirms the robustness and validity of the two identified clusters. Since these data-driven clusters lack inherent clinical interpretation, we statistically analyzed subgroup mortality. The GMM-derived α and β clusters exhibited the most significant mortality difference (32.09% vs 17.62%, respectively), indicating GMM's superior effectiveness for identifying patient subgroups with divergent mortality risks. In the chord diagram, both cluster α and cluster β consistently showed the widest chords with coagulation-inflammatory category (ribbons connect with these portions of the circle), indicating coagulation-inflammatory predictors are key defining features for the clustering. Independently, the radar plot also confirmed the coagulation-inflammatory category as the highest connection point. These findings were robustly replicated across all five unsupervised clustering methods (Fig. 2f, Supplementary Fig. 3).” (Page 5)

(2)Supplementing the Model Relevance and Clinical Significance of Markers.

Revised in the manuscript:

“At the model-level explanation scope, the SHapley Additive exPlanations (SHAP) analysis indicated that APTT, WBC, and patient age are important predictors of sepsis outcomes, regardless of the perspective (global, local, or hierarchical cumulative contribution). These factors significantly influence whether a patient's condition will deteriorate or improve, providing valuable guidance for clinicians to focus on these key indicators during patient assessment. The SHAP summary plot further quantified the contributions of predictors such as APTT to the prediction outcomes, highlighting the importance of predictor ranking in model prediction (Fig. 2g). The decision plot, with predicted values below -0.40 or above 0.40, showed how the contributions of predictors have different impacts on sepsis outcomes in specific patient populations. The Sankey diagram shows that as Transformer depth increases, the cumulative contributions of the eight predictors slightly increase yet remain generally stable, indicating consistent feature integration across layers. To optimize the model architecture, we conducted a systematic hyperparameter sensitivity analysis on the number of Transformer number of Transformer layers L and attention heads H . We chose $L=8$ to balance complexity and feature extraction, despite peak performance at $L=1\sim 7$. With M fixed at 8, we observed a similar pattern for attention heads H ,

with optimal performance at $H=1\sim 7$ before a decline at $H=8$ (Supplementary Fig. 5). Furthermore, the consistent ranking of predictor importance underscores the model's structural robustness and deterministic nature. These findings collectively affirm the model's reliability and interpretability in multivariate prediction tasks. The results consistently indicated that coagulation-inflammatory indicators such as APTT, WBC, and the patient's age were important predictors from various perspectives. Therefore, compared with the functional indicators of specific organs such as the liver, kidneys, and arterial blood gas, the coagulation-inflammatory indicators reflect the systemic or general state and can better reflect the systemic pathophysiological connotation of sepsis. They are also an important basis for reflecting and inducing functional disorders of other organs^{25, 26}.” (Page 6)

And new reference:

27.Tsantes, A. G. et al. Sepsis-Induced Coagulopathy: An Update on Pathophysiology, Biomarkers, and Current Guidelines. *Life* (Basel, Switzerland) 13, doi:10.3390/life13020350 (2023).

28.Williams, B., Zou, L., Pittet, J. F. & Chao, W. Sepsis-Induced Coagulopathy: A Comprehensive Narrative Review of Pathophysiology, Clinical Presentation, Diagnosis, and Management Strategies. *Anesthesia and analgesia* 138, 696-711, doi:10.1213/ane.0000000000006888 (2024)

4.Strengthening the Presentation of Validation Content Related to Transcriptomic Biomarkers

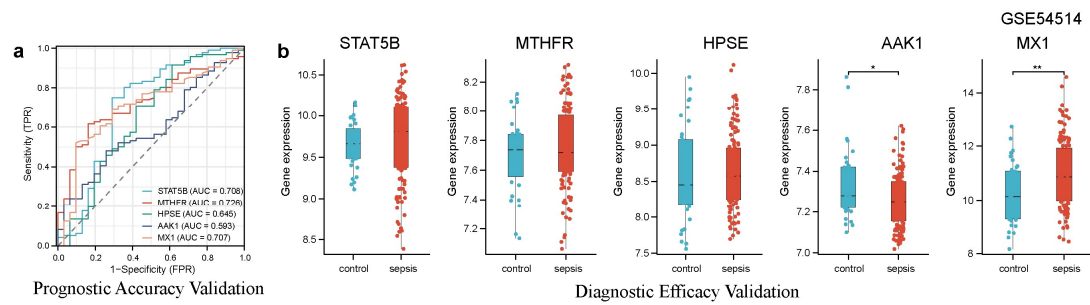
To validate the conclusion of the key role of coagulation-inflammatory markers in sepsis heterogeneity, we have identified additional transcriptomic biomarker evidence from coagulation-related genes (CRGs), inflammation-related genes (IRGs), and disseminated intravascular coagulation (DIC)-related genes. We have highlighted the clinical relevance of transcriptomic biomarkers in sepsis diagnosis and prognosis. Specifically, we discussed the diagnostic potential of CD59, SERPINB2, CFD, and P2RX1 as independent markers and provided guidance on their application for early detection and dynamic monitoring to assess treatment response and patient outcomes.

In the revised manuscript, we elaborated on the internal connection between these

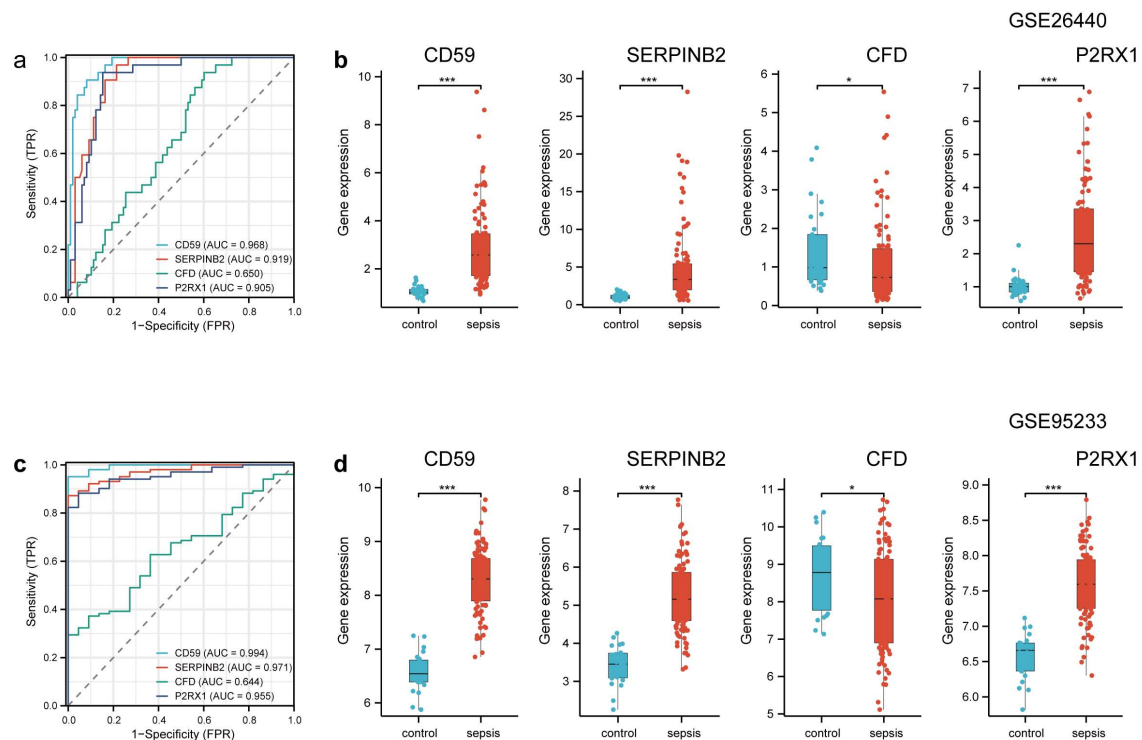
transcriptomic biomarkers and coagulation-inflammatory markers and added the related content in the Results section:

Revised in the manuscript:

“Our transcriptomic-level explanation provides complementary evidence for the critical role of coagulation-inflammatory markers in sepsis heterogeneity, initially identified through EHR-based variable and model explanations. We specifically examined coagulation-related genes (CRGs), inflammatory-related genes (IRGs), and disseminated intravascular coagulation (DIC)-related genes (Fig. 2h and Supplementary Fig. 6-7). This investigation reaffirmed the significance of these markers and identified five key prognostic biomarkers: STAT5B, MTHFR, HPSE, AAK1, and MX1. Patients stratified into a high-risk group based on these biomarkers exhibited significantly higher mortality. Notably, MTHFR, AAK1, and MX1 expression levels were elevated in this high-risk group, suggesting their influence on sepsis prognosis potentially through modulation of immune and coagulation pathways (Supplementary Table 8). The diagnostic efficacy of these five genes was further validated in external dataset GSE54514 (Supplementary Fig. 8). Furthermore, transcriptome analysis confirmed the diagnostic relevance of four genes: CD59, P2RX1, CFD, and SERPINB2 (Supplementary Table 9-10). These markers demonstrated robust performance as independent diagnostic biomarkers for sepsis. Their diagnostic efficacy was successfully validated in external datasets GSE26440 and GSE95233 (Supplementary Fig. 9). RT-PCR analysis of PBMCs from 29 sepsis patients and 11 healthy controls revealed significant differential expression of CFD and P2RX1 ($p < 0.05$), with greater variability observed in the sepsis group. Therefore, these transcriptomic findings further explained the important roles of coagulation-inflammation related indicators in the diagnosis and prognosis of sepsis from multiple perspectives.”(Page 6)



(New) Supplementary Figure 8. The prognostic and diagnostic efficacy of STAT5B, MTHFR, HPSE, AAK1, and MX1 in an external cohort. From GSE54514. a. prognostic AUC curves; b. Expression levels of STAT5B, MTHFR, HPSE, AAK1, and MX1. * $p < 0.05$, *** $p < 0.001$ vs. Healthy control.



(New) Supplementary Figure 9. The diagnostic efficacy of CD59, SERPINB2, CFD, and P2RX1 in external cohorts. From GSE26440, and GSE95233. a. and c., AUC curves; b. and d, Expression levels of CD59, SERPINB2, CFD, and P2RX1. * $p < 0.05$, *** $p < 0.001$

Q4: In terms of the reporting of the methods, here, less is perhaps more. The authors have presented a multitude of methods to support their results. Although in general, I do believe in the findings from a reviewer perspective, the large number of methods

employed with limited reporting makes it exacting for the reviewer to establish the correctness of the execution of these methods. To illustrate, in the text, the authors wrote “[...] radar plots of subtypes identified by unsupervised clustering methods, including GMM, MiniBatchKMeans, K-means, hierarchical agglomerative clustering (HAC), and Birch [...]”. The writing here presents more of a report of a collection of methods with limited ability for the reviewers to verify and perhaps add limited value to the paper other than to signal to the review team that extensive work has been put into the paper. A similar writing style is pervasive in the entire methods section, e.g., “[...] MIMIC-III and MIMIC-IV are merged, and SMOTE is employed to balance the class distribution [...]”

For example, what is the extent of SMOTE (percentage, number of neighbors used, etc.)? Even though I downloaded all the datasets in this submission, reading this article does not permit any reviewer to completely comprehend the inner workings of the analysis based on the reporting.

Response: We sincerely appreciate your detailed and constructive feedback regarding the reporting of our methods. We understand that the current presentation has made it challenging for you to verify the correctness of the method execution, and we addressed these concerns in the revised manuscript. And we hope that the new framework diagrams (Supplementary Fig.1 and Fig.2) listed above will help us present our methods and results more clearly.

1. Specifically, we **added Supplementary Figures 1 and 2** to provide an overview of the models, methods, biomarkers, clinical applications, and the research framework for clustering and subphenotype identification. As shown in Supplementary Figure 2, the unsupervised clustering methods served two key purposes: (1) selecting suitable clustering algorithms based on inter-cluster mortality differences for subphenotype identification, and (2) assessing the category stability across different clustering methods. We agree that the original description resembled a technical report, and have revised this section to improve clarity and readability.

Revised in the manuscript:

“Five different unsupervised methods-Gaussian Mixture Model (GMM), MiniBatchKMeans, K-means, Hierarchical Agglomerative Clustering (HAC), and Birch—generated highly aligned clusters, a consistency across distinct approaches that confirms the robustness and validity of the two identified clusters. Since these data-driven clusters lack inherent clinical interpretation, we statistically analyzed subgroup mortality. The GMM-derived α and β clusters exhibited the most significant mortality difference (32.09% vs 17.62%, respectively), indicating GMM's superior effectiveness for identifying patient subgroups with divergent mortality risks.(Fig. 2f, Supplementary Fig. 3).”(Page 5)

“This study employs GMM for subphenotype identification, as the patient subgroups derived from GMM show the most significant differences in mortality rates (shown in Section Explainability Analysis), indicating greater clinical relevance.”(Page 7)

2. We have thoroughly revised **Supplementary Method 1: Domain-Adaptive Generator for Fine-Tuning**. This section now includes detailed analyses of multi-center covariate distribution shifts, theoretical justifications, and comprehensive implementation procedures. The main additions cover:

(1) **Rationale for Domain Adaptation in Prognostic Modeling**, an explanation of why domain adaptation is crucial in handling data heterogeneity across multiple centers.

(2) **Theoretical Analysis of Covariate Distribution Shifts**, providing insights into the impact of such shifts on model performance and generalization;

(3) **Detailed Implementation Steps**, describing how domain adaptation is practically realized in our framework.

Additionally, in the implementation of the MMID-SMOTE algorithm, we have clearly specified the key parameters necessary:

Step 5: Customized SMOTE Sample Generation Synthetic samples are generated via SMOTE under these constraints, with the following parameters

optimized for medical data context.

Number of nearest neighbors: 4.

Minority class oversampling ratio: 1:1 (achieving balanced class distribution).

Random seed: Fixed for reproducibility.

Parallel jobs: 3 (accelerating computation).

Q5: *I suggest the authors do one of the two. 1) In the main paper, report a shorter list of analyses employed, but do that in a more comprehensive and clearer manner, complete with a STROBE or modified CONSORT diagram to showcase the data used. This presentation will lend more credibility to the results. If you want to showcase additional analyses reported here, maybe provide some additional supplement you might want to host on GitHub. These supplementary analyses may have minute computational errors or bugs—which is normally the case, but it does not fall within the purview of NComms review process. 2) Continue with your existing reporting of all the collection of analyses, but in your next review, you perhaps will have to consider providing more details of how the analyses for each technique is done, including the software codes in an executable format. This will also make the review process more exacting.*

Response: Thank you for your thoughtful suggestions regarding the presentation of our analyses.

STROBE (STrengthening the Reporting of OBservational studies in Epidemiology) is a guideline for clinical observational studies. Since our research is not a pure cohort clinical study (involving interdisciplinary, basic, and clinical research areas), yet it contains substantial clinical components, we have completed the STROBE checklist and improved the manuscript according to relevant guidelines.

While we appreciate the clarity with which you have outlined the two valuable approaches, and the following is also the content of the manuscript improvement work we have carried out according to Option 2 you provided, mainly further improving the methodology, results, and discussion sections, etc. You can also find them in the responses to the comments above.

We believe that reporting the full range of analyses conducted is essential for demonstrating the comprehensiveness of our study. In response to your feedback, we are fully committed to enhancing the level of detail in our methodology section for the following review. For each analytical technique employed, we provided an in-depth description of the procedures, including specific parameter settings, data preprocessing steps, and the rationale behind each choice.

To ensure transparency and reproducibility, we will also include executable codes for all analyses as supplementary materials. These codes are well-documented, with clear annotations explaining the functionality of each segment. We understand that this may make the review process more rigorous, but we view it as a necessary step to uphold the integrity of our research.

While we made every effort to ensure the accuracy of all our work, we also recognized that the scope of the Nature Communications review process may not allow for a detailed examination of these additional components. We, therefore, took extra care in validating our analyses and clearly indicated any exploratory or supplementary nature of certain parts of our work.

Thank you again for your guidance. We are confident that these improvements will strengthen our manuscript and facilitate a more thorough review.

Reviewer #1 (Remarks on code availability):

I have downloaded the datasets, but the current reporting make it onerous and not possible for one to review the analysis in detail.

Response: We deeply apologize for the inconvenience and difficulty you encountered while attempting to review the analysis in detail despite having downloaded the datasets. We have now included the relevant codes, along with detailed instructions and videos on how to run them. To promote transparency, reproducibility, and clinical applicability, we have made the following resources publicly available:

1. Interactive Web-based Platform (<http://smartsepsis.org.cn>): We developed an open-access, user-friendly online platform for sepsis subphenotyping and risk

stratification. This tool allows end-users, such as clinicians and researchers, to perform real-time subphenotype identification and risk level prediction by uploading their own data without requiring additional coding or technical expertise.

2. Source Code and Datasets (<https://github.com/zhuli19031218/SepsisFormer>): All raw data, preprocessed datasets, and fully documented source code are available in our GitHub repository. These resources enable independent verification, replication, and extension of our study.

3. Code Execution Tutorial (<https://doi.org/10.5281/zenodo.15634369>): A detailed step-by-step video tutorial is provided to guide users through the complete process of reproducing the main results presented in this study, ensuring ease of use even for non-technical users.

These provided resources allow for replication and application of our method. We believe these improve the paper's quality and transparency.

We sincerely hope the revised manuscript now meets with your approval.

Thank you once again for your time and effort. We truly appreciate your valuable suggestions, especially regarding the clinical explanations and overall writing, which have greatly enhanced the quality of our work.

Hope everything goes well with you!

Reviewer #2 (Remarks to the Author):

Q1: *What are the noteworthy results?*

This study has some key findings.

(A) High Prognostic Accuracy of SepsisFormer

- SepsisFormer (Transformer-based model) achieves an AUC of 0.9301, outperforming traditional risk scoring models like qSOFA, SOFA, and APACHE II.*

- The model demonstrates generalizability across multiple cohorts (MIMIC-III, MIMIC-IV, eICU-CRD, and a local ICU dataset).*

(B) SMART Risk Stratification Model Improves Sepsis Classification

- SMART (a scoring system based on seven coagulation-inflammatory markers and patient age) achieved an AUC of 0.7360, surpassing most established clinical scoring systems.*

- The study identifies four risk levels (Mild, Moderate, Severe, Dangerous), with clear mortality risk stratification (~5%, 15%, 30%, and 50% mortality rates, respectively).*

(C) Identification of Two Sepsis Subphenotypes (CIS1 and CIS2) with Different Outcomes

- CIS2 patients exhibited higher mortality rates (~32.23%) compared to CIS1 (25.84%), and a stronger inflammatory-coagulation response.*

- The study demonstrates that patients with CIS2 benefit more from anticoagulant therapy (e.g., heparin), while CIS1 shows moderate benefits.*

(D) Transcriptomic Analysis Identifies Sepsis Biomarkers

- Five transcriptomic biomarkers (STAT5B, MTHFR, HPSE, AAK1, MX1) were found to be significantly linked to sepsis prognosis.*

- The study suggests these biomarkers influence sepsis prognosis through immune and coagulation pathways.*

Response: We sincerely appreciate your positive comments on our study and the recognition of our unique contributions. We are very grateful for your constructive suggestions. We have carefully reviewed them and made point-to-point revisions. All the revised parts in the manuscript are marked in red. We hope for your approval.

Q2: Will the work be significant to the field and related fields?

Yes, this work is significant to the field.

Relevance to Sepsis Research:

- The study contributes to sepsis mortality prediction and risk stratification, both of which are essential for clinical decision-making in ICUs.
- The identification of coagulation-inflammatory subtypes (CIS1 and CIS2) aligns with ongoing efforts to better phenotype sepsis patients.

Relevance to Machine Learning & Explainable AI in Healthcare:

- SepsisFormer demonstrates that deep learning models can achieve superior predictive performance while maintaining interpretability through XAI methods.
- SMART scoring provides a structured risk assessment tool that could be clinically deployed.

Potential Applications in Other Fields:

- The approach can be extended to other critical conditions (e.g., septic shock, ARDS, multi-organ failure) where heterogeneity complicates prognosis.
- The explainability techniques used here could be applied to other AI-driven healthcare models.

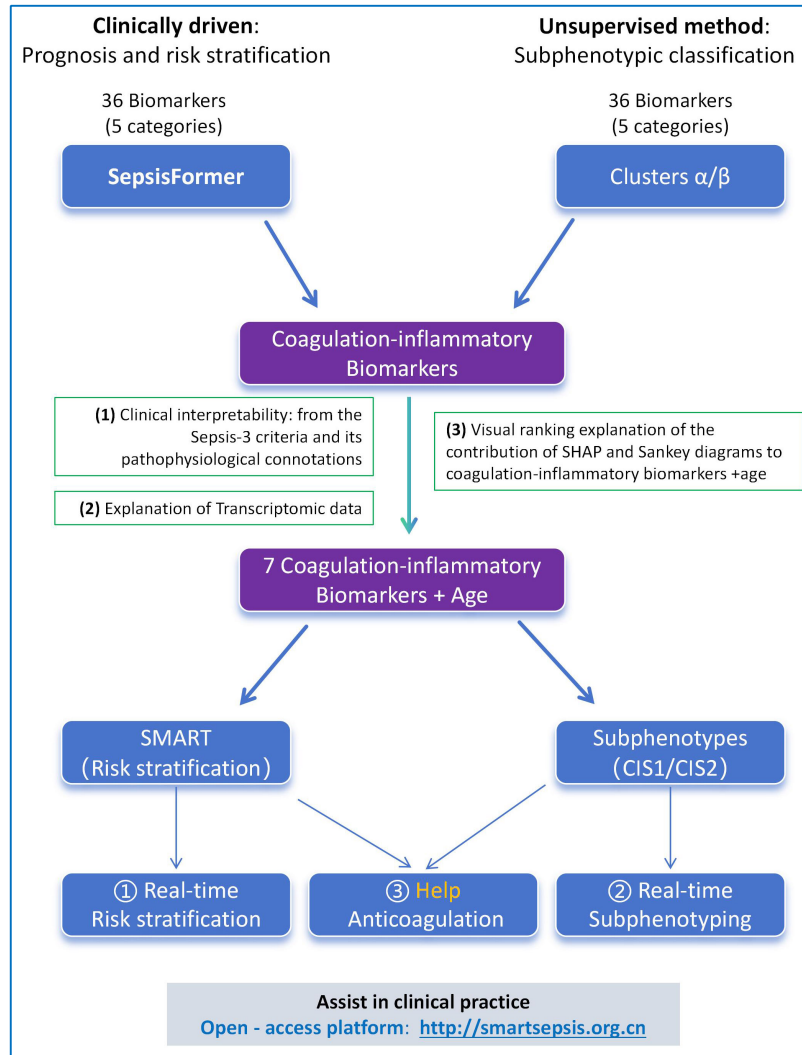
Comparison to Existing Literature:

SepsisFormer's AUC of 0.9301 is notably higher than previously reported AI-based sepsis models (e.g., prior deep learning models in literature typically range from 0.85–0.91).

Risk stratification studies, such as those defining MARS1-4 or α , β , γ , δ subtypes, have not explicitly linked coagulation-inflammatory markers to treatment response.

However, the study does not compare CIS1 and CIS2 to other existing sepsis subtyping frameworks (e.g., MARS1-4, SRS1-2, hyper-/hypo-inflammatory states).

Response: Thank you very much for your constructive comments. CIS1 and CIS2 subphenotypes are one of the three important achievements of this study (SepsisFormer, SMART, and Subphenotype). Our subphenotypic classification is capable of distinguishing between high or low systemic inflammatory response Index (SIRI), prognosis of death, and differences in the benefits of anticoagulant treatment.



(New) Supplementary Figure 1. Integrated framework: An overview of models, methods, biomarkers, and clinical applications in our study.

Indeed, risk stratification studies, such as those defining MARS1-4 or α , β , γ , δ subtypes, have not explicitly linked coagulation-inflammatory markers to treatment response.

The previous work (MARS1-4, SRS1-2, hyper-/hypo-inflammatory states, et al.) has been done very significantly. Our results can be regarded as being inspired by their work, or we can say that we are very grateful for their work. Only by standing on their shoulders can we make further progress on the basis of subtype classification. Furthermore, through SepsisFormer and clinical interpretation, transcriptomic interpretation, etc., we identified coagulation-inflammation indicators and patient age as the indicator combination for our risk stratification and subphenotypic

classification, which also inspired us to analyze the differences in their anticoagulation benefits. The results show that there are indeed differences between them. Moreover, anticoagulant therapies such as heparin are also a focus of great concern in the clinical practice of sepsis at present. However, there is no quantifiable or precise study worldwide to propose which types of sepsis patients or when anticoagulant therapy can benefit them. Therefore, this is also one of the important contributions of this study.

Here, we need to express our sincere gratitude once again to **Professor Craig Coopersmith**, the former chair of the Society of Critical Care Medicine in the United States, for his guidance on this research.

The main purpose of our article is to develop and validate a new risk stratification and subphenotypic classification tool, as well as an analysis of the sepsis population benefiting from anticoagulant therapy based on this. The reasons why our study did not involve substantive comparisons with the previous MARS1-4/ SRS1-2/ α , β , γ , δ / hyper-/hypo-inflammatory states, etc., are as follows:

Firstly, as a newly developed tool, it's challenging to compare so many tools in one study. Some well-established tools didn't conduct extensive comparisons in the early stages of their development, which could be partly due to differences in research objectives and practical workload considerations. For example, MARS 1-4 developed by the team of Professor Tom van der Poll (*Lancet Respir Med.* 2017 Oct; 5(10):816-826), α , β , γ , δ by the team of Professor Christopher W. Seymour (*JAMA.* 2019 May 28;321(20):2003-2017; *Crit Care.* 2023 Jun 15; 27(1):236), and SRS1-2 by the team of Professor Julian C. Knight. (*Lancet Respir Med.* 2016 Apr; 4(4):259-71).

Secondly, because MARS 1-4 is an endotypic study, where they have proposed typing suggestions for eight transcriptional genes, this approach requires a relatively large number of transcriptomic tests separately. The subtype classification of α , β , γ , and δ also involves more than 30 indicators. Presently, even with maximal effort to extend research time in this aspect, operational limitations would still hinder the progress, therefore significantly impacting the publication timeline of our results. We earnestly hope for your understanding, dear professor.

Thirdly, precisely because of the above reasons, the team of Prof. Christopher W. Seymour and Prof. Lonneke A. van Vught recently conducted A comparative study of various subtypes (*Intensive Care Med* (2023) 49:1360–1369). They compared ARDS subtypes (Hyper-/ Hypo-inflammatory) vs. MARS and SRS subtypes, and SENCEA subtypes, it was found that the overlap ratio between them was not ideal. This might still be related to a combination of factors such as the main goals of the development of each subtype classification, the different AI algorithms or models selected, and the differences in the pathophysiological connotations reflected by each indicator. So, we cannot say that our classification must have a good overlap with theirs. However, we have simplified the detection indicators with advanced algorithms, launched the SMART scorecard, developed an open-access web-based sepsis subphenotype and SMART platform that enables clinicians to real-time interactively evaluate septic patients' risk level and subphenotype (<http://smartsepsis.org.cn>), and also put forward the viewpoint of the beneficiary population of anticoagulant therapy. All these are the results of innovation and forward development based on previous research. In addition, we believe that with the rapid development of this field, subsequent research will surely build upon and learn from the research achievements and experiences of our predecessors and ourselves, and will once again bring forth new ideas. Subsequent research will also gradually bridge the gap of differences in the classification of various subtypes. Science is moving forward. We are looking forward to a better future, and this day is sure to come.

Last but not least, although our study did not conduct a comparison with the various subtypes mentioned above, we also discussed in detail the results and differences of these subtypes in the discussion (Paragraphs 3 and 4 of the discussion section). We believe that this can at least provide some references for our readers.

[REDACTED]

Above figure refers to: [Intensive Care Med \(2023\) 49:1360–1369.](#)

Q3: Does the work support the conclusions and claims, or is additional evidence needed?

Most claims are supported by strong experimental data.

Well-supported claims:

SepsisFormer outperforms existing scoring systems → Supported by ROC curves, AUC analysis, external validation across multiple datasets.

SMART provides a clinically relevant stratification system → Clear association between SMART's risk levels and increasing mortality.

CIS2 patients benefit more from anticoagulants → Kaplan-Meier survival analysis demonstrates significant mortality reduction in CIS2 patients receiving anticoagulation.

Less-supported claims:

Q3.1 *CIS1/CIS2 are novel sepsis subtypes → The study does not compare them to previous subtyping models (e.g., MARS1-4, α - δ).*

Recommendation:

Benchmark CIS1/CIS2 against prior subtyping frameworks.

Response: Thanks for your valuable comment on our research.

We are pleased that our findings have the potential to contribute meaningfully to ongoing discussions, and we firmly believe that our work holds great significance for the field of sepsis research, as well as for its intersections with machine learning and explainable artificial intelligence (AI) in the healthcare domain. Our study offers practical tools like SepsisFormer for mortality prediction (with an AUC of 0.9301, surpassing prior models) and SMART for risk stratification, while identifying coagulation-inflammatory subtypes (CIS1 and CIS2) to enable personalized treatment. The methodologies can be applied to other critical conditions, and the XAI techniques used with SepsisFormer have broader utility.

As we presented the new framework diagram above (Supplementary Fig. 1), this study primarily aimed to achieve three goals: SepsisFormer/SMART/subphenotypic classification and the analysis of differences in the benefits of anticoagulant therapy based on these classifications. Meanwhile, we developed an open-access web-based sepsis subphenotype and SMART platform that enables clinicians to interactively evaluate septic patients' risk level and subphenotype (<http://smartsepsis.org.cn>).

Nevertheless, we admit that we haven't compared CIS1 and CIS2 with other established sepsis subtyping frameworks. The main reasons are as stated above (please see the response to Q2 comment). Meanwhile, in the discussion section, we provided a comparison and analysis of subtype classification. Hope to gain an in-depth understanding and approval.

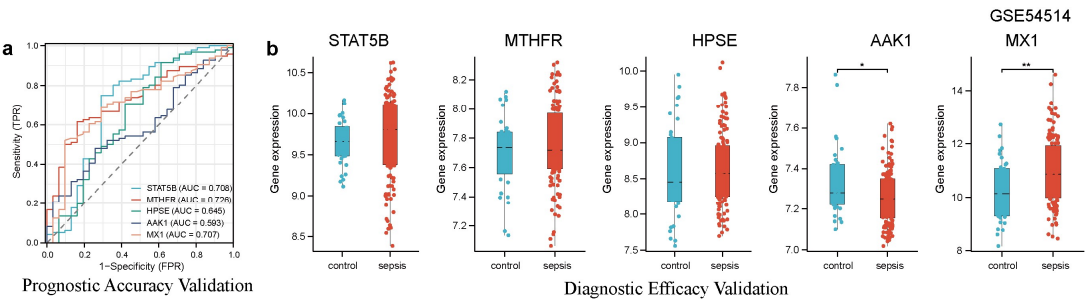
Q3.2 *Transcriptomic biomarkers are clinically relevant for sepsis prognosis → No external validation of transcriptomic findings in an independent dataset.*

Recommendation:

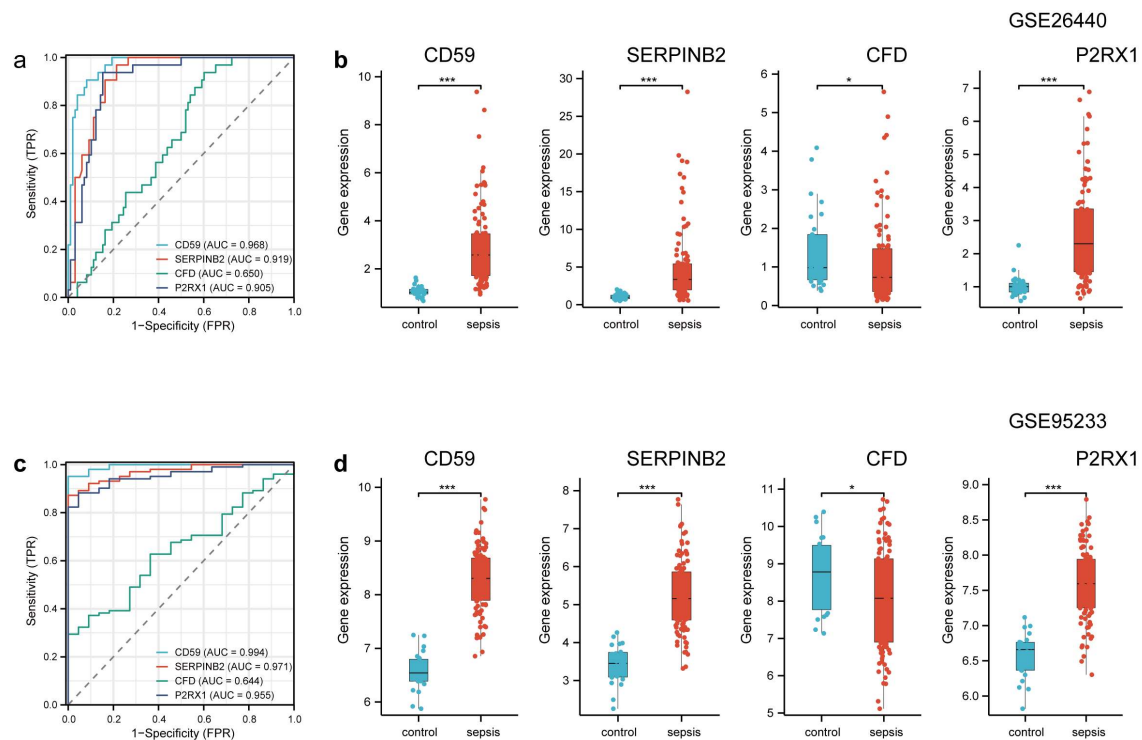
Validate transcriptomic biomarkers in a completely independent dataset.

Response: Thank you for acknowledging that our study adheres to best practices in machine learning for medicine. We wholeheartedly agree with your assessment regarding the need for clarifications and additional validations.

1. Transcriptomic Biomarker Validation: Recognizing the critical importance of external validation in biomarker research, we rigorously evaluated the five transcriptomic biomarkers (STAT5B, MTHFR, HPSE, AAK1, MX1) identified in our study using an independent sepsis cohort from the GSE54514 dataset. Critically, we confirmed their prognostic efficacy and further evaluated their diagnostic performance. ROC curve analysis demonstrated that the biomarkers' diagnostic predictive ability (AUC) closely aligned with our original dataset. Significant overexpression of AAK1 and MX1 ($p < 0.05$) was additionally observed in sepsis patients compared to controls. While methodological harmonization across studies was maintained where feasible, these outcomes, particularly the consistent diagnostic performance and differential expression patterns, collectively validate the biological relevance of these biomarkers, with detailed AUC metrics visualized in Supplementary Figure 8.



(New) Supplementary Figure 8. The prognostic and diagnostic efficacy of STAT5B, MTHFR, HPSE, AAK1 and MX1 in an external cohort. From GSE54514. a. prognostic AUC curves; b. Expression levels of STAT5B, MTHFR, HPSE, AAK1 and MX1. * $p < 0.05$, *** $p < 0.001$ vs. Healthy control.



(New) Supplementary Figure 9. The diagnostic efficacy of CD59, SERPINB2, CFD, and P2RX1 in external cohorts. From GSE26440, and GSE95233. a. and c., AUC curves; b. and d, Expression levels of CD59, SERPINB2, CFD, and P2RX1. * $p < 0.05$, *** $p < 0.001$

In the revised manuscript, we elaborated on the internal connection between these transcriptomic biomarkers and coagulation-inflammatory markers and added the related content in the Results section:

Revised in the manuscript:

“Our transcriptomic-level explanation provides complementary evidence for the critical role of coagulation-inflammatory markers in sepsis heterogeneity, initially identified through EHR-based variable and model explanations. We specifically examined coagulation-related genes (CRGs), inflammatory-related genes (IRGs), and disseminated intravascular coagulation (DIC)-related genes (Fig. 2h and Supplementary Fig. 6-7). This investigation reaffirmed the significance of these markers and identified five key prognostic biomarkers: STAT5B, MTHFR, HPSE, AAK1, and MX1. Patients stratified into a high-risk group based on these biomarkers exhibited significantly higher mortality. Notably, MTHFR, AAK1, and MX1 expression levels were elevated in this high-risk group, suggesting their influence on

sepsis prognosis potentially through modulation of immune and coagulation pathways (Supplementary Table 8). The diagnostic efficacy of these five genes was further validated in external dataset GSE54514 (Supplementary Fig. 8). Furthermore, transcriptome analysis confirmed the diagnostic relevance of four genes: CD59, P2RX1, CFD, and SERPINB2 (Supplementary Table 9-10). These markers demonstrated robust performance as independent diagnostic biomarkers for sepsis. Their diagnostic efficacy was successfully validated in external datasets GSE26440 and GSE95233 (Supplementary Fig. 9). RT-PCR analysis of PBMCs from 29 sepsis patients and 11 healthy controls revealed significant differential expression of CFD and P2RX1 ($p < 0.05$), with greater variability observed in the sepsis group. herefore, these transcriptomic findings further explained the important roles of coagulation-inflammation related indicators in the diagnosis and prognosis of sepsis from multiple perspectives.” (Page 6)

2. Prospective Validation of SepsisFormer: Regarding SepsisFormer, we understand that prospective validation in real-time ICU settings is crucial for translating our findings into clinical practice. Our current retrospective analysis across multiple cohorts (MIMIC-III, MIMIC-IV, eICU-CRD, and a local dataset) showcases the model’s generalizability. We have also conducted real-time validation specifically for SMART, which is better suited for real-time applications. We monitored 40 patients in real time, and the results were consistent with the model's predictions. We included this additional validation data in the main text of the revised manuscript or as supplementary materials (Supplementary case materials 1 and 2). In future work, we plan to collaborate with clinical partners to deploy SepsisFormer in ICUs, collecting data on model predictions and actual patient outcomes in real time. This will allow us to evaluate the model’s performance under dynamic clinical conditions, including its timeliness in detecting high-risk patients and its impact on clinical decision-making.

Revised in the manuscript:

“Last but not least, clinicians could conduct real-time risk stratification and subphenotypic classification of sepsis based on the SMART scorecard or our

open-access sepsis subphenotype and SMART platform (<http://smartsepsis.org.cn>). Here, we conducted risk stratification, phenotypic classification, and prognosis prediction for 40 sepsis patients locally admitted from March 21, 2025, to April 27, 2025 (external observational verification only) (Fig. 4e-h, and Supplementary case materials 1 and 2). The results showed that the proportions of patients in the four risk levels of mild, moderate, severe, and dangerous were 12.5%, 30%, 32.5%, and 25%, respectively (Fig. 4e), and their 28-day actual mortality rates were 0%, 16.7%, 38.5%, and 90% respectively (Fig. 4f). The overall mortality rate of the CIS1 subphenotype was 33.3%, significantly lower than 53.8% of CIS2, and the trend was the same at different risk levels (Fig. 4g and h). The mortality rate at the dangerous level (this external observational cohort) was a little higher than the predicted rate of the model in this study (approximately 50%), as well as simultaneously increased the overall mortality rates of CIS1 and CIS2, which might be related to the small sample size, but the overall trend was consistent. Therefore, clinicians can utilize the SMART scorecard or our open-access platform to conduct real-time risk classification and subphenotyping of patients, enabling objective and accurate assessment of sepsis patients to intervene as early as possible and improve their prognosis.” (Page 9)

Q4: *Are there any flaws in the data analysis, interpretation, and conclusions?*

Overall, the methodology is robust, but a few concerns require attention.

Q4.1 *External Validation of SepsisFormer Needs Clarification*

How the data were split for training, validation, and testing is unclear.

Recommendation: Clearly define cross-validation strategy and how model generalizability was assessed.

Response: Thank you for highlighting the areas that require clarification. Your feedback has been invaluable in helping us improve the clarity and rigor of our work. We acknowledge that the current manuscript lacks sufficient detail in this section.

1. **To address this concern**, we incorporated the following clarifications in the revised version. We detailed the partitioning of datasets (MIMIC-III, MIMIC-IV, eICU-CRD, and the local ICU dataset). The pretraining, fine-tuning, and validation

procedures, utilizing these diverse datasets, are also explained. Furthermore, we outline considerations for maintaining representativeness across subsets (e.g., ensuring balanced class distributions for mortality outcomes). This information has been incorporated into the methods section.

Revised in the manuscript:

“(e) Pretraining Procedure. Cohort 1 (MIMIC-III and eICU-CRD) was partitioned 7:3 for training and internal testing. From this training data, the pretrained SepsisFormer model was initialized, establishing its core architecture, weights, and hyperparameters. SMOTE was applied for data augmentation to address class imbalance and enhance model robustness during this phase. (f) Fine-tuning Procedure. Subsequently, for domain adaptation and enhanced generalizability, SepsisFormer underwent fine-tuning. Cohort 1 served as the source domain, with fine-tuning specifically targeting external validation cohorts: Cohort 2 (MIMIC-IV) and Cohort 3 (Local ICU) as the target domains. A Domain-adaptive Generator was employed for data augmentation during this transfer learning stage, optimizing performance in new clinical center.” (Page 14-15)

2. To evaluate generalizability, we tested SepsisFormer on multiple independent cohorts with different patient demographics and clinical practices. In the revised manuscript, we presented additional analyses to quantify generalizability. For example, we reported calibration plots and discrimination metrics (e.g., AUC) for each external dataset, demonstrating consistent performance across diverse settings. We also discussed any performance differences and potential reasons for these variations. We also revised the titles of the figures and tables to clarify methodological comparisons. For example, **key revisions** are:

“**Supplementary Table 6.** Performance comparison of the pre-trained SepsisFormer and baseline models for prognostic prediction on the internal test set (Cohort 1: 7,789 septic patients from MIMIC-III and eICU-CRD; 70% training, 30% test).”

“**Supplementary Table 7.** Performance comparison of the fine-tuned SepsisFormer and baseline models for prognostic prediction on the two external validation cohorts (Cohort 2: 4,191 septic patients from MIMIC-IV; and Cohort 3:428 septic patients from a local ICU), using 10-fold cross-validation.”

Q4.2 SMART's Performance Varies Across Datasets

SMART's AUC ranges from 0.6222 to 0.7360—why does it perform worse in some cohorts?

Recommendation: Provide dataset-level breakdown and discuss factors influencing variability.

Response: Thank you for pointing out the variability in SMART's performance. The AUC range of 0.6222 to 0.7360 for SMART does not reflect performance across different cohorts. Instead, these values derive from our local data, where various scoring models were evaluated. As shown in Fig. 3d, the highest AUC (0.7360) corresponds to SMART. These results suggest that the SMART outperforms the existing models in external validation. To clarify this issue, we revised the statement as follows:

Revised in the manuscript:

“In the local ICU cohort, the SMART demonstrated the highest predictive accuracy, with an AUC of 0.7360. For the five established scoring systems, including SOFA, qSOFA, LIP, APACHE II, and SIRS, the AUCs are 0.6833, 0.6441, 0.6431, 0.6222 and 0.5428 respectively (Fig. 3d). ” **(Page 8)**

Q4.3 Statistical Comparisons Are Missing for Model Performance

No DeLong's test or confidence intervals are reported to compare AUCs of different models.

Recommendation: Perform statistical comparisons to confirm that SepsisFormer is significantly better than traditional methods.

Response: Thank you for highlighting the absence of statistical comparisons for

model performance. In the revised manuscript, we addressed this issue by conducting comprehensive statistical analyses.

We supplemented our analysis by performing DeLong's test to statistically compare the AUCs of different models, confirming that SepsisFormer significantly outperforms those methods. The revised manuscript now includes a detailed description in the **Statistical Analysis** of the **Methods**, legends for **Figure 2**. Detailed statistical results are also provided on the **Source Code and Datasets website**.

Revised in the manuscript:

“DeLong's tests were performed to statistically compare the AUCs of different models using `Delong_test` from the `MLstatkit.stats` package, in conjunction with `roc_auc_score` from `scikit-learn`.” (**Page 16**)

“ROC curves illustrating the prognostic prediction ability of SepsisFormer using 36 sepsis predictors is superior to the based models on the dataset MIMIC-III ($p<0.01$, except SepsisFormer vs. LSTM $p=0.08$,) and MIMIC-IV ($p<0.01$, except SepsisFormer vs. GRU $p=0.41$) . All p-values by DeLong's test. ” (**Legends for Figure 2, Page 24**)

Q5: *Is the methodology sound? Does the work meet the expected standards in your field?*

Yes, the study follows best practices for machine learning in medicine, but it requires clarifications.

Areas for Improvement:

Q5.1 *No external validation of transcriptomic biomarkers → Validation in an independent cohort is needed.*

Response: Details are provided in our responses to Q3.2.

Q5.2 *No prospective validation of SepsisFormer in real-time ICU settings.*

Response: Thank you very much for your comment. It has been conducive to us. We have already conducted relevant supplementary research work.

1. Prospective Validation of SepsisFormer: Regarding SepsisFormer, we

understand that prospective validation in real-time ICU settings is crucial for translating our findings into clinical practice. Our current retrospective analysis across multiple cohorts (MIMIC-III, MIMIC-IV, eICU-CRD, and a local dataset) showcases the model's generalizability. Therefore, SepsisFormer's reliability and its central role in identifying key coagulation-inflammatory biomarkers make it indispensable for developing the final SMART model and deriving subphenotyping results. As the SMART model and subphenotyping represent this study's core findings with clear clinical translation application value, we performed external validation through prospective, real-time cohort analyses to verify their performance on SMART risk stratification and subphenotype classification, as well as prognosis prediction.

New Supplementary Figure 1. Integrated framework has showed above. (See the response to Q2)

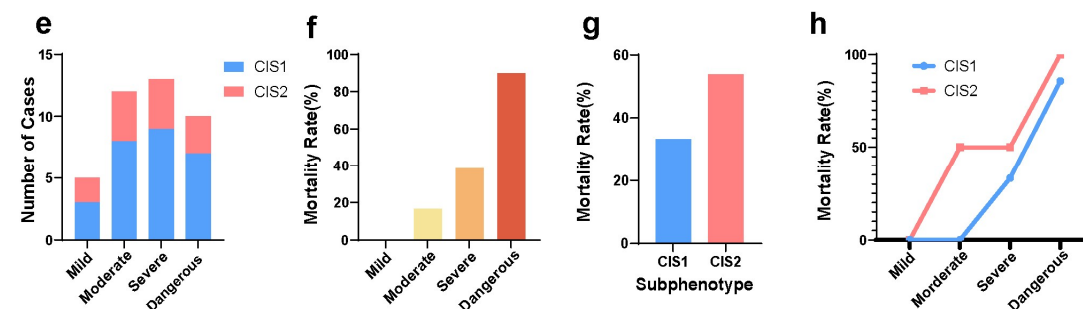
Due to our open-access web-based sepsis subphenotype and SMART platform (<http://smartsepsis.org.cn>), which is better suited for real-time applications. Here, we have added a prospective, real-time, external cohort validation study. We monitored 40 patients in real-time (the first day of being enrolled in this cohort), and the results were consistent with the model's predictions, as follows:

Revised in the manuscript:

“Furthermore, for the real-time and external validation of the risk stratification and subphenotypic classification, with the approval of the Ethics Committee of Chongqing University Affiliated Central Hospital (Chongqing Emergency Medical Center) (ID: 2025-55), a prospective observational study on sepsis risk stratification and subphenotypic classification via our risk stratification and subphenotype platform was conducted. The endpoint event was to observe the actual 28-day mortality rate and obtain the SMART score and subphenotype at the time of enrollment. After obtaining written informed consent, a total of 40 patients with sepsis were enrolled from March 24, 2025, to April 28, 2025.” (Page 13-14)

“Last but not least, clinicians could conduct real-time risk stratification and subphenotypic classification of sepsis based on the SMART scorecard or our open-access sepsis subphenotype and SMART platform (<http://smartsepsis.org.cn>).

Here, we conducted risk stratification, phenotypic classification and prognosis prediction for 40 sepsis patients locally admitted from March 21, 2025, to April 27, 2025 (external observational verification only) (Fig.4e-h, and Supplementary case materials 1 and 2). The results showed that the proportions of patients in the four risk levels of mild, moderate, severe, and dangerous were 12.5%, 30%, 32.5% and 25% respectively (Fig. 4e), and their 28-day actual mortality rates were 0%, 16.7%, 38.5% and 90% respectively (Fig. 4f). The overall mortality rate of the CIS1 subphenotype was 33.3%, significantly lower than 53.8% of CIS2, and the trend was the same at different risk levels (Fig. 4g and 4h). The mortality rate at the dangerous level (this external observational cohort) was a little higher than the predicted rate of the model in this study (approximately 50%), as well as simultaneously increased the overall mortality rates of CIS1 and CIS2, which might be related to the small sample size, but the overall trend was basically consistent. Therefore, clinicians can utilize the SMART scorecard or our open-access platform to conduct real-time risk classification and subphenotyping of patients, enabling objective and accurate assessment of sepsis patients to intervene as early as possible and improve their prognosis.” (Page 9)



Additional subgraphs in Revised Figure 4.

In future work, we plan to collaborate with multi-center clinical partners to deploy SepsisFormer in ICUs, collecting data on model predictions and actual patient outcomes in real time. This will allow us to further evaluate the model’s performance under dynamic clinical conditions, including its timeliness in detecting high-risk patients and its impact on clinical decision-making.

Q6: *Is there enough detail provided in the methods for the work to be reproduced?*
The methodology is well-documented but requires additional clarification in some areas.

Details that are clear and reproducible:

** Feature selection process for SMART.*

** SHAP-based explainability analysis.*

** Survival analysis methodology.*

Details that require clarification:

Q6.1 *How were CIS1 and CIS2 validated against existing subphenotypes?*

Recommendation:

Clarify subphenotype validation and external benchmarking.

Response: Thank you for your thorough assessment of our methodology. Your feedback regarding areas that need clarification is invaluable, and we address these points comprehensively here.

Regarding the validation of CIS1 and CIS2 subphenotypes, In the revision, we will explicitly describe how these subphenotypes were defined. We acknowledge the need for more context. Nevertheless, we admit that we have not compared CIS1 and CIS2 with other established sepsis subtyping frameworks. The main reasons are as stated above (**please see the response to Q2 comment**). Meanwhile, in the discussion section, we provided a comparison and analysis of subtype classification. Hope to gain an in-depth understanding and approval; thank you very much.

Q6.2 *What hyperparameters were used in training SepsisFormer? And how was model performance compared across datasets (cross-validation strategy)?*

Recommendation:

Provide hyperparameter details for SepsisFormer.

Response: Thank you very much for your constructive recommendations.

We provide hyperparameter details for SepsisFormer used in training, including learning rate:0.0010, batch size: 5000, number of epoch: 1400, dropout rate: 0.1000, number of attention head: 8, and depth of the Transformer architecture:8. This

transparency will enable other researchers to replicate our model.

In the revised manuscript, we clearly defined the training, testing, and external validation cohorts, and provided a detailed description of the model's pretraining and finetuning procedures, as well as the cross-validation strategy. Further details are provided in our response to Q4.1.

Revised in the manuscript:

“The hyperparameters used for training SepsisFormer include a learning rate of 0.0010, a batch size of 5000, a dropout rate of 0.1000, 1400 training epochs, eight parallel self-attention heads, and an eight integrated Transformer architecture (Supplementary Table 15).” **(Page 5)**

Thank you once again for your valuable time and feedback. Your insightful recommendations on specific issues were particularly appreciated. The manuscript has been thoroughly revised in light of these suggestions, which has significantly enhanced the quality of the work.

We hope that the revised manuscript now meets with your approval.

Wishing you all the best!

Reviewer #3 (Remarks to the Author):

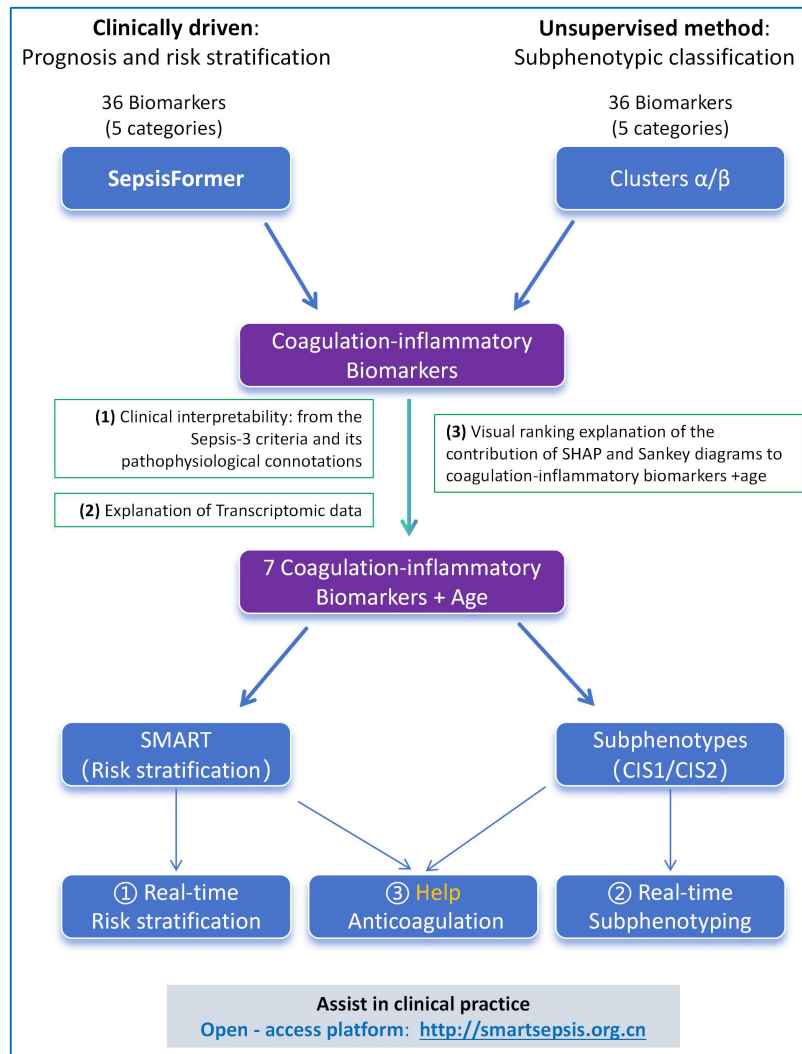
Q1: *What are the noteworthy results?*

- A prognosis model for sepsis that leverages Transformer models and domain adaptation technique that achieves high predictive accuracy and a scorecard model for risk stratification.

- The actual subphenotypes and risk levels classified by the findings from the model experiments, and how these actually capture patient heterogeneity. Although, I could appreciate this contribution later in the paper.

- An in depth analysis of and evidence for the heterogeneity of sepsis patients based on subphenotypes and risk levels.

Response: We sincerely appreciate the reviewer's positive comments on the significance of our study and the recognition of our contributions. We have carefully considered the reviewer's constructive suggestions for improvement and have taken them into account during the revision process. To further aid understanding, we've added Supplementary Figure 1.



(New) Supplementary Figure 1. Integrated framework: An overview of models, methods, biomarkers, and clinical applications in our study.

Q2: Will the work be of significance to the field and related fields? How does it compare to the established literature? If the work is not original, please provide relevant references.

Q2.1 -The performance of existing AI prediction/prognosis models is mentioned in the introduction but does not include the techniques used in those models, which can help the reader to appreciate how the proposed models in this work contribute and differ from existing literature (in terms of input data, model choice, training paradigms, etc.).

Response: Thank you for your valuable suggestion. We agree that providing details about the techniques used in existing models would offer a better context for understanding the contributions and differences of our proposed models. In the revised manuscript, we expanded the introduction to briefly describe a total of 256 sepsis prediction models from 2016–2023, including machine learning models, deep learning models, input data, and code.

Revised in the manuscript:

“A total of 256 AI-based sepsis prediction models from 73 studies (2016–2023, n=457,932) showed a pooled AUC of 0.825 (95% CI: 0.809–0.840)¹⁸. Models mainly include machine learning approaches (DT, LR, SVM, GLM, NB) and neural networks (MLP, LSTM, CNN, GRU, RETAIN, Dipole). Public datasets (e.g., MIMIC-III/IV, eICU, Computing in Cardiology) were used in 53% of studies. However, only 21.9% performed external validation; data-sharing transparency was critically limited—only three studies disclosed data access; and no studies released code¹⁸. Meanwhile, although Transformer-based models (e.g., RETAIN, BEHRT, Med-BERT)¹⁹ have achieved strong performance in EHR-driven disease risk prediction, their use in sepsis remains relatively limited. On the other hand, in clinical practice, several well-established prognostic warning score systems (SOFA¹, APACHE II, LODS²⁰, qSOFA¹, SIRS¹) remain benchmarks. In an analysis of 148,907 EHRs of suspected infection cases, the area under the receiver operating characteristic curve (AUC) for patients admitted to the ICU ranged from 0.64 to 0.75 for existing scoring systems¹⁴. In our previous study, we developed the LIP scoring system, which incorporates lymphocyte count, INR, and procalcitonin as a simple sepsis screening tool, achieving 92.8% sensitivity and 94.1% specificity. The LIP tool is well-suited for rapid clinical screening and is particularly beneficial in resource-limited settings.”

(Page 3-4)

Q2.2 - *In the domain adaptive generation, the method for oversampling is claimed to be novel but also says that it uses existing domain adaptation methods. It is not clear what the novelty of the method is. Besides, the domain adaptation is briefly mentioned in the abstract and discussed again in the results under the Prognostic prediction*

performance of SepsisFormer based on coagulation–inflammatory markers header.
The last paragraph of the introduction can include this component, as it seems
relevant considering the heterogeneity and class imbalance problems it addresses.

Response: Thank you for your valuable suggestion.

We introduce MMID, a novel domain adaptive generation method. Unlike established techniques such as mean-teacher, whitening, and moment matching, MMID is a newly developed approach. We evaluated MMID's performance against these methods and a non-adaption baseline (serving as an ablation study). The comparative results are detailed in Figure 2d and Supplementary Table 8. We have revised the original manuscript to clarify that MMID is a novel contribution, not an existing method.

As MMID represents a cross-domain strategy employing SMOTE, we have revised its nomenclature to MMID-SMOTE for improved specificity and recognition of its underlying technique. In the revised manuscript, we rewrite the generator in the prognostic model as follows:

Revised in the manuscript:

“Domain-adaptive generator for fine-tuning. To address the class imbalance and distribution heterogeneity identified through our analysis of mortality outcomes and multi-center covariate distribution shifts, we integrated a domain-adaptive generator module into the prognostic modeling framework for fine-tuning. We specifically implemented and compared several state-of-the-art domain adaptation methods, including Mean–teacher⁵², Whitening⁵³, and Moment Matching⁵⁴, to reduce inter-center distributional discrepancies. Crucially, we also propose MMID-SMOTE, an innovative and clinically practical data augmentation strategy. This novel method incorporates statistical moment alignment and min-max interval constraints. MMID-SMOTE ensures that the synthetic samples generated are both statistically robust and clinically pertinent, thereby substantially enhancing the model's generalizability to unobserved target domains while maintaining clinical reliability and applicability.” (Page 14)

Reference:

52. Tarvainen A, Valpola H. Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results. *Advances in neural information processing systems* 30, 1195–1204 (2017).

53. Roy S, Siarohin A, Sangineto E, Bulò SR, Sebe N, Ricci E. Unsupervised Domain Adaptation Using Feature-Whitening and Consensus Loss. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (2019).

54. Peng X, Bai Q, Xia X, Huang Z, Saenko K, Wang B. Moment Matching for Multi-Source Domain Adaptation. In: *Proceedings of the IEEE/CVF international conference on computer vision* (2019).

We've rewritten the Domain-adaptive Generator in **Supplementary Method 1** and provided a more explicit rationale for this module. Additionally, visual analyses have been incorporated to illustrate **class imbalance** and **heterogeneity across multiple centers**.

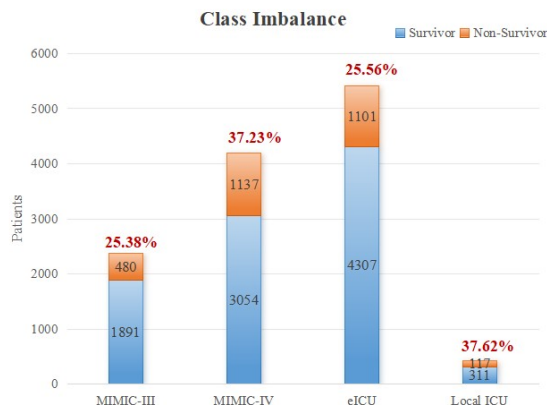
Revised in the Supplementary Method 1:

A. Domain-adaptive Generator for fine-tuning.

Rationale for Domain Adaptation in Prognostic Modeling. In multi-center Electronic Health Records (EHR) studies, class imbalance within single centers and distribution heterogeneity across multiple clinical sites are prevalent, especially in sepsis prognosis prediction tasks. Existing data augmentation methods, such as the Synthetic Minority Over-sampling Technique (SMOTE), primarily address intra-center class imbalance without considering inter-center distributional shifts. To bridge this gap, we first theoretically justify the necessity of domain adaptation by analyzing class imbalance in mortality outcomes and distribution divergence across clinical centers. These domain discrepancies can degrade model performance when transferring knowledge from one clinical center (source domain) to another (target domain), making domain-adaptive strategies an essential component for robust prognostic modeling.

The statistical chart of eICU, Local ICU, MIMIC III, and MIMIC IV is shown in Reviewer-only Figure. All datasets exhibit significant class imbalance between

positive samples (non-survivors) and negative samples (survivors), with mortality rates of 25.38%, 37.23%, 25.56%, and 37.62% respectively. The number of non-survivors was significantly lower than that of survivors.



Reviewer-only Figure. Class imbalance in mortality of multiple clinical centers.

We conducted multi-center covariate distribution shift analyses to clearly demonstrate the significant distribution shifts in laboratory measurements observed across different medical centers. Kolmogorov-Smirnov (KS) non-parametric tests were utilized to assess the consistency of distributions between two medical centers. Results indicate significant distributional shifts ($p < 0.0001$) for the 7 laboratory measurements between every pair of centers across eICU, Local ICU, MIMIC III, and MIMIC IV (see Supplementary Figure for Method). The 7 coagulation-inflammatory laboratory measurements exhibit significant class imbalance within individual centers and notable distribution shifts across different centers. Meanwhile, these clinical markers also share common feature patterns among these centers, indicating their suitability for domain adaptation approaches to address domain discrepancy (Supplementary Table 2 and 3).

Considering the heterogeneity and commonalities of multi-center laboratory measurements, this paper introduces a method combining domain adaptation and SMOTE to alleviate class imbalance and distribution bias. This ultimately enhances model generalization to unobserved target domains. We assume that the sepsis predictors in source domain ($\mathcal{S}$) and target domain ($\mathcal{T}$) share a common feature space, but their data distributions differ $p(\mathcal{X}^{\mathcal{S}}) \neq p(\mathcal{X}^{\mathcal{T}})$. Domain adaptation aims to reduce

this discrepancy by optimizing:

$$\min_{\theta} \mathcal{L}_s(\theta) + \lambda \times \mathcal{D}(p(\mathcal{X}^s), p(\mathcal{X}^T)) \quad (1)$$

where $\mathcal{L}_s(\theta)$ denotes the supervised loss on the source domain, and $\mathcal{D}$ measures the distribution discrepancy between the source and target domains, with the balancing parameter λ .

Implementation of Existing Domain Adaptation Methods. This study designates a combined dataset of 7,789 septic patients from MIMIC-III and eICU-CRD, Cohort 1, as the source domain. The target domain included two external datasets from previously unobserved centers: Cohort 2 with 4,191 septic patients from MIMIC-IV and the smaller Cohort 3, with 428 septic patients from a local ICU. Each patient's sepsis predictors are derived from the five distinct classification dimensions. To address domain discrepancy in the mortality prediction model, we incorporated several state-of-the-art domain adaptation techniques during model fine-tuning phase.

(1) Mean-Teaching. Employs a teacher-student architecture to enforce prediction consistency under perturbations across domains. (2) Whitening Transformation. Aligns the feature distributions by normalizing covariance structures between source and target domains. (3) Moment Matching: Minimizes the statistical moment differences (e.g., means, variances) between feature representations across domains. Thus, these methods were applied to synthesize domain-adaptive training samples for the fine-tuning phase of the pre-trained sepsis prognosis prediction model.

A Novel Clinically Practical Domain Adaptation Method MMID-SMOTE. Despite the effectiveness of existing domain adaptation approaches, their high computational cost, complex optimization, and resource requirements limit their direct applicability in clinical environments. To overcome this, we propose a novel and lightweight cross-domain data augmentation method, Maximum and Minimum Interval Difference-based SMOTE (MMID-SMOTE). This method integrates statistical moment analysis, domain adaptation, and clinical interpretability constraints into the classic SMOTE framework, ensuring that the synthetic data generated are both statistically valid and clinically meaningful.

Step 1: Statistical Moment Calculation and Distribution Profiling

For each prognostic predictors, compute its origin moments (mean $\mu_X = \frac{1}{n} \sum_{i=1}^n X_i$, variance $\sigma_X^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \mu_X)^2$) and central moments (skewness, kurtosis) to fully capture those distributional characteristics across the source domain S and target domain T . These moments reveal both location and dispersion, providing the statistical foundation for cross-domain feature alignment to reduce distributional shift.

Step 2: Clinical Regularization to Address Heteroscedasticity

Considering the biological variability in medical data (e.g., laboratory test results), features exhibiting excessive variance (e.g., $\sigma^2 > 100$) are subject to a logarithmic transformation to stabilize distribution and mitigate heteroscedasticity without compromising the internal correlation structure:

$$X' = \log_{10}(X + 1) \quad (2)$$

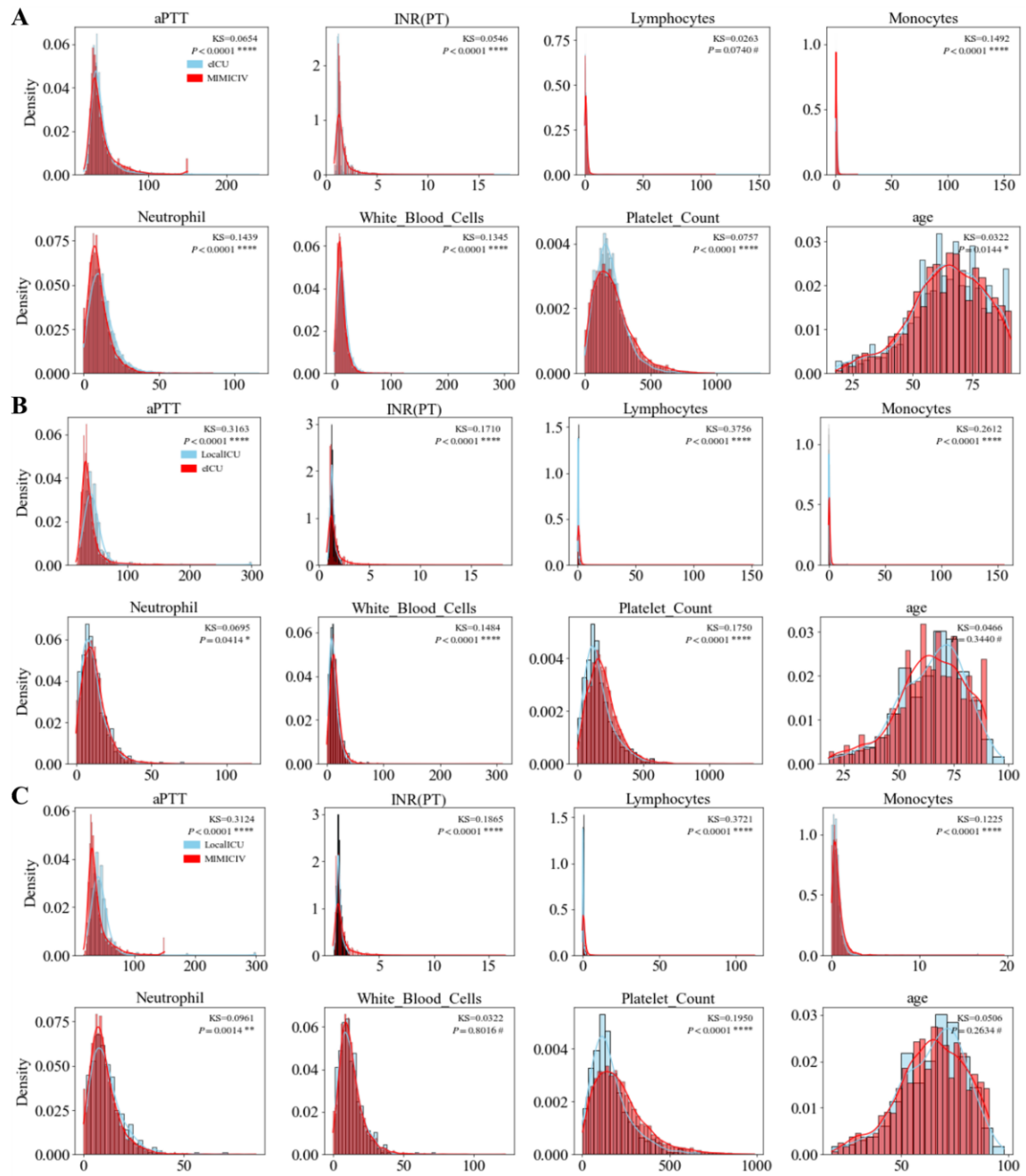
In this study, this adjustment was necessary only for the White Blood Cell (WBC) count, which showed extreme dispersion across centers—a known clinical variability marker in sepsis patients.

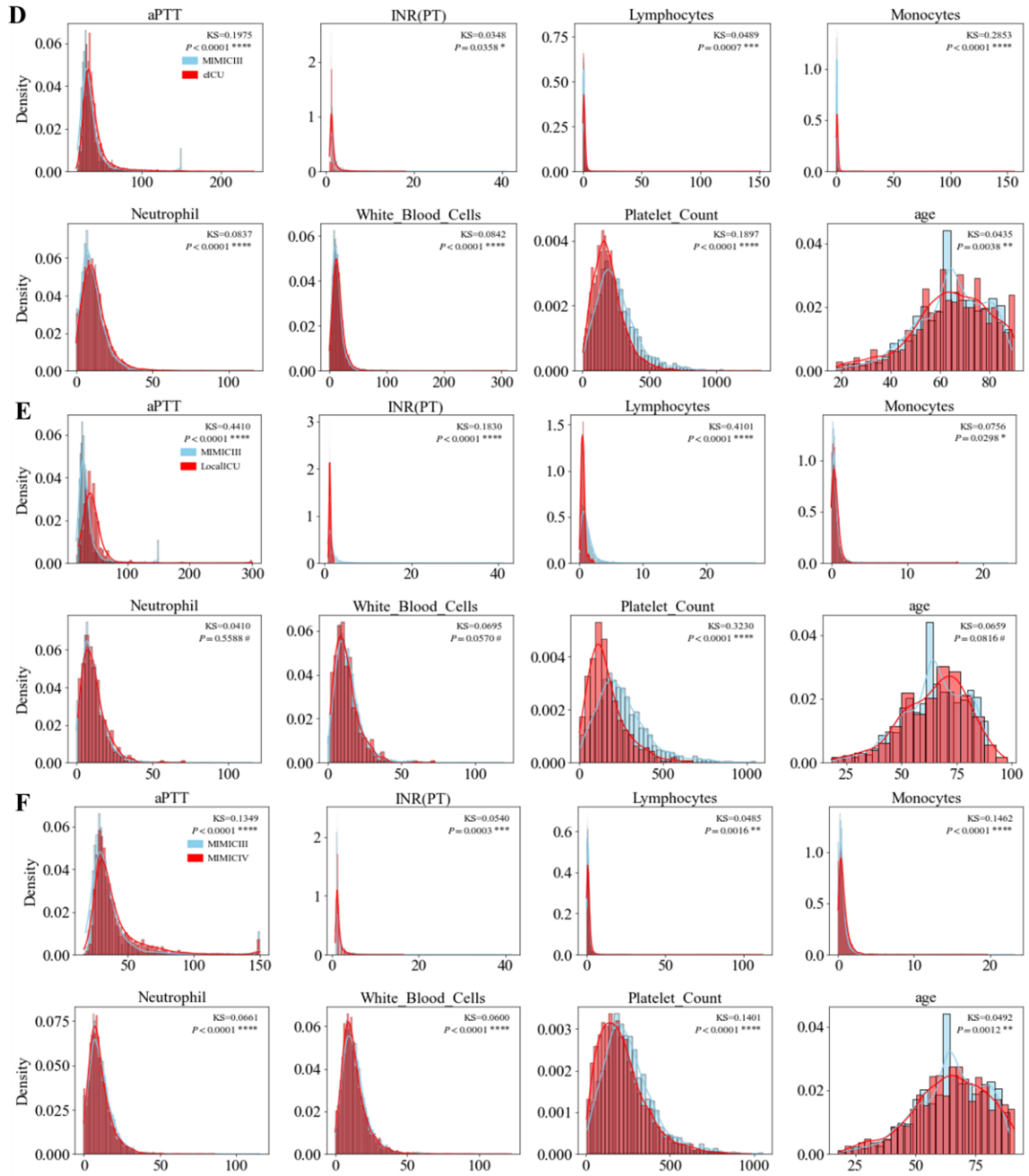
Step 3: Moment-Based Cross-Domain Projection and Alignment

To reduce distributional divergence between domains, MMID-SMOTE employs moment normalization and projection, aligning the target domain T data into the statistical space of the source domain S :

$$X_T^{\text{aligned}} = \frac{X_T - \mu_T}{\sigma_T} \times \sigma_S + \mu_S \quad (3)$$

This operation ensures domain-invariant feature learning, prevents the generation of outlier samples, and aligns the data distributions—a fundamental requirement in domain adaptation theory for machine learning models applied to heterogeneous clinical data.





Supplementary Figure for Method. Multi-center covariate distribution shift analyses.
 Data pairs used for these analyses are A, eICU and MIMICIV. B, Local ICU and eICU.
 C, Local ICU and MIMICIV. D, MIMICIII and eICU. E, MIMICIII and eICU. F, MIMIC
 III and MIMIC IV.

Step 4: Min-Max Interval Constraint Definition for SMOTE

Given that the KS test relies on the maximum vertical distance of the empirical cumulative distribution function (ECDF) $D = \max \|F_1(x) - F_2(x)\|$, and as evidenced by Supplementary Tables 3 and 4, the data exhibit substantial range discrepancies (e.g., maximum and minimum values). We define a strict min-max interval constraint for each predictor to maintain clinical plausibility.

$$X_{\min} = \min(X_S, X_T), \quad X_{\max} = \max(X_S, X_T) \quad (4)$$

Synthetic instances must satisfy:

$$X_{\text{new}} \in [X_{\min}, X_{\max}] \quad (5)$$

This constraint respects both statistical boundaries and medical interpretability, ensuring that no biologically implausible values are introduced into the dataset—a key consideration in real-world clinical AI applications.

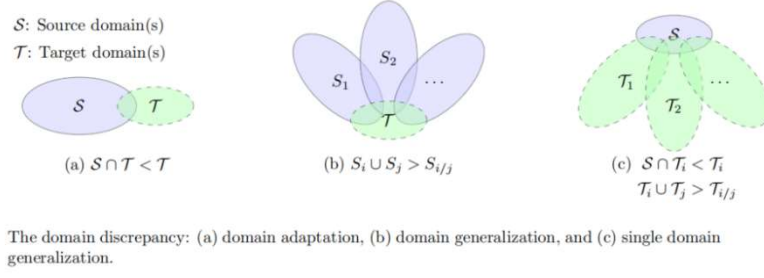
Step 5: Customized SMOTE Sample Generation

Finally, synthetic samples are generated via SMOTE under these constraints, with the following parameters optimized for medical data context.

Number of nearest neighbors : 4. Minority class oversampling ratio: 1:1 (achieving balanced class distribution). Random seed (random_state): Fixed for reproducibility. Parallel jobs: 3 (accelerating computation). Each new sample is computed as:

$$X_{\text{new}} = X_i + \delta \times (X_j - X_i) \quad (6)$$

where X_i and X_j are minority class instances within the $k=4$ nearest neighbors and $\delta \sim U(0,1)$. The synthetic point is only accepted if $X_{\text{new}} \in [X_{\min}, X_{\max}]$. This ensures the synthetic data remain both statistically valid and clinically feasible.



Reviewer-only Figure. Sepsis laboratory measurements have commonalities across centers, making them suitable for domain adaptation of domain discrepancy.

We have rewritten Figure 2d and Supplementary Table 8 to more clearly demonstrate the performance advantages of MMID-SMOTE. The comparative baseline methods, such as Mean Teacher, Whitening, and Moment Matching, all represent the current state-of-the-art in the field. The results of the ablation experiments further confirmed the importance of domain adaptive generation.

Revised in the manuscript:

“As shown in Fig. 2d and Supplementary Table 7, MMID-SMOTE significantly outperformed other domain adaptation techniques and the ablation experiment (no-adaptation baseline) by substantially improving the model's cross-cohort generalization capability. AUC, accuracy, sensitivity, specificity, and F1-score were used to evaluate the model's predictive performance. The ablation study demonstrates that MMID-SMOTE achieves superior performance over the no-adaptation baseline across all five evaluation metrics, highlighting the importance of the domain-adaptive generation module in enhancing prediction accuracy. In terms of AUC, accuracy, sensitivity, and F1-score, MMID-SMOTE consistently outperformed three state-of-the-art domain adaptation methods. ACU results show Mean-teacher and Whitening underperformed a no-adaptation baseline, with Moment Matching ranking second. Mean-teacher struggles with low-quality pseudo-labels, while Whitening's mean/covariance adjustments failed to boost cross-domain performance. Moment Matching, in contrast, matches second-order statistics. This highlights the importance of domain adaptation method selection. Our proposed MMID-SMOTE achieved superior prognostic prediction across performance in different clinical cohorts. As an

XAI tool, SepsisFormer offers reliable, sensitive, and interpretable support for clinical decision-making using a minimal set of routine markers. ” (Page 7)

“Fig.2 d, ROC curves of prognostic prediction performance based on coagulation–inflammatory markers from local ICU data, comparing MMID-SMOTE with various domain-adaptive approaches and a no-adaptation baseline.” (Figure 2 Legends Page 25)

In the Introduction and Supplementary Method 1, we explicitly addressed class imbalance and data heterogeneity.

Revised in the manuscript:

“ Although progress has been made, some limitations still exist: (1)....(2) Despite demonstrating superior performance, AI models encounter two primary challenges in clinical application: constrained predictive performance and generalization capability, largely attributable to class imbalance and multi-center data heterogeneity; concurrently, the "black-box" nature of model impedes clinicians comprehension and trust in their decision-making.” (Page 4)

Q2.3- The transcriptomic biomarker evidence analysis and study of anticoagulant treatment effect seem crucial for better understanding sepsis, reveal insights about heterogeneity, and validate algorithmic model development. However, I think that the presentation of these approaches can be clarified because it wasn't clear what experiments actually took place in this study or were using measurements from existing public datasets (E.g., the Heparin vs control studies, or PCR studies), and how the biomarkers were actually selected.

Response: Thank you for the comments on the transcriptomic biomarker analysis and anticoagulant treatment effect study. We appreciate the opportunity to clarify the approaches presented in the manuscript. It should be noted that we integrated the transcriptomic-level perspective solely for mechanistic analysis and did not incorporate it into the development of tools such as SepsisFormer and SMART; these tools rely exclusively on seven coagulation-inflammatory markers and age for analysis. The following is a detailed response to your concerns:

1. Transcriptomic Biomarker Analysis.

Analysis	Data source	Type	Location in Manuscript
GSE65682	Prognostic gene discovery (28-day mortality)	Sepsis: 760 patients Healthy controls: 42	• Fig. 2h, Supp. Fig. 6,
GSE54514	External validation of prognostic biomarkers (STAT5B, MTHFR, HPSE, AAK1, MX1)	Septic non-survivors: 9 Septic survivors: 26 Healthy controls: 18	• Supp. Fig. 8 (labeled as "external cohort from GSE54514")
GSE26440	External validation of diagnostic biomarkers (CD59, SERPINB2, CFD, P2RX1)	Septic shock: 98 children Healthy controls: 32	• Supp. Fig. 9a,b
GSE95233	External validation of diagnostic biomarkers (CD59, SERPINB2, CFD, P2RX1)	Septic shock: 51 Healthy controls: 22	• Supp. Fig. 9c,d

The biomarker selection process involved two main steps. In Step 1, gene screening was conducted to identify diagnostic and prognostic biomarkers. For diagnostic biomarkers (CD59, SERPINB2, CFD, P2RX1), they were first identified from the differentially expressed genes (DEGs) between sepsis patients and controls in the GSE65682 dataset, then overlapped with coagulation- and inflammation-related genes (CRGs/IRGs) retrieved from KEGG and GeneCards databases, and finally validated in external GEO datasets (GSE26440 and GSE95233).

For prognostic biomarkers (STAT5B, MTHFR, HPSE, AAK1, MX1), they were derived from the DEGs between survivors and nonsurvivors in GSE65682 (Supplementary Fig. 6a), overlapped with disseminated intravascular coagulation (DIC)-related genes (Supplementary Fig. 6c), and the final genes were selected through LASSO-Cox regression (Supplementary Fig. 6e). In Step 2, further gene screening was performed using reverse transcription-quantitative polymerase chain reaction (RT-qPCR) on a local cohort consisting of 29 sepsis patients and 11 controls, which confirmed the differential expression of CFD and P2RX1 (Fig. 2h, violin plots; Supplementary Table 9). Meanwhile, STAT5B, HPSE, AAK1, and MX1 were only computationally validated using GEO datasets.

2. Anticoagulant Treatment Effect (e.g., Heparin vs. Control)

Retrospective Analysis of Observational Data: The analysis of anticoagulant effects (heparin) on sepsis outcomes was performed using retrospective clinical data from the MIMIC-IV EHRs (n=4,191 patients: 946 heparin-treated vs. 3,245 controls). Method: Patients were stratified into SMART risk levels (mild/ moderate/ severe/ dangerous) and CIS subphenotypes. The heparin cohort comprised patients receiving heparin for ≥ 3 consecutive days (n=946), whereas the control cohort included those not receiving heparin (n=3,245). The primary outcome, 28-day survival, was analyzed using Kaplan-Meier and Cox models (Fig. 4a,c). However, this study did not involve in vitro or in vivo experiments for anticoagulant treatment. The focus was on clinical data-driven insights into treatment patterns and their association with biomarker profiles.

Q3: *Does the work support the conclusions and claims, or is additional evidence needed?*

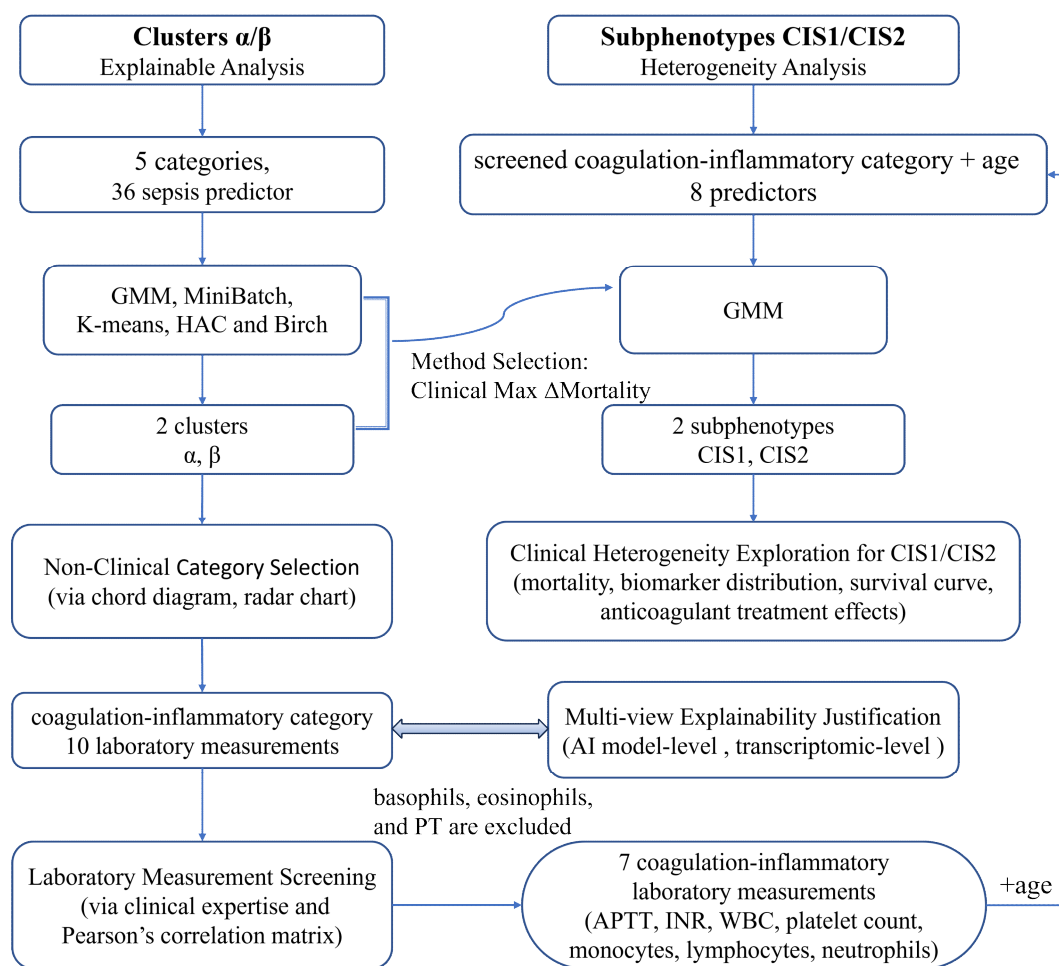
Q3.1 *As I was reading the paper, I struggled to understand the support for some of the claims in the results section since it wasn't clear the criteria used for doing further classifications or analyses. For example, how does further categorizing the predictors into clinically relevant ranges (not clear how these are defined) facilitate clinical decision making? And then, why further analyzing the two subtypes across the five key categories of sepsis predictors? This comment is also related to the clarity presenting the methods.*

Response: Thank you for your valuable feedback. I understand the result and method sections lacked clarity regarding the criteria for further classifications or analyses.

We appreciate the reviewer's insightful feedback. While our aim was to categorize 36 predictors into clinically relevant ranges (low, normal, high) based on sepsis expert consensus for clinical reference, we acknowledge that this section did not effectively enhance clinical decision-making or cohere with the overall manuscript. Therefore, we have proceeded with the removal of this content from the main body and eliminated Supplementary Table 2.

Further analysis of the two clusters across the five key categories of sepsis predictors aimed to pinpoint critical categories, specifically highlighting the coagulation-inflammatory category. We have systematically replaced all instances of 'subtypes α and β ' with 'clusters α and β ' throughout the manuscript. Concurrently, this section and Fig. 2f have been rewritten. Here, we delineate the difference and relationships concerning clusters α/β and subphenotypes CIS1/CIS2 in Supplementary Figure: 'Research Framework for Cluster and Subphenotype Analyses'.

For more information on the differences between clusters α/β and subphenotypes CIS1/CIS2, please refer to our response to Q5.3.



(New) Supplementary Figure 2. Research framework for cluster and subphenotype.

Revised in the manuscript:

“Clusters α and β were automatically derived using unsupervised clustering methods using 36 sepsis predictors, requiring no prior information (e.g., mortality, disease

outcomes, or treatment medications), based on the optimal number of clusters (Supplementary Table 4). Five different unsupervised methods—Gaussian Mixture Model (GMM), MiniBatchKMeans, K-means, Hierarchical Agglomerative Clustering (HAC), and Birch—generated highly aligned clusters, a consistency across distinct approaches that confirms the robustness and validity of the two identified clusters. Since these data-driven clusters lack inherent clinical interpretation, we statistically analyzed subgroup mortality. The GMM-derived α and β clusters exhibited the most significant mortality difference (32.09% vs 17.62%, respectively), indicating GMM's superior effectiveness for identifying patient subgroups with divergent mortality risks. In the chord diagram, both cluster α and cluster β consistently showed the widest chords with coagulation-inflammatory category (ribbons connect with these portions of the circle), indicating coagulation-inflammatory predictors are key defining features for the clustering. Independently, the radar plot also confirmed the coagulation-inflammatory category as the highest connection point. These findings were robustly replicated across all five unsupervised clustering methods (Fig. 2f and Supplementary Fig. 3).” (Page 5)

“This study employs GMM for subphenotype identification, as the patient subgroups derived from GMM show the most significant differences in mortality rates (shown in Section Explainability Analysis), indicating greater clinical relevance.” (Page 7)

Q3.2 From my understanding and interpretation, I agree with the paper that explainable AI was used to uncover patient heterogeneity, I am not sure if the model SepsisFormer was built to be explainable. Many parts use modules from Transformers, a complex deep learning architecture, that as mentioned, can be beneficial for modeling the complex relationships between the predictors. In fact, the explainability technique used in the model directly is a post-hoc technique and this can be acknowledge for clarity. Besides, it is mentioned that the model is heterogeneity-aware because of the domain-adaptive generator, but this sounds like a data augmentation strategy for training.

Response: We sincerely thank the reviewer for this insightful comment.

We fully acknowledge that both the variable-level and model-level explainability analyses employed in our study are inherently post-hoc. To improve clarity, we have clarified this aspect in the revised manuscript. Furthermore, in response to **Q3.1** and **Q5.3**, we have consistently replaced 'subtypes α and β ' with 'clusters α and β '.

Revised in the manuscript:

“Multi-view explainability analyses were conducted from three perspectives: two post-hoc approaches (cluster-informed EHR variable-level and AI model-level), and a transcriptomic-level analysis inspired by multi-omics principles.” (Page 14-15)

We agree that the domain-adaptive generator is indeed a strategy for training, and we have explicitly clarified its role as such in the revised manuscript. However, it is fundamentally different from data augmentation methods. We have addressed this point in our response to **Q2.2**. Here, we provide further clarification. Both domain-adaptive generation and data augmentation (e.g., SMOTE) are used to improve model generalization, but they serve different goals and operate differently. We summarize their differences below.

	Data Augmentation	Domain-Adaptive Generation
Primary Goal	Improve generalization within a single domain	Improve performance across different domains
Domain Scope	Typically single domain or center	Typically multiple domains or centers
Data Modification	Creates variations of existing data	Transfers knowledge between domains
Domain Shift Focus	Addresses variance within the same distribution	Addresses variance between different distributions
Labeled Target Data	Not required (operates on source data)	Can range from none to some labeled data in target domain
Transfer Learning	Not a direct subset of Transfer Learning	A subfield of Transfer Learning
Common Techniques	Geometric/photometric transforms, noise injection	Domain distribution alignment, domain mapping, invariant feature learning, adversarial training

Q3.3 Claims around improving patient outcomes in the introduction: this has to be evaluated through the application in clinical decision-making and translational research.

Response: We agree this is crucial for the lack of clinical implementation evidence. We replaced the illustration with “Our work, therefore, offers a novel set of simple, real-time executable tools for sepsis heterogeneity, demonstrating considerable potential to significantly enhance sepsis clinical practice globally, particularly in resource-constrained healthcare settings.” in the **Abstract** section (**Page 2**)/ and “Overall, this work, by combining XAI and coagulation-inflammatory markers, deeply explores sepsis heterogeneity and develops high-performance, real-time tools for clinical practice.” in the **Introduction** section. (**Page 4**)

A prospective, ethics committee-approved study has evaluated the clinical impact of these tools by comparing pre-implementation and post-implementation outcomes, including mortality rates and ICU length of stay (**Fig4 e-h**).

Revised in the manuscript:

“e-f, External validation of real-time SMART risk stratification and subphenotype classification conducted using our open-access platform (<http://smartsepsis.org.cn>). e, The real-time (day 1) risk stratification and distribution of the 40-patient cohort. The number of patients in the four risk levels of mild, moderate, severe and dangerous were 5 (CIS1, 3 and CIS2, 2), 12 (CIS1, 8 and CIS2, 4), 13 (CIS1, 9 and CIS2, 4) and 10 (CIS1, 7 and CIS2, 3). f, The mortality rates of different risk levels were 0%, 16.7%, 38.5% and 90% respectively. g, The mortality rates of CIS1 and CIS2 were 33.3% and 53.8%. h, The mortality rates of patients with different CIS classifications at the four risk levels were as follows: at the mild level, both CIS1 and CIS2 had a mortality rate of 0; at the moderate level, CIS1 had a mortality rate of 0 and CIS2 had a rate of 50%; at the severe level, CIS1 had a rate of 33.3% and CIS2 had a rate of 50%; and at the risk level, CIS1 had a rate of 85.7% and CIS2 had a rate of 100%.” (**Figure legends, page 29**)

Q3.4 What is meant by clinically relevant in this statement: “XAI method achieved clinically relevant sensitivity and accuracy in prognostic prediction with only eight

routine markers.”

Response: Thanks for the clarification. In this context, "clinically relevant" refers to performance thresholds that would meaningfully alter clinical decision-making, primarily encompassing AUC, specificity, and sensitivity.

Revised in the manuscript:

“Using only seven coagulation-inflammation biomarkers and age, SepsisFormer achieved high prognostic performance, with AUCs of 0.8558 in internal testing (in Cohort 1, specificity: 0.7398; sensitivity: 0.9264) and 0.8596/0.8364 in external validations (Cohorts 2/3). DeLong’s test confirmed its superiority over all baseline models in Cohort 1 (all $p < 0.01$) and most baseline models in the external cohorts, demonstrating the model’s robustness and generalizability (Fig. 2e; Supplementary Tables 5,6).” (Page 7)

Q4: *Are there any flaws in the data analysis, interpretation and conclusions? - Do these prohibit publication or require revision?*

Q4.1 *Even though the components of SepsisFormer along with their functionality are explained, the process for making these algorithmic choices is not clear. Was there any ablation study to select the parameters or assess the contribution of some parts (e.g., the domain adaptation)? This will provide stronger evidence for the models proposed in this paper.*

Response: We sincerely thank the reviewer for this insightful comment. We agree that a more detailed explanation of our algorithmic choices and the empirical evidence supporting them would significantly strengthen the paper. In the revised manuscript, we elaborate on the following points:

Contribution of Domain Adaptation Module (Ablation Studies). To provide stronger evidence for the proposed model, we quantify the impact of the domain adaptation module on model performance. This demonstrates its necessity and effectiveness in mitigating domain shift and improving generalization. Please refer to our response to **Q2.2** for detailed revisions.

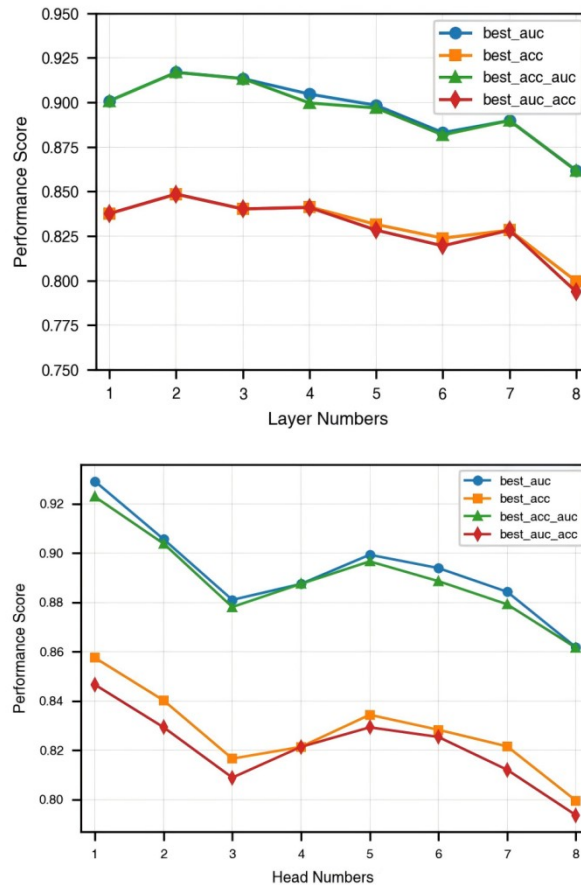
Transformer Encoder. We included supplementary experiments, examining the

effects of varying the number of Transformer layers and self-attention heads. The corresponding changes in performance illustrate the rationale behind these choices. Additionally, we have added justifications for choosing the Transformer architecture in the *Introduction* section. Please refer to our response to **Q5.2**.

To determine the optimal architecture for our deep learning-based prognostic model, we conducted a systematic hyperparameter sensitivity analysis on its core components: the number of Transformer layers (M) and the number of self-attention heads (N) within each layer. The experiment was performed on the training set, cohort1. First, we evaluated the impact of varying the number of layers on model performance. The results indicated that the model achieved excellent and stable performance when M was between 1 and 7, with a slight decline observed at $M=8$. Considering that a deeper model has greater potential for feature extraction and to strike a balance between model complexity and generalization, we selected an 8-layer architecture. Subsequently, with M fixed at 8, we further investigated the effect of the number of attention heads (N) ranging from 1 to 8. A similar trend was observed, where performance was superior for N between 1 and 7, followed by a drop at $N=8$. The corresponding performance curves are illustrated in the new Supplementary Figure.

Revised in the manuscript:

“To optimize the model architecture, we conducted a systematic hyperparameter sensitivity analysis on the number of Transformer layers L and attention heads. We chose $L=8$ to balance complexity and feature extraction, despite peak performance at $L=1\sim 7$. With M fixed at 8, we observed a similar pattern for attention heads (H), with optimal performance at $H=1\sim 7$ before a decline at $H=8$ (Supplementary Fig. 5). Furthermore, the consistent ranking of predictor importance underscores the model's structural robustness and deterministic nature. These findings collectively affirm the model's reliability and interpretability in multivariate prediction tasks.” (Page 6)



(New) Supplementary Figure 5. Hyperparameter sensitivity analysis of the number of Transformer layers and attention heads on model prognostic performance.

Statistical Comparison of Performance. We supplemented our analysis by performing DeLong’s test to statistically compare the AUCs of different models, confirming that SepsisFormer significantly outperforms the eight deep learning and machine learning methods. We implemented the test using Delong_test from the MLstatkit.stats package, in conjunction with roc_auc_score from scikit-learn. The revised manuscript includes detailed descriptions of the statistical methods and results in the **Statistical Analysis** of the **Methods** section, along with legends for **Figure 2**.

Revised in the manuscript:

“DeLong’s tests were performed to statistically compare the AUCs of different models using Delong_test from the MLstatkit.stats package, in conjunction with roc_auc_score from scikit-learn.” (**Page 16**)

“d, ROC curves of prognostic prediction performance based on

coagulation-inflammatory markers from local ICU data, comparing MMID-SMOTE with various domain-adaptive approaches and a no-adaptation baseline. e, The AUC for prognostic prediction demonstrates the superior performance of SepsisFormer with coagulation-inflammatory markers over the baseline models in Cohort 1 (all $p < 0.01$), Cohort 2 ($p < 0.01$, except SepsisFormer vs. LSTM, $p = 0.91$), and Cohort 3 ($p < 0.05$, except SepsisFormer vs. GPT, $p = 0.16$, and SepsisFormer vs. RF, $p = 0.60$). All p-values by DeLong's test. Cohort 1 with 7789 septic patients from MIMIC-III and eICU-CRD, cohorts 2 with 4,191 sepsis patients from MIMIC-IV and Cohort 3 with 428 patients from a local ICU.” (Legends for Figure 2, Page 24)

Hyperparameters. We have added details regarding the hyperparameters used for training our model SepsisFormer. Specifically, the learning rate is set to 0.0010, the batch size is 5000, the dropout rate is 0.1000, and 1400 training epochs.

Q4.2 *The discussion section currently presents medical literature and definitions in a list-like format. While this demonstrates a strong understanding of sepsis from a clinical perspective (which is valuable for algorithm development and interpretation of the results), the manuscript would benefit from a more analytical approach. I recommend expanding the discussion to include comparative analyses of the different models for the prognosis and risk stratification task, the different domain adaptation strategies used, or any other pertinent comparisons.*

Response: Thank you very much for your constructive suggestions. Just as we mentioned at the very beginning, we utilized two new framework diagrams (Supplementary Figures 1-2) to help us understand our research work and content more quickly. Regarding the discussion section, we did indeed lack comparisons of existing subtype classifications or models before. Based on the opinions of other reviewers, we have already been discussing the relevant comparisons of some certificates.

Revised in the manuscript:

“One of the important studies that directly applied transcriptomic data to the

subtyping of sepsis came from the MARS consortium in 2017⁴⁴. They enrolled a total of 787 cases of sepsis patients in a discovery cohort and two validation cohorts. Through machine learning and analysis of differentially expressed genes, sepsis was classified into four subtypes, MARS1-4. Among them, MARS1 had the poorest prognosis with a mortality rate as high as 35%. This classification at the level of gene expression is also named endotype. In 2019, Seymour CW *et al.*⁷ conducted phenotypic analysis on 20,189 patients with sepsis using statistics, machine learning and simulation tools. 29 variables, including cardiovascular, hematopoietic, hepatic, coagulation-inflammatory, neurological, pulmonary, renal, and so on, were used. Finally, they divided the patients into four phenotypes: α , β , γ , and δ . Among them, the mortality rate of phenotype α was the lowest at approximately 5%, while the mortality rates of type β , γ and δ were 13%, 24% and 40% respectively. From the perspective of mortality rates, our risk stratification results are similar to those of the Seymour CW's phenotypic classification. However, we revealed the heterogeneity of sepsis from two aspects: the stratification of risks and the subtypes. Moreover, the clinical indicators we use are fewer and more beneficial for clinical practice, which not only reduces the consumption of patients' blood samples but also saves costs, and it is also conducive to real-time dynamic assessment. Furthermore, the Seymour CW team has not yet developed a scoring system or a shared platform that can be universally implemented by clinicians. Most importantly, although many sepsis subtypes have been published at present, due to the different goals, the use of different clinical indicators, different machine learning models, or black-box algorithms, there are varying degrees of differences among these subtypes, ultimately resulting in low comparability or overlap among them (like MARS1-4, SRS1-2, Hyper or hypo inflammatory, and SENECA subtypes)⁸, and no shared classification tools or platforms have been simultaneously proposed. However, it is still necessary to further conduct comparative studies on our classification and the existing subtype classifications.” (Page 11)

Reference as following:

44. Classification of patients with sepsis according to blood genomic endotype: a

prospective cohort study. Lancet Respir Med. 2017 Oct;5(10):816-826.

7. Derivation, Validation, and Potential Treatment Implications of Novel Clinical Phenotypes for Sepsis. JAMA. 2019 May 28;321(20):2003-2017.

8. Uncovering heterogeneity in sepsis: a comparative analysis of subphenotypes. Intensive Care Med (2023) 49:1360–1369.

Q4.3 *Considering the multiple insights found in this study, the paper is lacking an overall conclusion about the approach followed. This can also be beneficial to reinforce the main message and contribution of the paper.*

Response: Thank you very much for your constructive comments and recommendations. Yes, we do need to present the conclusion, which we have supplemented in the main text. Please refer to the end of the discussion section in the manuscript.

Revised in the manuscript:

“In summary, this study introduces the prognostic model SepsisFormer, the automated risk stratification tool SMART, and the subphenotypes CIS1/CIS2 to characterize sepsis heterogeneity. Through multi-level explainability analyses, age and seven routine coagulation-inflammatory markers were identified as key predictors for sepsis diagnosis and prognosis. SepsisFormer achieved an AUC of 0.9301, outperforming state-of-the-art models, while SMART reached an AUC of 0.7360, exceeding conventional clinical scores and effectively stratifying mortality risk. Notably, CIS2 patients showed higher mortality and distinct coagulation-inflammatory profiles compared to CIS1. Integrating subphenotyping with risk stratification uncovered heterogeneity in anticoagulation treatment effects, supporting more precise and safer therapeutic decision-making in sepsis management. Patients with moderate/severe levels or CIS2 get more substantial benefits from anticoagulant treatment. An open-access, web-based platform (<http://smartsepsis.org.cn>) facilitates real-time risk stratification and subphenotype identification using only seven low-cost, routinely available coagulation-inflammatory biomarkers. Its simplicity, accessibility, and practical applicability make it a promising tool for improving sepsis management

worldwide, particularly in resource-constrained healthcare settings. (Page 12)

Reference: Global, regional, and national sepsis incidence and mortality, 1990-2017: analysis for the Global Burden of Disease Study. Lancet. 2020 Jan 18;395(10219):200-211.

Q5: Is the methodology sound? Does the work meet the expected standards in your field?

In general, the selected methods seem appropriate for the problems to be addressed in the paper. However, I found the following concerns:

Q5.1 In the introduction it says that the methods use only routine laboratory blood measurements and how this provides an alternative to blood transcriptomic-based methods. Therefore, I found confusing when later in the paper an analysis of transcriptomic biomarkers is introduced (which involved additional methodological steps). Please clarify if I misunderstood this part.

Response: Thank you for your valuable feedback. You have astutely identified a point of potential confusion in our manuscript, and we sincerely apologize for the lack of clarity in our initial manuscript.

As stated in our introduction, the core objective of this research is to develop a sepsis prognostic tool that relies **exclusively on routine, low-cost laboratory blood measurements** to ensure its accessibility and utility in a real-world clinical setting. We wish to formally clarify that our prognostic model development, validation, and proposed clinical application workflow **do not, at any stage, involve the use of costly and time-intensive transcriptomic data.**

The transcriptomic analysis you refer to was designed as a separate, *post-hoc mechanistic validation module*. It is not part of the predictive pipeline. Instead, its purpose was to provide multi-omic evidence to explore and validate the **biological plausibility** of the coagulation-inflammatory markers selected by our model, thereby strengthening the mechanistic underpinnings of our findings.

To present this logic unequivocally, our research path can be summarized as follows:

1. **Prediction Model Development:** We leveraged machine learning to construct a sepsis prognostic model based on 35 routine blood measurements.
2. **Key Feature Selection:** Through model explainability analysis, we identified seven critical coagulation-inflammatory markers that were pivotal for prediction, enabling the creation of a more parsimonious and effective model for sub-phenotyping and risk stratification.
3. **Biological Validation:** Subsequently, we employed an independent transcriptomic analysis to elucidate, from a mechanistic standpoint, why the coagulation-inflammatory pathways represented by these seven routine markers hold such a central role and high prognostic utility in the pathophysiology of sepsis.

To thoroughly resolve the ambiguity in the revised manuscript, we have undertaken four systematic modifications:

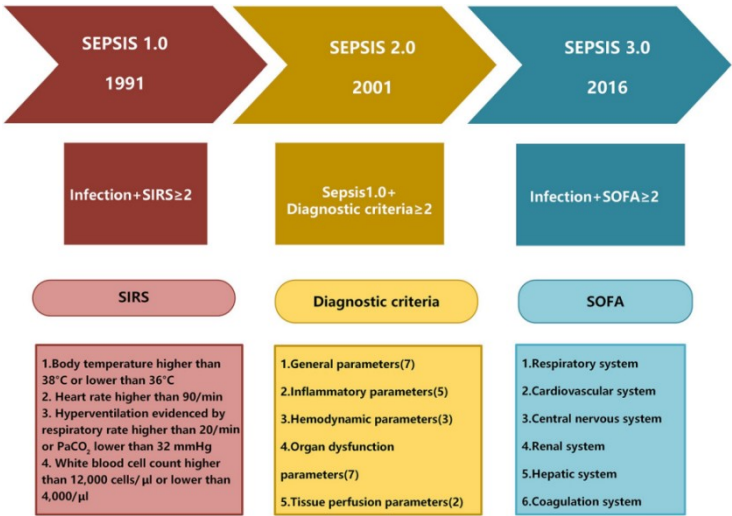
1. **Addition of Supplementary Figure 1:** A new schematic overview of the study design visually delineates "Prognostic Model SepsisFormer" and "Explainability Analyses" as two distinct yet complementary modules, explicitly positioning the transcriptomic analysis within the latter.
2. **Addition of Supplementary Figure 2:** This figure details the feature-selection funnel from 35 initial predictors to the seven key coagulation-inflammatory markers and illustrates how transcriptomic analysis provides robust corroborating evidence for their biological significance.
3. **Reinforcement of Transcriptomic Validation Rigor (Supplementary Figures 8, 9):** To further bolster the robustness of our explanatory findings, we have incorporated external dataset validation for the key transcriptomic biomarkers, confirming their stability and performance.
4. **Systematic Revision of the Manuscript Narrative:** We have systematically rewritten relevant descriptions in the Introduction and Results sections to consistently emphasize the explanatory role of the transcriptomic analysis and to affirm that our final proposed model is based entirely on routine laboratory markers, underscoring its clinical actionability.

We are confident that these comprehensive revisions fully address the issue you raised. Thank you once again for your incisive comment, which has been instrumental in enhancing the rigor and clarity of our work.

***Q5.2** The analyses presented in the paper are extensive, but the rationale for each type is missing. For example, why does it matter to see the contribution of the five categories of predictors? And their associations? Why is it important to analyze the relationships between the 1-N transformer heads and lab measurements? Including these justifications can also clarify the choice for the visualization and/or analysis technique.*

Response: Thank you for your feedback. It is crucial for improving the paper's completeness and readability. In the revised manuscript, we will ensure that the rationale and clinical/research justifications for these analytical methods are clearly, concisely, and adequately explained and supported by sufficient evidence.

Sepsis, a complex and heterogeneous syndrome, is evaluated in this study using 36 predictors across five categories, which align with the latest Sepsis-3.0 definition (excluding the central nervous system) and comprehensively reflect the disease state. However, the practical clinical utility of all 36 predictors is often limited by several significant factors. Therefore, our further analysis involving clustering across these five key categories serves critical clinical research objectives.



Reviewer-only Figure. Categories in Sepsis Definition

We have thoroughly revised this issue as follows.

(1) We have elaborated on the research aims of explainability analyses.

Revised in the manuscript:

“The 35 laboratory measurements collectively capture multi-organ dysfunction spanning five categories: coagulation-inflammatory, hepatic, renal, blood gas, and oxygen transport (Erythrocyte), while largely conforming to the connotation of Sepsis-3 criteria (excluding neurological markers). However, their full clinical adoption is hindered by prohibitive costs, large blood volume requirements, and implementation barriers in resource-constrained settings. To overcome these challenges, explainability analyses were performed across five categories to achieve two clinical goals: (a) Identify a mechanistically relevant, and clinically feasible subset of laboratory measurements within one category for real-time and cost-effective application; (b) Select an efficient unsupervised clustering method to uncover clinically meaningful sepsis subphenotypes, essential for understanding disease heterogeneity and guiding clinical insights.” (Page 5)

(2) We've elaborated the cluster-informed EHR variable-level explainability.

Revised in the manuscript:

“Multi-view explainability analyses were conducted from three perspectives: two post-hoc approaches (cluster-informed EHR variable-level and AI model-level) and a transcriptomic-level analysis inspired by multi-omics principles. To select key categories that effectively mirror sepsis pathophysiology and are practical for clinical application, we employed unsupervised clustering methods, including Gaussian Mixture Model (GMM), MiniBatchKMeans, K-means, Hierarchical Agglomerative Clustering (HAC), and Birch. Chord diagrams and radar plots enhance the interpretability of unsupervised clustering results by revealing the variable patterns defining each cluster. Clinical expertise and Pearson's correlation matrix were employed to analyze the associations between individual laboratory measurements within each category.” (Page 14-15)

(3) The rationale for selecting the seven coagulation-inflammatory biomarkers and age is detailed in the **new Supplementary Figure 2: 'Research Framework for Cluster and Subphenotype Analyses.'** Further details are provided in our responses to **Q3.1** , **Q5.2** and **Q5.3**. We have added a comprehensive summary regarding the selection of laboratory measurements. Further details are provided in our responses to **Q5.4**.

Why is it important to analyze the relationships between the I-N transformer heads and lab measurements?

Response: In fields such as medicine and bioinformatics, the Sankey diagram is a widely used visualization tool. The Sankey diagram in Figure 2g illustrates the cumulative contribution and ordering of eight predictors across different Transformer depths, with $N \in [1, 8]$. Here, N denotes the number of cascaded Transformer layers, rather than the number of self-attention heads, which is represented by M . In this study, both N and M are 8.

Two key observations can be drawn from the Sankey diagram:

(1) As the number of Transformer layers increases, the overall contribution of the eight predictors shows a slight upward trend, indicating enhanced feature integration capacity with deeper architectures. However, the overall variation remains modest, suggesting structural stability;

(2) For any fixed N , the contribution ranking of the predictors remains highly consistent, reflecting strong repeatability and robustness in the model's feature selection process. This implies that model performance is governed by deterministic structural mechanisms rather than stochastic fluctuations.

The Sankey diagram provides visual and empirical support for the model's stability and consistency in feature utilization as network depth increases, reinforcing its reliability and interpretability in medical or multivariate predictive tasks.

Revised Manuscript:

“The Sankey diagram shows that as Transformer depth increases, the cumulative contributions of the eight predictors slightly increase yet remain generally stable, indicating consistent feature integration across layers.” (**Page 6**)

Q5.3 Not clear how the subtypes alpha and beta in the variable-level explainability analysis differ from the subphenotypes if they are both identified based on distinctions in mortality rates.

Response: Thank you for your valuable feedback. It is crucial to clarify the two conceptual groupings presented in our study.

In the revised manuscript, we further emphasize that the "clusters α and β " and "subphenotypes CIS1 and CIS2" aren't defined by differences in mortality rates. Instead, they were automatically derived using unsupervised clustering methods that require no prior knowledge. Specifically, the observed difference in mortality between "clusters α and β ," "CIS1 and CIS2" emerged after unsupervised clustering and was found to be consistent with clinical observations, thereby supporting the clinical relevance of these clusters. Differential mortality analysis of clusters α/β guided subphenotype clustering methodology (GMM is selected).

We acknowledge that the term "subtypes" may cause confusion. However, we acknowledge that the term "subtypes" may cause confusion, particularly given the established use of "subtypes α , β, γ, δ " derived from EHR data. To improve clarity, we have replaced all instances of 'subtypes α and β ' with 'clusters α and β ' throughout the manuscript. According to the widely accepted definition, a subphenotype refers to "A distinct subgroup within a phenotype that can be reliably discriminated from other subgroups based on a set or pattern of observable or measurable properties." In this context, categorizing CIS1 and CIS2 as subphenotypes is appropriate and justified.

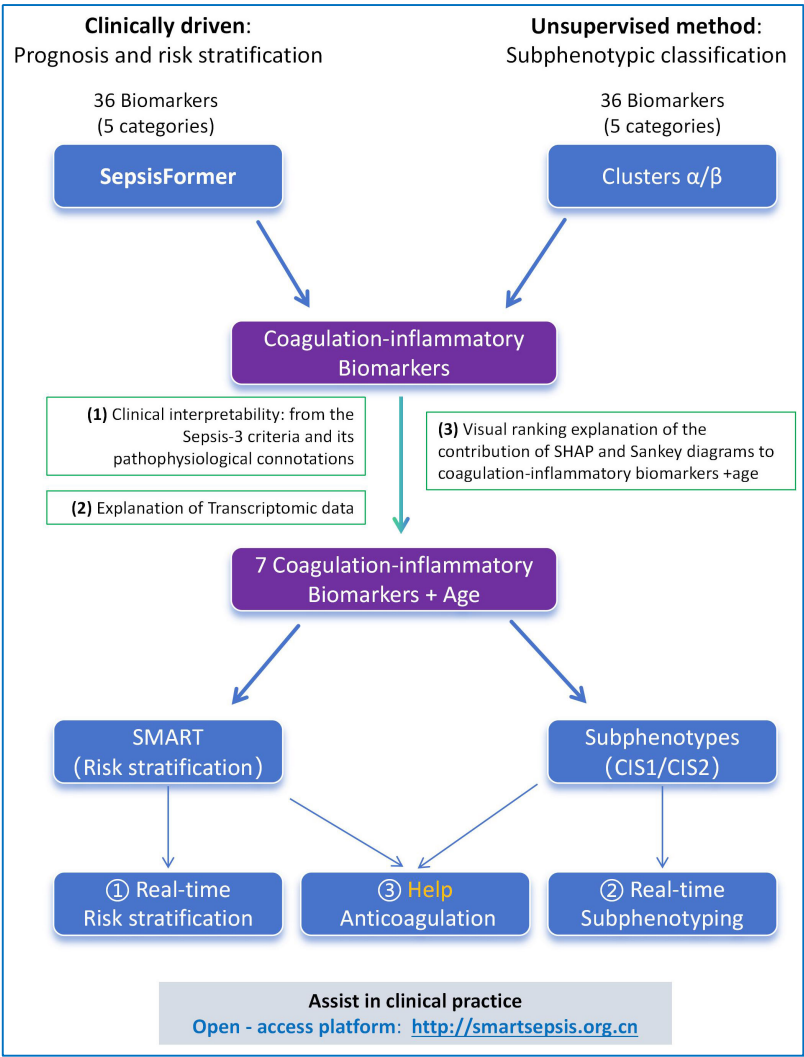
For further explanation and revision regarding clusters α and β as well as subphenotypes CIS1 and CIS2, please refer to our response to Q3.1.

Reference

1. Antcliff, DB, Burrell A, Boyle AJ et al (2025). Sepsis subphenotypes, theragnostics and personalized sepsis care. Intensive Care Medicine, 1-13.
2. Gordon AC, Alipanah-Lechner N, Bos LD et al (2024) From ICU syndromes to ICU subphenotypes: consensus report and recommendations for developing precision medicine in ICU. American Journal of Respiratory and Critical Care Medicine 2:155–166.

Q5.4 Throughout the paper, it is stated that eight markers were identified (7 coagulation-inflammatory measures + age) and used for prognosis modeling, risk stratification, and further identifying subphenotypes, but the rationale for selecting this subset of predictors is unclear. The use of routine coagulation-inflammatory are more than seven measures.

Response: The rationale for selecting the seven coagulation-inflammatory biomarkers and age is detailed in the new Supplementary Figure: 'Research Framework for Cluster and Subphenotype Analyses.' We have rewritten the entire Explainability Analysis of Coagulation–Inflammatory Dysfunction section (Pages 5-7). Further details are provided in our responses to Q3.1, Q5.2, and Q5.3.



(New) Supplementary Figure 1. Integrated framework: An overview of models, methods, biomarkers, and clinical applications in our study.

Q5.5 *For the evaluation of the risk stratification model, it could be beneficial to clarify what ground truth is used for the evaluation of this task.*

Response: Thank you very much for your constructive suggestions. The comment you mentioned is precisely the problem we are researching and aiming to solve. Our goal is to integrate multi-omics information through advanced AI algorithm tools and provide a tool or platform with practical clinical significance that can be operated in real-time by clinicians, thereby improving the prognosis of patients with sepsis.

We have made very detailed revisions in the interpretability section of the manuscript. Please refer to pages 5 to 9 of the revised manuscript. The efficacy of SMART risk stratification was verified to be very close and robust in MIMIC-III, MIMIC-IV, eICU, and local ICU. (Fig. 3e)

Moreover, the efficacy of SMART risk stratification is very close and robust in the verification results of MIMIC-III, MIMIC-IV, eICU, and local ICU. Meanwhile, its prognostic predictive efficacy is due to the currently widely used SOFA, qSOFA, A, LIP, APACHE II, and SIRS. The AUCs of them are 0.7360 (SMART), 0.6833, 0.6441, 0.6431, 0.6222 and 0.5428, respectively. The SMART scoring system achieved the highest predictive value.

Lastly, combining the comments of other reviewers, we promptly initiated a prospective external real-time validation study of the cohort, with the aim of assisting clinicians in presenting a practical risk stratification and subphenotypic classification system. We have supplemented it in both the methodology section and the results section.

Due to our open-access web-based sepsis subphenotype and SMART platform (<http://smartsepsis.org.cn>), which is better suited for real-time applications. Here, we have added a prospective, real-time, external cohort validation study. We monitored 40 patients in real-time (the first day of being enrolled in this cohort), and the results were consistent with the model's predictions, as follows:

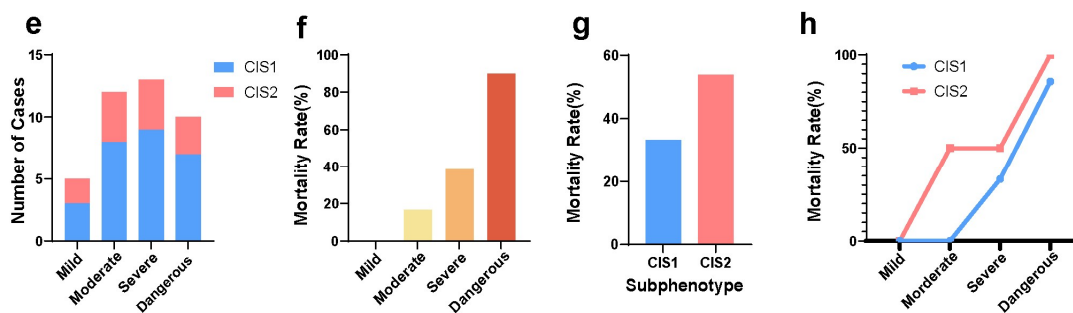
Revised in the manuscript:

“Furthermore, for the real-time and external validation of the risk stratification and subphenotypic classification, with the approval of the Ethics Committee of

Chongqing University Affiliated Central Hospital (Chongqing Emergency Medical Center) (ID: 2025-55), a prospective observational study on sepsis risk stratification and subphenotypic classification via our risk stratification and subphenotype platform was conducted. The endpoint event was to observe the actual 28-day mortality rate and obtain the SMART score and subphenotype at the time of enrollment. After obtaining written informed consent, a total of 40 patients with sepsis were enrolled from March 24, 2025, to April 28, 2025.” (Page 13-14)

“Last but not least, clinicians could conduct real-time risk stratification and subphenotypic classification of sepsis based on the SMART scorecard or our open-access sepsis subphenotype and SMART platform (<http://smartsepsis.org.cn>). Here, we conducted risk stratification, phenotypic classification and prognosis prediction for 40 sepsis patients locally admitted from March 21, 2025, to April 27, 2025 (external observational verification only) (Fig. 4e-h, and Supplementary case materials 1 and 2). The results showed that the proportions of patients in the four risk levels of mild, moderate, severe, and dangerous were 12.5%, 30%, 32.5% and 25% respectively (Fig. 4e), and their 28-day actual mortality rates were 0%, 16.7%, 38.5% and 90% respectively (Fig. 4f). The overall mortality rate of the CIS1 subphenotype was 33.3%, significantly lower than 53.8% of CIS2, and the trend was the same at different risk levels (Fig. 4g and 4h). The mortality rate at the dangerous level (this external observational cohort) was a little higher than the predicted rate of the model in this study (approximately 50%), as well as simultaneously increased the overall mortality rates of CIS1 and CIS2, which might be related to the small sample size, but the overall trend was basically consistent. Therefore, clinicians can utilize the SMART scorecard or our open-access platform to conduct real-time risk classification and subphenotyping of patients, enabling objective and accurate assessment of sepsis patients to intervene as early as possible and improve their prognosis.” (Page 9)

Therefore, multiple data show that SMART risk stratification is robust, clinically significant, and has real-time clinical operability, which is not possessed by the vast majority of current risk stratification and typing tools.



Additional subgraphs in Revised Figure 4.

Q6: Is there enough detail provided in the methods for the work to be reproduced?

Considering the details and codes provided in the supplementary materials, probably yes. However, I found additional experiments, like fine-tuning or pertaining the prognosis models, which was never mentioned in the main manuscript.

Response: We are very grateful for the reviewer's feedback, which acutely identified a significant omission in the main body of our manuscript. Although Figure 1 clearly illustrates the complete framework of SepsisFormer model, depicting both its pre-training (pink lines) and fine-tuning (green lines) processes, the corresponding textual description was indeed missing. We have now rectified this issue by providing a comprehensive elaboration of the model's architecture within the main text.

1. We have added descriptions of both the pre-training and fine-tuning procedures.

Revised in the manuscript:

“(e) Pretraining Procedure. Cohort 1 (MIMIC-III and eICU-CRD) was partitioned 7:3 for training and internal testing. From this training data, the pre-trained SepsisFormer model was initialized, establishing its core architecture, weights, and hyperparameters. SMOTE was applied for data augmentation to address class imbalance and enhance model robustness during this phase. (f) Fine-tuning Procedure. Subsequently, for domain adaptation and enhanced generalizability, SepsisFormer underwent fine-tuning. Cohort 1 served as the source domain, with fine-tuning specifically targeting external validation cohorts: Cohort 2 (MIMIC-IV) and Cohort 3 (Local ICU)

as the target domains. A Domain-adaptive Generator was employed for data augmentation during this transfer learning stage, optimizing performance in the new clinical center. The details of SepsisFormer can be found in [Supplementary Method 1.](#)”
(Page 14)

2. We've rewritten the Domain-adaptive Generator in **Supplementary Method 1** and provided a more explicit rationale for this module. Additionally, visual analyses have been incorporated to illustrate **class imbalance** and **multi-center data heterogeneity**. Further details are provided in our responses to [Q2.2](#).

3. We have revised the title in Supplementary Tables 6 and 7 to more clearly indicate pre-training and fine-tuning.

Revised in the Supplementary Info:

“**Supplementary Table 6.** Performance comparison of the pre-trained SepsisFormer and baseline models for prognostic prediction on the internal test set (Cohort 1: 7,789 septic patients from MIMIC-III and eICU-CRD; 70% training, 30% test).”

“**Supplementary Table 7.** Performance comparison of the fine-tuned SepsisFormer and baseline models for prognostic prediction on the two external validation cohorts (Cohort 2: 4,191 septic patients from MIMIC-IV; and Cohort 3: 428 septic patients from a local ICU), using 10-fold cross-validation.”.

Reviewer #3 (Remarks on code availability):

Code was not available in the submission. Some pseudo codes were presented in the supplementary materials.

Response: We sincerely apologize for the inconvenience and difficulty you encountered while attempting to review the analysis in detail despite having downloaded the datasets. We have now included the relevant codes, along with detailed instructions and videos on how to run them. To promote transparency,

reproducibility, and clinical applicability, we have made the following resources publicly available:

1. Interactive Web-based Platform (<http://smartsepsis.org.cn>): We developed an open-access, user-friendly online platform for sepsis subphenotyping and risk stratification. This tool allows end-users, such as clinicians and researchers, to perform real-time subphenotype identification and risk level prediction by uploading their own data without requiring additional coding or technical expertise.

2. Source Code and Datasets (<https://github.com/zhuli19031218/SepsisFormer>): All raw data, preprocessed datasets, and fully documented source code are available in our GitHub repository. These resources enable independent verification, replication, and extension of our study.

3. Code Execution Tutorial (<https://doi.org/10.5281/zenodo.15634369>): A detailed step-by-step video tutorial is provided to guide users through the complete process of reproducing the main results presented in this study, ensuring ease of use even for non-technical users.

These provided resources allow for replication and application of our method. We believe these improve the paper's quality and transparency.

We're especially grateful for your insightful comments regarding the resolution of methodological concerns, the clarification of transcriptomic data interpretation, and the distinction between subtypes and subphenotypes. These revisions have significantly strengthened the scientific rigor and overall presentation of our study.

We believe these comprehensive revisions will meet with your approval.

Hope everything goes great for you!

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4 **Explainable AI-driven heterogeneity using coagulation–inflammatory markers**
5 **improves prognosis prediction, risk stratification, and anticoagulant treatment**
6 **effects for sepsis**

7
8 Corresponding Author: Niu Bailin

9
10 **This file contains all reviewer reports followed by all author rebuttals in order.**
11

12 Version 2:

13 Reviewer comments:

14 **Reviewer #1 (Remarks to the Author):**
15

16 In my previous review I have raised the following points:

- 17 1. The clarity and framing of the main message/ contribution of the paper
18 2. Issues relating to the reporting of the methodology and sample
19 3. Greater transparency towards the methods employed (e.g. datasets and codes)
20

21 Glad to share that these three issues were addressed in this latest revision.
22

23 The results of these paper are noteworthy as indicated in my earlier review and the ability to translate
24 the algorithm developed into a clinical profiling score has field implications.
25

26 Good effort of showcasing the execution of the codes in the videos.
27
28
29

30 **Reviewer #2 (Remarks to the Author):**
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32 The authors have adequately addressed the concerns, and the manuscript is now suitable for
33 publication.
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35
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37 **Reviewer #3 (Remarks to the Author):**
38

39 The main concerns were addressed in the revision, including the performance comparisons to
40 existing models, including both of the main model and the domain-adaptation strategy.
41 Implementation details and definitions were also provided for the latter method, which helped to
42 understand how it differs from existing techniques. However, in the description of the domain

adaptation in the supplementary method, the justification that analyzed class imbalance in mortality outcomes is not theoretical but rather empirical I think, since you based the analysis on your data samples. Further methodological details were provided for the transcriptomic analysis and study of HTEs effect. Ablation studies are also presented and clarifications around techniques used and claims made throughout the manuscript.

Besides, the authors extended the interpretation of the explainability analysis and how this information is useful to understand the heterogeneity of the disease and relevant biomarkers for sepsis prognosis. They also commented on the implications of having a model that predicts risk-stratified levels and the corresponding analyses.

In the results, there is connection between findings and medical relevance (e.g., found importance of coagulation-inflammatory indicators vs functional indicators of specific organs and how the former better reflects the pathophysiology of sepsis), and an application use case of the tool available online was demonstrated.

The paper still needs some organization edits, especially in the presentation of the results:

- Some analyses are listed but then presented later. E.g., 1) the correlation between predictors mentioned in line 145 and then presented in line 176, 2) finding a subset of laboratory measurements within one category (part (a) in line 155) but the clustering results of part (b) are presented first, followed by the correlation analysis.

- Improve the introduction of the transcriptomic-level explanations and transcriptome analysis: It is clarified that this was to show the important roles of coagulation-inflammation related indicators, but there are some aspects that can be clarified. where was the gene information gathered from? Briefly mention the methods involved in the paragraph in line 207.

- It is still unclear to me how the SEVEN predictors were selected. In the model-level explainability only 3 are mentioned (APTT, WBC, and age) and the importance of coagulation-inflammation predictors, but there at least 10 from what I can see in Figure 2c and 8 in Figure 2g. The overall selection process was illustrated in a figure in the response letter, but I may be missing some details to fully understand the criteria for picking those 7 variables. I also noticed that the functions of these 7 coagulation-inflammation markers is discussed in detail in the paragraph starting in line 362 and they are part of the online platform.

- Domain adaptation comparisons are introduced out of the blue in page 7. Explained the results using comparisons to other methods.

- Revise claim about an XAI tool: Sepsis former on its own is not an XAI tool. The analysis used post-hoc XAI tools to understand contributors to predictions and relationships between inputs.

- The use of unsupervised clustering is better explain to find the subphenotypes, but I am still confused about why there are two analyses for this, i.e., first to identify alpha and beta groups using 36 predictors and then CIS1 and CIS2 using only the seven coagulation-inflammation predictors?

As another reviewer mentioned, not all analyses have to be part of the manuscript because it becomes hard to appreciate what are the main take aways.

- Regarding the radar plot in the assessment of HTEs: It seems like all predictors are included in this analysis. I thought that the SMART model only used 8 predictors. Please clarify why variables from the other categories were included and the purpose of this analysis. As another reviewer mentioned, not all analyses have to be part of the manuscript because it becomes hard to appreciate what are the main take aways.

Discussion:

Overall, the claims in the discussion are supported by the results: how explainability analyses were done to study sepsis heterogeneity, how the different patient populations stratified by subphenotype and risk levels had different clinical characteristics and mortality rates, and the insights from the combined analysis of HTEs' effect.

The clarifying sentence at the end of the third paragraph of the discussion can be included in the paragraph opening to mention the transition to a new analysis.

Included other works that have done subtyping, how the findings and techniques differ, and acknowledge that comparisons are hard between identified subphenotypes.

I suggest beginning the paragraph in line 400 with this sentence before you contrast the results with other works that have done subtyping: "Another strength is the ability to distinguish subphenotypes and risk levels of septic patients using only coagulation-inflammatory markers and patient age" . And then in the next paragraph discuss the effects of HTEs for the subgroups.

Minor comments:

Introduce some acronyms in the main paper for clarity: HTE in the introduction, MMID-SMOTE in the results, the machine learning abbreviations (even if they are well known).

Line 425 "and so on" may not be clear for all readers and sounds informal.

Reviewer #3 (Remarks on code availability):

Even though the code is provided, the documentation can be improved and a step by step description on what codes to run to generate what results. The README file does not have enough instructions and does not mention which dependencies are needed. The names of the scripts can also be more informative. For the transcriptomic analysis results, the folder only contains data files, so I wonder if the code to recreate these results is missing.

Response to Reviewers' Comments

We are incredibly grateful to the reviewers for their insightful and constructive comments, which have been very helpful in further improving the manuscript and enhancing its robustness, logic, and presentation of work highlights. The following is our detailed and thorough response to each comment one by one. We hope to address your concerns and obtain your approval.

Reviewer #1 (Remarks to the Author):

In my previous review I have raised the following points:

1. The clarity and framing of the main message/ contribution of the paper
2. Issues relating to the reporting of the methodology and sample
3. Greater transparency towards the methods employed (e.g. datasets and codes)

Glad to share that these three issues were addressed in this latest revision.

The results of these paper are noteworthy as indicated in my earlier review and the ability to translate the algorithm developed into a clinical profiling score has field implications.

Good effort of showcasing the execution of the codes in the videos.

Response: Thank you for your valuable time and insightful suggestions, which have helped us improve the quality of the article. We also appreciate your recognition of our efforts.

Reviewer #2 (Remarks to the Author):

The authors have adequately addressed the concerns, and the manuscript is now suitable for publication.

Response: We appreciate your approval of the revised manuscript, and we are especially grateful for your constructive suggestions which have enhanced our academic proficiency throughout this revision process.

Reviewer #3 (Remarks to the Author):

Q1: The main concerns were addressed in the revision, including the performance comparisons to existing models, including both of the main model and the domain-adaptation strategy. Implementation details and definitions were also provided for the latter method, which helped to understand how it differs from existing techniques. However, in the description of the domain adaptation in the supplementary method, the justification that analyzed class imbalance in mortality outcomes is not theoretical but rather empirical I think, since you based the analysis on your data samples. Further methodological details were provided for the transcriptomic analysis and study of HTEs effect. Ablation studies are also presented and clarifications around techniques used and claims made throughout the manuscript.

Response: Thanks for your feedback on the revised manuscript. We are pleased to learn that the main concerns have been addressed, and that the further methodological details provided for the transcriptomic analysis, the study of HTEs effect, as well as the ablation studies, have met your expectations.

The presence of class imbalance in mortality outcomes among sepsis patients is a widely held consensus, substantiated by empirical analyses of clinical data. In this study, we statistically analyzed the mortality rates within our utilized datasets, the results of which corroborate this imbalance. It is crucial to emphasize that while analyses constitutes an empirical investigation based on specific datasets, the methodology we propose is not necessarily limited in its applicability to these datasets alone. We appreciate you highlighting the need to clarify this. We revised this section to explicitly characterize the analysis as empirical as following.

Revised in the manuscript:

1. “The presence of class imbalance in mortality outcomes among sepsis patients is a widely held consensus, substantiated by empirical analyses of clinical data^{4, 5, 8}. ”

(Line 107-109)

2. Supplementary Method 1. Rationale for Domain Adaptation in Prognostic Modeling.

“Class imbalance in sepsis mortality outcomes is a well-established empirical

finding. In this study, the statistical chart of eICU, Local ICU, MIMIC III, and MIMIC IV datasets exhibit significant class imbalance between positive samples (non-survivors) and negative samples (survivors), with mortality rates of 25.38%, 37.23%, 25.56%, and 37.62% respectively. The number of non-survivors was significantly lower than that of survivors.”

Moreover, to maintain manuscript conciseness, we omitted detailed enumeration of supporting studies from the main text. Actually, the research we present here reports mortality rates that are largely consistent with the findings of our study.

(1) A total of 14,607 sepsis patients were screened and the overall 30-day mortality rate was 21.0%.

Ref: Cao, Bingbing, et al. Non-linear relationship between baseline mean arterial pressure and 30-day mortality in patients with sepsis: a retrospective cohort study based on the MIMIC-III database. *Annals of Translational Medicine* 10.16: 872 (2022).

(2) In cohorts of patients with suspected sepsis infection (training: n=12,473; validation: n=8,256), the four sub-phenotypes exhibited distinct mortality rates: Group A (28%), Group B (13%), Group C (32%), and Group D (27%).

Ref 4 in the manuscript. Bhavani SV, et al. Development and validation of novel sepsis subphenotypes using trajectories of vital signs. *Intensive Care Med* 48, 1582-1592 (2022).

(3) In some countries, sepsis mortality is being reported at 20–30%.

Ref 5 in the manuscript. Hotchkiss RS, Moldawer LL, Opal SM, Reinhart K, Turnbull IR, Vincent JL. Sepsis and septic shock. *Nat Rev Dis Primers* 2, 16045 (2016).

(4) In a cohort of 2,499 sepsis patients, mechanical ventilation was common within 24 hours of ICU admission, and in-hospital mortality was 30.1%.

Ref 8 in the manuscript. van Amstel RBE, et al. Uncovering heterogeneity in sepsis: a comparative analysis of subphenotypes. *Intensive Care Med* 49, 1360-1369 (2023).

Q2: Besides, the authors extended the interpretation of the explainability analysis and how this information is useful to understand the heterogeneity of the disease and

relevant biomarkers for sepsis prognosis. They also commented on the implications of having a model that predicts risk-stratified levels and the corresponding analyses.

Response: Thanks for your thorough review and recognizing the enhancement we have made to the explainability analysis and the discussions around disease heterogeneity, sepsis prognostic biomarkers, as well as the implications of our risk-stratified prediction model. We are pleased that these extensions have resonated with your review. Your acknowledgment reinforces that this effort has been effective.

Q3: In the results, there is connection between findings and medical relevance (e.g., found importance of coagulation-inflammatory indicators vs functional indicators of specific organs and how the former better reflects the pathophysiology of sepsis), and an application use case of the tool available online was demonstrated.

Response: Thank you for recognizing both the connections between our findings and their medical relevance, and the practical application of our online tool. We also appreciate your acknowledgment of how this work bridges computational insights with clinical applicability.

Q4: The paper still needs some organization edits, especially in the presentation of the results:

- Some analyses are listed but then presented later. E.g., 1) the correlation between predictors mentioned in line 145 and then presented in line 176, 2) finding a subset of laboratory measurements within one category (part (a) in line 155) but the clustering results of part (b) are presented first, followed by the correlation analysis.

Response: We are very grateful to you for the insightful comments on the organization of our manuscript. We fully agree that a clear presentation is crucial for accurately conveying our research findings. Following your valuable suggestions, we revised the Results section to improve its narrative logic and readability, enabling readers to more easily follow our analytical framework and findings.

As you correctly pointed out, the correlation analysis of predictors was mentioned twice. This was not a repetition, but rather addressed two distinct stages of our analysis.

The first mention, with results presented in Fig. 2c, pertains to the correlation analysis of the entire set of all 36 predictors. The second mention, with results in Fig. 2f, refers to the correlation analysis conducted on the 8 core predictors. As Fig. 2f already displays the correlation analysis results for the eight predictors, NO modifications were made.

We've reordered 'the clustering results' to be Part (a) and 'finding a subset of laboratory measurements within one category' to be Part (b) for better alignment with our figures and tables.

The results of the Sankey diagram have been relocated.

Revised in the manuscript:

1. "interrelationships of all 36 predictors were explained via a correlation network diagram, with Pearson correlation coefficients and corresponding *p* values (Fig. 2c)."

(Line 150)

3. to achieve two clinical goals: (a) Select an efficient unsupervised clustering method.....; (b) Identify a sepsis mechanism relevant and clinically feasible subset of laboratory measurements.....(Line 161-164)

4. The results of the Sankey diagram "The results consistently indicated that coagulation-inflammatory indicators such as APTT, WBC, and the patient's age were important predictors from various perspectives." have been relocated to Line 207-208.

5. "A clinically relevant scorecard (Table 1) was developed for SMART based on medical knowledge, allowing clinicians to easily and directly calculate a patient's risk score." (Line 299).

Q5: Improve the introduction of the transcriptomic-level explanations and transcriptome analysis: It is clarified that this was to show the important roles of coagulation-inflammation related indicators, but there are some aspects that can be clarified. where was the gene information gathered from? Briefly mention the methods involved in the paragraph in line 207.

Response: Thank you for your constructive feedback. We have revised the paragraph

in line 207 to clarify the gene information sources paragraph. These changes enhance transparency and align with your suggestions. Thank you for guiding us to improve the manuscript.

Revised in the manuscript:

“We retrieved the sepsis expression profile from the Gene Expression Omnibus (GEO, <https://www.ncbi.nlm.nih.gov/geo/>), selecting dataset GSE65682 for analysis. DIC-related genes were sourced from the GeneCards (<https://www.genecards.org/>) and DisGeNET (<https://www.disgenet.org/>) databases. Genes overlapping between these disseminated intravascular coagulation (DIC)-related gene sets and differentially expressed genes (DEGs) were defined as DIC-related DEGs. To clarify their core biological functions and underlying mechanisms, we performed KEGG and GO enrichment analyses on these genes using the clusterProfiler package (version 3.14.3) in R.” (Line 221-227)

Q6: It is still unclear to me how the SEVEN predictors were selected. In the model-level explainability only 3 are mentioned (APTT, WBC, and age) and the importance of coagulation-inflammation predictors, but there at least 10 from what I can see in Figure 2c and 8 in Figure 2g. The overall selection process was illustrated in a figure in the response letter, but I may be missing some details to fully understand the criteria for picking those 7 variables. I also noticed that the functions of these 7 coagulation-inflammation markers is discussed in detail in the paragraph starting in line 362 and they are part of the online platform.

Response: Thank you for the helpful suggestion. We recognize that the justification for our predictor selection could be stated more clearly. Accordingly, we have revised the third paragraph of the 'Explainability analysis of coagulation-inflammatory dysfunction' section. It now better explains why we selected 7 of the 10 predictors—excluding Prothrombin Time (PT), basophils, and eosinophils—based on medical knowledge and a correlation matrix analysis. To further improve clarity, we have also elaborated on these points and added a summary sentence to conclude the paragraph. We believe that we have adequately addressed this question.

Figure 2g's SHAP and Sankey diagram ranks APTT, WBC, and Age as the top three predictors from various perspectives, and the other 8 sub-variables are also very important. We report this ranking objectively—without implying selective inclusion.

Revised in the manuscript:

“Due to PT tests extrinsic, APTT intrinsic coagulation pathways and INR is a standardized value calculated from PT results, we retained INR and APTT while excluding PT. Meanwhile, basophils and eosinophils exhibit weak or negative correlation coefficients with other predictors, rendering them impractical to reflect the risk status of patients. These two types of WBC, influenced by external factors like allergies, display inconsistent behavior and limited prognostic value in sepsis, with their mechanisms remain unclear^{23, 24}. To focus our analysis on the most robust and clinically relevant markers, we excluded PT, basophils, and eosinophils. The final coagulation-inflammatory markers analyzed in this study comprises 8 variables: APTT, INR, lymphocytes, monocytes, neutrophils, WBC, PLT and age.” (Line 188-196)

Q7: Domain adaptation comparisons are introduced out of the blue in page 7. Explained the results using comparisons to other methods.

Response: Thank you for your valuable feedback. In response to your comment, we have revised the manuscript to provide a clearer context for the introduction of domain adaptation techniques. Specifically, we have added an explanation regarding the challenge of generalizing SepsisFormer across different cohorts due to heterogeneity in patient characteristics. Additionally, we have clarified the comparison of MMID-SMOTE with other techniques, highlighting its superior performance and the rationale behind its selection for improving model generalization. We believe these changes enhance the clarity and flow of the manuscript.

Revised in the manuscript:

“Despite strong performance within each cohort, SepsisFormer’s ability to generalize across different clinical settings remains a key challenge, as patient characteristics can differ significantly between datasets. To address this issue, we

incorporated methods to enhance the model’s adaptability, allowing it to adjust to these differences and maintain high accuracy in new settings.

As shown in Fig. 2d and Supplementary Table 7, Maximum and Minimum Interval Difference-based Synthetic Minority Oversampling Technique (MMID-SMOTE) significantly outperformed other domain adaptation techniques and the ablation experiment (no-adaptation baseline) by substantially improving the model's cross-cohort generalization capability. MMID-SMOTE performed better than other approaches across multiple evaluation metrics, including AUC, accuracy, sensitivity, specificity, and F1-score. This is demonstrated through an ablation study, which shows that MMID-SMOTE outperformed the no-adaptation baseline and other state-of-the-art methods. The Mean-teacher and Whitening methods showed poorer performance, with Mean-teacher struggling with low-quality pseudo-labels and Whitening’s adjustments failing to improve generalization. In contrast, Moment Matching showed second-place performance by focusing on second-order statistics, but still did not surpass MMID-SMOTE in any metric. This highlights the importance of selecting the right method for improving model performance across diverse clinical settings. Our findings show that MMID-SMOTE improves SepsisFormer’s ability to predict outcomes and adapt to new clinical environments. As a post-hoc XAI model, SepsisFormer offers reliable and interpretable support for clinical decision-making, utilizing a minimal set of routine biomarkers.” (Line 254-271)

Q8: Revise claim about an XAI tool: Sepsis former on its own is not an XAI tool. The analysis used post-hoc XAI tools to understand contributors to predictions and relationships between inputs.

Response: Thank you for your insightful and constructive comment. Your keen observation has prompted us to reconsider the precision of the terminology used in our manuscript, and we are grateful for this opportunity to improve its quality. We fully agree with you, SepsisFormer is not, in itself, an XAI tool. The explainability presented in our paper is achieved by applying post-hoc analytical techniques.

We would like to clarify that our work establishes an **analytical framework** that

couples the high-performance predictive with post-hoc XAI methodologies. In the XAI field, it is common practice to refer to this "model + explanation" composite as a unified whole. This convention aims for concise and clear communication, emphasizing that the model has been rendered interpretable.

Our proposed terminology is well-grounded in the current literature. A key distinction in the XAI field is drawn between ante-hoc and post-hoc paths to interpretability (Retzlaff et al., 2024). Ante-hoc refers to using inherently transparent models (e.g., linear regression, decision trees mentioned in the INTRODUCTION), whereas post-hoc methods are specifically designed to explain already-trained black-box models. Our approach, which falls into the latter category, aims to confer explainability upon an opaque model. Therefore, designating the resulting model-explanation entity as a "**post-hoc XAI model**" is both appropriate and logical.

This practice is supported by precedent in top-tier journals. For instance, Lauritsen et al. (2020) in Nature Communications refer to their system, holistically as an "explainable AI early warning score (xAI-EWS) system for sepsis." This precedent provides strong support for our rationale in referring to the combination of SepsisFormer and post-hoc analysis as "**post-hoc XAI model SepsisFormer**." Furthermore, a significant body of literature is dedicated to systematically surveying (Bhati et al., 2025) and evaluating (Bello et al., 2025; Belaid et al., 2022) various post-hoc XAI methods. The shared premise of these studies is that applying such methods allows an otherwise opaque model to be effectively explained, thereby becoming an examinable "explainable system."

In response to your valuable feedback and to enhance the manuscript's precision, we will implement the following revisions:

1. Upon its first introduction, we will explicitly state that SepsisFormer is a predictive model and that we employ post-hoc XAI techniques to interpret it.

Revised in the manuscript:

“we developed a Transformer-based prognostic model (SepsisFormer) interpreted via post-hoc XAI techniques,.....” (Line 111-112)

2. Throughout the remainder of the manuscript, we will consistently use the more

precise term "**post-hoc XAI model SepsisFormer**".

We are confident that these revisions will make our manuscript clearer, more accurate, and fully aligned with the current academic standards in the XAI field. Thank you again for your professional guidance.

Reference

[1] Bhati, Deepshikha, et al. "A Survey of Post-Hoc XAI Methods from a Visualization Perspective: Challenges and Opportunities." IEEE Access (2025).

[2] Retzlaff, Carl O., et al. "Post-hoc vs ante-hoc explanations: xAI design guidelines for data scientists." Cognitive Systems Research 86 (2024): 101243.

[3] Bello, Marilyn, et al. "The level of strength of an explanation: A quantitative evaluation technique for post-hoc XAI methods." Pattern Recognition 161 (2025): 111221.

[4] Belaid, Mohamed Karim, et al. "Do we need another explainable AI method? Toward unifying post-hoc XAI evaluation methods into an interactive and multi-dimensional benchmark." arXiv preprint arXiv:2207.14160 (2022).

[5] Lauritsen, Simon Meyer, et al. "Explainable artificial intelligence model to predict acute critical illness from electronic health records." Nature communications 11.1 (2020): 3852.

Q9: The use of unsupervised clustering is better explain to find the subphenotypes, but I am still confused about why there are two analyses for this, i.e., first to identify alpha and beta groups using 36 predictors and then CIS1 and CIS2 using only the seven coagulation-inflammation predictors? As another reviewer mentioned, not all analyses have to be part of the manuscript because it becomes hard to appreciate what are the main take aways.

Response: We sincerely thank you for this insightful comment and for highlighting a crucial aspect of our analytical workflow. We understand your concern regarding the clarity and potential redundancy of the two-part analysis. In our first revision, we added Supplementary Figures 1 and 2 to explain our rationale. We'd like to further elaborate

on the distinct, sequential logic behind our approach, as we believe it's fundamental to the rigor and interpretability of our findings.

Our two-step clustering process was not designed as two separate attempts to identify subphenotypes, but rather as a **principled, data-driven framework moving from broad, unbiased exploration to a focused, clinically explainable model.**

Step 1: Unbiased, Hypothesis-Generating Exploration (Alpha and Beta Clusters)

The primary purpose of the first analysis, using all 36 predictors, was to serve as a crucial exploratory and hypothesis-generating step. Our objectives here were threefold:

1. To avoid a priori bias. By starting with a comprehensive set of 35 predictors derived from the Sepsis-3 criteria (SOFA based), we allowed the data itself to reveal the most dominant biological signals driving patient heterogeneity, rather than pre-supposing the importance of a specific pathway (**Line 138-142**). A direct analysis using only seven coagulation-inflammatory markers could have been criticized for being subjective. This initial step provides an objective, data-driven justification for our subsequent focus on coagulation and inflammation.

2. To establish analytical feasibility. This analysis confirmed that meaningful, distinct patient clusters (alpha and beta) did exist within our dataset and that the Gaussian Mixture Model (GMM) was a suitable method for the identification. See “This study employs GMM for subphenotype identification, as the patient subgroups derived from GMM show the most significant differences in mortality rates (shown in Section Explainability Analysis), indicating greater clinical relevance.” (**Line 275-277**)

3. To identify the most informative features. The key output of this step was the discovery that coagulation-inflammatory markers were the most significant drivers distinguishing the alpha and beta groups.

Step 2: Definitive, Interpretable Subphenotype Modeling (CIS1 and CIS2)

The findings from Step 1 directly informed our primary, definitive analysis. We proceeded with a focused model using only the seven key coagulation-inflammatory predictors for the following critical reasons:

1. To enhance interpretability and clinical utility. A subphenotype defined by a parsimonious set of seven mechanistically-linked markers is far more interpretable and

clinically actionable than a "black-box" phenotype derived from a complex combination of 36 variables. This allows clinicians to understand the underlying pathophysiology and potentially use these markers for patient stratification.

2. To increase model robustness and reduce noise. Using all 36 predictors risks falling into the "curse of dimensionality," where noise from less relevant variables can obscure the true signal and lead to less stable clusters. By concentrating on the most informative markers identified in Step 1, we use a more robust and dimensionally reduced model that is more likely to be generalizable (Fig3a).

3. To define biologically coherent subphenotypes. The final subphenotypes, CIS1 and CIS2, are not just statistically derived clusters; they are defined by a core biological process, which is a well-recognized and critical component of sepsis pathophysiology. This biological coherence is the core strength of our findings (Fig3).

In summary, the first analysis provides the necessary justification for the second, moving from a wide, unbiased search to a deep, precise conclusion. We contend that removing the first step would weaken the scientific foundation of our work, as it would obscure the data-driven rationale for selecting final set of predictors.

To address your concern about clarity, we have substantially revised the **Methods** and **Results** sections of our manuscript. We have explicitly framed the two analyses as "exploratory clusters" and "definitive subphenotypes" and have detailed the rationale presented above. We believe these revisions now make the logic of our analytical strategy transparent, allowing the reader to appreciate the main takeaways of our study.

Q10: Regarding the radar plot in the assessment of HTEs: It seems like all predictors are included in this analysis. I thought that the SMART model only used 8 predictors. Please clarify why variables from the other categories were included and the purpose of this analysis. As another reviewer mentioned, not all analyses have to be part of the manuscript because it becomes hard to appreciate what are the main takeaways.

Response: Thank you for this important comment. You are absolutely correct that the SMART model includes only 8 selected predictors. The radar plot (Figure 4b), however,

was not derived from the model but was constructed as a descriptive visualization to show how different physiological systems - such as hepatic, renal, and inflammatory— vary across risk levels.

In clinical research, such radar plots are particularly valuable as they allow for integrated visualization of multi-system involvement, which is often essential when evaluating complex conditions involving systemic dysfunction, such as sepsis, multi-organ failure, or advanced chronic disease. In our context, the plot helps reveal consistent patterns of systemic deterioration across risk levels, providing additional clinical insight beyond the predictors used in the model.

The intent was not to imply that these additional variables were involved in risk stratification or predictive modeling, but to provide clinicians with a more intuitive understanding of the biological profiles associated with varying levels of treatment benefit. In this context, the radar plot serves a purely exploratory and interpretive role. To maintain focus in the main manuscript, we have clarified the purpose of the radar plot and retained it in the main figures due to its interpretive value.

Furthermore, we presented similar figures from other studies to demonstrate the utility of radar plots in the context of heterogeneity analysis within the biomedical field. For instance, refer to Figure 2c in Xue, Ruidong, et al. "Liver tumour immune microenvironment subtypes and neutrophil heterogeneity." *Nature* 612.7938 (2022): 141-147.

[REDACTED]

Revised in the manuscript:

“As shown in Fig. 4b, the radar plot demonstrates that risk stratification based solely on coagulation-inflammatory markers captures a broader pattern of multi-organ dysfunction. Except for the erythrocyte category, laboratory values across all other

systems increase consistently with risk level, peaking in the dangerous group. The consistent upward trend supports the use of coagulation-inflammatory markers as indicators of systemic severity and their utility in clinical risk stratification.”

(Line 335-340)

Q11: Discussion:

Overall, the claims in the discussion are supported by the results: how explainability analyses were done to study sepsis heterogeneity, how the different patient populations stratified by subphenotype and risk levels had different clinical characteristics and mortality rates, and the insights from the combined analysis of HTEs’ effect.

Response: Thank you. We appreciate you recognizing that our discussion's claims are supported by the results.

Q12: The clarifying sentence at the end of the third paragraph of the discussion can be included in the paragraph opening to mention the transition to a new analysis.

Response: We've revised the discussion as suggested. The clarifying sentence “Although transcriptomic data are not direct items for SMART scores and CIS typing, they further enhance the explanation of the importance of coagulation-inflammatory markers in diagnosis and prognosis assessment.” has been moved to the beginning of the third paragraph (Line 404-406) to clearly signal the transition to the new analysis.

Q13: Included other works that have done subtyping, how the findings and techniques differ, and acknowledge that comparisons are hard between identified subphenotypes.

Response: Thanks for your feedback. We've fully addressed your comments by adding the requested discussion on other subtyping works, detailing how our findings and techniques differ, and acknowledging the challenges in comparing identified subphenotypes. We're pleased to confirm that these revisions have met your satisfaction.

Q14: I suggest beginning the paragraph in line 400 with this sentence before you contrast the results with other works that have done subtyping: “Another strength is

the ability to distinguish subphenotypes and risk levels of septic patients using only coagulation-inflammatory markers and patient age” . And then in the next paragraph discuss the effects of HTEs for the subgroups.

Response: We really appreciate your insightful suggestion on strengthening the framing of our findings. We've now incorporated the proposed sentence. This addition, placed right before our comparative discussion with existing subtyping literature, significantly sharpens the focus on our contribution.

Dear Professor, please forgive us. It might be due to the WORD version or other reasons that the line numbers you provided are slightly different from the actual line numbers in our WORD document. However, we have understood your comments and made the necessary revisions at the appropriate places.

Revised in the manuscript:

1. “Another strength is the ability to distinguish subphenotypes and risk levels of septic patients using only coagulation-inflammatory markers and patient age.” (Line 424-425)

2. “The anticoagulant drug heparin may offer potential benefits in sepsis management, with clinical outcomes varying across different risk level and subphenotypes.” (Line 448-449)

Q15: Minor comments:

Introduce some acronyms in the main paper for clarity: HTE in the introduction, MMID-SMOTE in the results, the machine learning abbreviations (even if they are well known).

Line 425 “and so on” may not be clear for all readers and sounds informal.

Response: We're grateful for your insightful feedback. Your comments significantly improved our manuscript's clarity and readability. We've carefully addressed all your suggestions, including the minor ones, and believe these revisions have truly strengthened the paper. We appreciate the time and effort you dedicated to our work.

Revised in the manuscript:

1. “HTE refers to the phenomenon where the same treatment can have different effects (beneficial, neutral, or even harmful) on different patients.” (Line 63-64)

2. “Models mainly include machine learning approaches (Decision Tree, Logistic Regression, Support Vector Machine, Generalized Linear Model, Naïve Bayes) and neural networks (Multilayer Perceptron, Long Short-term Memory, Convolutional Neural Network, Gated Recurrent Unit, and two attention-based explainable models: RETAIN and Dipole).” (Line 87-90)

3. “Maximum and Minimum Interval Difference-based Synthetic Minority Oversampling Technique (MMID-SMOTE) significantly”(Line 258-259)

4. “Twenty-nine variables, including cardiovascular..... pulmonary, and renal systems were used.” (Line 433)

Reviewer #3 (Remarks on code availability):

Even though the code is provided, the documentation can be improved and a step by step description on what codes to run to generate what results. The README file does not have enough instructions and does not mention which dependencies are needed. The names of the scripts can also be more informative. For the transcriptomic analysis results, the folder only contains data files, so I wonder if the code to recreate these results is missing.

Response: We sincerely thank you for your valuable and constructive feedback, which has significantly helped us improve the clarity and reproducibility of our work.

In response to your suggestions, we have comprehensively revised the README.md file to include a detailed step-by-step guide that walks users through the entire process of generating our results. To address the lack of dependency information, we have added a dedicated installation section to simplify the environment setup process. We also agree that the script names could be more informative, we've updated README file in <https://github.com/zhuli19031218/SepsisFormer/blob/main/README.md>. To facilitate quick access to the information, we have included the README file in the Appendix of this reply letter.

Furthermore, we apologize for the oversight regarding the transcriptomic analysis code; we have now added a new `transcriptomics_analysis` folder containing all

the necessary four R scripts, six datasets and all videos to reproduce those findings.
(https://github.com/zhuli19031218/SepsisFormer/tree/main/transcriptomics_analysis;
<https://doi.org/10.5281/zenodo.15634368> Version3.0).

The updated files on these websites include:

- (1) 4 R scripts for the analysis of diagnostic markers;
- (2) 6 datasets, which had already been provided in the previous revised version, including: GEO-GSE26440 (diagnostic markers),GEO-GSE54514 (prognostic markers),GEO-GSE65682 (diagnostic markers),GEO-GSE65682 (prognostic markers),GEO-GSE95233 (diagnostic markers),LocalICUNB-qPCR
- (3) 24 Code Execution Videos, including: 3h_1_Forest plot, 3h_2_Diagnostic ROC, 3h_3_Risk factor graph and survival curve, S6a_volcano plot, S6b_Venn Diagram, S6c_The PPI network of intersection genes, S6d_heat map, S6e_GO biological pathway, S6fgh, S6i_Lasso regression, S7a_Volcano plot,S7b_GSEA enrichment analyses, S7c_Venn diagram, S7d_The PPI network of intersection genes, S7e_Lasso regression, S7f_Forest plot, S7g_ROC curves, S7h_nomogram, S7i_ROC curves, S7j_Calibration curve, S7k_DCA curve, S8_The prognostic and diagnostic efficacy in external cohort, S9_1_Group comparison and ROC curve, S9_2_Group comparison and ROC curve.

We are confident that these extensive revisions have made our code repository far more user-friendly, transparent, and reproducible. We are also grateful for the opportunity to make these improvements.

Our team is deeply grateful for your constructive suggestions and well-researched recommendations. Your expert feedback is essential to strengthening our manuscript.

Wishing you all the best.



zhuli19031218 / SepsisFormer



<> Code

Issues

Pull requests

Actions

Projects

Wiki

Security



main

SepsisFormer / README.md



zhuli19031218 Update README.md

a0058fb · 18 minutes ago



471 lines (336 loc) · 16.3 KB

SepsisFormer and SMART

This repository contains the implementation of the paper "Explainable AI-driven heterogeneity using coagulation–inflammatory markers improves prognosis prediction, risk stratification, and anticoagulant treatment effects for sepsis". The code includes all major data, models, algorithms, and experimental settings described in the paper, enabling researchers to reproduce and verify our results. Enjoy~~~

Sepsis, a leading cause of hospital mortality, is characterized by substantial heterogeneity, hindering the development of effective and interpretable prognostic and stratification methods. To address this challenge, we developed an explainable prognostic model (SepsisFormer, a transformer-based deep neural network with an enhanced domain-adaptive generator) and an automated risk stratification tool (SMART, a scorecard consistent with medical knowledge). In a multi-center retrospective study of 12,408 sepsis patients, SepsisFormer achieved high predictive accuracy (AUC: 0.9301, sensitivity: 0.9346, and specificity: 0.8312). SMART (AUC: 0.7360) surpassed most established scoring systems. Based on SMART, four risk levels (mild, moderate, severe, dangerous) can be identified by using seven coagulation-inflammatory routine laboratory measurements and patient age, and the corresponding mortality is approximately 5%, 15%, 30%, and 50%, respectively. Meanwhile, two subphenotypes (CIS1 and CIS2) can be classified through unsupervised GMM, and CIS2 has a worse survival prognosis. Notably, patients with moderate/severe levels or CIS2 derive more significant benefits from anticoagulant treatment. In conclusion, explainable artificial intelligence (SepsisFormer) drives risk stratification and subphenotypic classification of sepsis, which help to guide anticoagulant treatment. Our work, therefore, offers a novel set of simple, real-time executable tools for sepsis heterogeneity, demonstrating considerable potential to significantly enhance sepsis clinical practice globally, particularly in resource-constrained healthcare settings.

NOTE: The Source Data for the 36 and 8 sepsis markers is provided in the "figure 2ab" folder, which contains the datasets used for Figures 2a, 2b, and 2e. The underlying code for these figures is identical; however, the input datasets differ: Figures 2a and 2b use 36 sepsis markers, while Figure 2e uses 8 sepsis markers.

Code Guide

This repository contains comprehensive code and data for generating all figures presented in the manuscript. This documentation provides detailed step-by-step instructions to facilitate the reproduction of all results reported in the study.



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Environment Requirements

Python Dependencies

```
pip install numpy pandas scikit-learn torch torchvision
pip install matplotlib seaborn plotly networkx
pip install shap shapely scipy statsmodels
pip install jupyter notebook
pip install igraph python-igraph
pip install lifelines
```



Core Dependency Versions

- Python $\geq$ 3.7
- PyTorch $\geq$ 1.8.0
- scikit-learn $\geq$ 0.24.0
- pandas $\geq$ 1.3.0
- numpy $\geq$ 1.21.0

Optional Software

- **GraphPad Prism:** For statistical chart generation in Figure 4a4c
- **Origin:** For clustering visualization charts in Figure 2f
- **R Language:** For statistical analysis (if using R scripts)

Project Structure

```
figure/
├── 2a2b/                # Figure 2a2b: Multi-model comparative
analysis
│   ├── model2/          # Deep learning model code
│   │   ├── Transformer.py # Transformer model implementation
│   │   ├── model_LSTM.py  # LSTM model implementation
│   │   ├── model_GRU.py   # GRU model implementation
│   │   ├── model_gpt.py   # GPT model implementation
│   │   ├── train.py       # Training script
│   │   ├── main_mimic.py  # Main execution script
│   │   └── results/       # Model results
│   ├── data/            # Data files
│   └── ml_logs/          # Machine learning logs
├── 2c/                  # Figure 2c: Network analysis visualization
│   ├── data/            # Node and edge data
│   ├── log/             # Execution logs
│   └── video/           # Generated videos
├── 2d/                  # Figure 2d: Domain adaptation analysis
│   ├── models/          # Machine learning model code
│   ├── logs/            # Experimental results
│   ├── roc_domain_adaptation.ipynb
│   └── video/
├── 2f/                  # Figure 2f: Clustering visualization
│   ├── Chord Diagram/   # Chord diagram
│   ├── Heatmap/         # Heatmap
│   └── Radar Chart/     # Radar chart
├── 2g/                  # Figure 2g: SHAP analysis
│   ├── data/            # Data files
│   ├── logs/            # Model logs
│   ├── SepsisFormer_shap_8.py
│   ├── shap_8 (2).ipynb
│   └── video/
├── 3c3h/                # Figure 3c3h: Hierarchical clustering
│   └── data/            # Data files
```



```

|   ├── log/                # Clustering logs
|   └── video/              # Clustering videos
├── 3d/                     # Figure 3d: 3D visualization
|   ├── data/              # Data files
|   ├── 3d.ipynb           # 3D visualization code
|   └── video/
├── 4a4c/                   # Figure 4a4c: Cox model
|   ├── 4a/                # New scoring analysis
|   ├── 4c/                # Subphenotype analysis
|   └── *.ipynb            # Cox analysis code
├── 4b/                     # Figure 4b: Radar chart
|   ├── data/              # Radar chart data
|   └── video/
└── 4d/                     # Figure 4d: Risk ratio
    ├── 风险比.ipynb
    └── video/

```



Execution Procedures

Figure 2a2b: Multi-Model Comparative Analysis

Objective: Compare the performance of different deep learning models (Transformer, LSTM, GRU, GPT) on sepsis prediction tasks

Primary Files:

- 2a2b/model12/main_mimic.py : Main execution script
- 2a2b/model12/train.py : Training and evaluation script
- 2a2b/model12/Transformer.py : Transformer model implementation
- 2a2b/model12/model_LSTM.py : LSTM model implementation
- 2a2b/model12/model_GRU.py : GRU model implementation
- 2a2b/model12/model_gpt.py : GPT model implementation
- 2a2b/data/SepsisFormer.py : SepsisFormer model implementation

Data Files:

- 2a2b/data/8/ : Dataset with 8 features
- 2a2b/data/36/ : Dataset with 36 features

Execution Steps:

1. Environment Preparation:

```
cd 2a2b/model12
```



2. Train Transformer Model:

```
python main_mimic.py --model_name Transformer --factors 8 --lr 0.004 --
```



3. Train LSTM Model:

```
python main_mimic.py --model_name Lstm --factors 8 --lr 0.004 --epoch 5
```



4. Train GRU Model:

```
python main_mimic.py --model_name GRU --factors 8 --lr 0.004 --epoch 50
```



5. Train GPT Model:

```
python main_mimic.py --model_name GPT --factors 8 --lr 0.004 --epoch 50
```



Model Parameter Specifications:

- `--model_name` : Select model type (Transformer/Lstm/GRU/GPT)
- `--factors` : Number of input features (8 or 36)
- `--lr` : Learning rate
- `--epoch` : Number of training epochs
- `--batch_size` : Batch size
- `--pretrain` : Whether to use pre-trained models
- `--loadmodel` : Path to pre-trained model

Model Architecture:

- **Transformer**: Multi-head attention mechanism, 8-layer depth
- **LSTM**: Long Short-Term Memory network, 2-4 layers
- **GRU**: Gated Recurrent Unit, 2-4 layers
- **GPT**: Generative Pre-trained Transformer

Output Results:

- Model performance metrics (AUC, Accuracy, F1-score, MCC)
- Training logs and TensorBoard visualization
- ROC curve plots

- Model comparison result tables

Expected Performance (based on 8 features):

- **GRU:** AUC ~0.645, Accuracy ~0.611
- **LSTM:** AUC ~0.641, Accuracy ~0.611
- **Transformer:** AUC ~0.630, Accuracy ~0.600
- **GPT:** Performance pending evaluation

Figure 2c: Network Analysis Visualization

Objective: Generate network relationship diagrams among sepsis patient features

Generation Method: Online generation using HiPlot website

Data Files:

- `2c/data/节点数据.csv` : Network node information
- `2c/data/连线数据-*.csv` : Edge data at different thresholds

Execution Steps:

1. Access HiPlot website (<https://hiplot.com.cn/>)
2. Upload node data and edge data files
3. Select network graph visualization type
4. Configure parameters:
 - Node label column: `media`
 - Node color column: `weight`
 - Node size column: `media.type`
 - Edge width column: `weight`
 - Layout style: Circular layout
5. Generate network analysis diagram

Parameter Specifications:

- Node label column: `media`
- Node color column: `weight`
- Node size column: `media.type`
- Edge width column: `weight`
- Layout style: Circular layout

Output: Network analysis video files

Reference: Please refer to the operation steps in `2c/video/Network_Igraph.mp4`

Figure 2d: Domain Adaptation Analysis

Objective: Analyze domain adaptation effects between different data sources

Primary Files:

- `2d/models/domain_adaptation.py` : Domain adaptation algorithm implementation
- `2d/models/随机森林.py` : Random Forest model
- `2d/models/逻辑回归.py` : Logistic Regression model
- `2d/roc_domain_adaptation.ipynb` : ROC analysis

Execution Steps:

1. Prepare source and target domain data
2. Execute domain adaptation algorithm:

```
cd 2d/models
python domain_adaptation.py
```



3. Train machine learning models:

```
python 随机森林.py
python 逻辑回归.py
python SVM.py
python SGD.py
```



4. Execute ROC analysis:

```
jupyter notebook roc_domain_adaptation.ipynb
```



Domain Adaptation Methods:

- `mean_teacher` : Mean teacher method
- `whitening` : Whitening method

Output: ROC curve plots and domain adaptation effect comparison

Figure 2f: Clustering Visualization

Objective: Generate multiple visualization charts for clustering results

Generation Method: Generated using Origin software

Content Included:

- **Chord Diagram** (Chord Diagram/): Display relationships between clusters
- **Heatmap** (Heatmap/): Feature correlation heatmap

- **Radar Chart** (Radar Chart/): Cluster feature radar chart

Data Files:

- 2f/Chord Diagram/data/雷达图与和弦图数据.xlsx
- 2f/Heatmap/data/36.xlsx
- 2f/Radar Chart/*.oggu : Origin project files

Execution Steps:

1. Open .oggu files using Origin software
2. Import corresponding data files
3. Generate chord diagram to display cluster relationships
4. Calculate feature correlations and generate heatmap
5. Create radar chart to display cluster features

Output: Chord diagram, heatmap, and radar chart video files

Software Requirements: Origin software

Figure 2g: SHAP Feature Importance Analysis

Objective: Analyze feature importance of SepsisFormer model using SHAP method

Primary Files:

- 2g/SepsisFormer_shap_8.py : SepsisFormer model implementation
- 2g/shap_8 (2).ipynb : SHAP analysis code

Execution Steps:

1. Load pre-trained SepsisFormer model
2. Prepare test data
3. Execute SHAP analysis:

```
cd 2g
jupyter notebook "shap_8 (2).ipynb"
```



Model Architecture:

- Transformer-based architecture
- Multi-head attention mechanism
- Feed-forward neural network

Output: SHAP feature importance plots and interpretability analysis

Figure 3c3h: Hierarchical Clustering Analysis

Objective: Perform hierarchical clustering analysis on sepsis patients

Data Files:

- 3c3h/data/mimic4_level_subphenotype_heparin.csv

Execution Steps:

1. Load patient data
2. Execute hierarchical clustering algorithm
3. Generate clustering results at different levels
4. Visualize clustering dendrogram

Clustering Methods:

- Hierarchical Clustering
- Subphenotype Clustering

Output: Hierarchical clustering dendrogram and subphenotype analysis videos

Figure 3d: 3D Visualization

Objective: Generate 3D visualization of sepsis patient features

Primary Files:

- 3d/3d.ipynb : 3D visualization code
- 3d/a.html : Interactive 3D chart

Execution Steps:

1. Load patient data
2. Execute 3D visualization:

```
cd 3d
jupyter notebook 3d.ipynb
```



Visualization Types:

- 3D scatter plots
- Interactive 3D charts
- Feature space visualization

Output: 3D visualization plots and interactive HTML files

Figure 4a4c: Cox Proportional Hazards Model

Objective: Analyze prognostic prediction capabilities of different scoring systems and subphenotypes

Generation Method: Generated using GraphPad Prism software

Primary Files:

- 4a4c/cox比例_mimic4_heparin.ipynb : Heparin-related Cox analysis
- 4a4c/cox比例_mimic4_8_subphenotype_level1.ipynb : Subphenotype Cox analysis



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Raw



- 4a4c/score_hierarchy_mimic4_*.csv : Scoring data at different levels

Execution Steps:

1. Execute Cox analysis to obtain data:

```
cd 4a4c
jupyter notebook cox比例_mimic4_heparin.ipynb
jupyter notebook cox比例_mimic4_8_subphenotype_level1.ipynb
```



2. Use GraphPad Prism software:

- Open 4a4c/4a/新评分.pzfx file
- Open 4a4c/4c/亚表型.pzfx file
- Import analysis result data
- Generate statistical charts

Statistical Methods:

- Cox proportional hazards model
- Survival analysis
- Risk stratification

Output: Cox regression result plots and statistical reports

Software Requirements: GraphPad Prism (<https://www.graphpad.com/features>)

Figure 4b: Radar Chart Analysis

Objective: Generate radar charts for multi-dimensional features

Data Files:

- 4b/data/雷达图_环状注释11.24.R : R script data

Execution Steps:

1. Execute radar chart script using R language
2. Generate circular annotation radar chart

Output: Radar chart PDF files and videos

Figure 4d: Risk Ratio Analysis

Objective: Calculate and visualize risk ratios for different factors

Primary Files:

- 4d/风险比.ipynb : Risk ratio calculation code

Execution Steps:

1. Calculate risk ratios for various factors
2. Generate risk ratio forest plot:

```
cd 4d
jupyter notebook 风险比.ipynb
```



Output: Risk ratio forest plot and statistical reports



Result Files

Each figure folder contains the following types of output files:

- **Video Files** (video/): Dynamic visualization results
<https://doi.org/10.5281/zenodo.15634368>
- **Log Files** (log/): Execution logs and intermediate results
- **Data Files** (data/): Raw and preprocessed data



Important Notes

1. **Data Paths:** Ensure all data file paths are correct
2. **Dependency Versions:** Recommend using specified dependency versions
3. **Computational Resources:** Some analyses (e.g., SHAP) require substantial computational resources
4. **Memory Requirements:** Large datasets may require sufficient memory

Technical Support

If encountering issues during execution, please check:

1. Python environment and dependency package versions
2. Data file paths and formats
3. Error messages in execution logs

Explainable AI Unravels Sepsis Heterogeneity via Coagulation-Inflammation Profiles for Prognosis and Stratification

Corresponding Author: Niu Bailin

Version 3:

Reviewer #3 (Remarks to the Author):

Overall, the revision provides satisfactory responses to the comments I included in the previous iteration. Language clarifications were made in the method's description and justifications (for domain adaptation, correlation analyses, clustering, and explainability). Improvements were made in the organization of the results. The paper now explains the selection of variables for risk models and other analyses.

Response: We sincerely thank you for your positive evaluation of our revision. We are glad that the clarifications and improvements we made in the methodology, justifications, and results organization have addressed your comments, and we truly appreciate your recognition of our efforts.

Reviewer #3 (Remarks on code availability):

The documentation of the repository has been improved with more clear instructions on how to run the code.

Response: We are truly grateful for your acknowledgment. Thank you very much for recognizing our improvements in the repository documentation and the clearer instructions provided for running the code.

Our team is deeply grateful for your feedback. Best wishes.